

TECHNICAL MANUAL

AF22M

UV FLUORESCENT SULFUR DIOXIDE ANALYZER

- JUNE 2010 -



Environnement s.a.
L'instrumentation de l'environnement

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EPA EQUIVALENCY DESIGNATION

Environnement S.A. Model AF22M SO2 Analyzer *Automated Equivalent Method: EQSA-0802-149*

“Environnement S.A Model AF22M UV Fluorescence Sulfur Dioxide Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 10 °C to 35 °C, with a 5-micron PTFE sample particulate filter, with a response time setting of 11 (Automatic response time), with the automatic “ZERO-REF” cycle ON and set for activation every 24 hours, and with or without either of the following options: Permeation oven, Rack mount / slides.

[*Federal Register*: Vol. 67, page 57811, 09/12/02]

CHAPTER 1

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1 GENERAL INFORMATION - CHARACTERISTICS

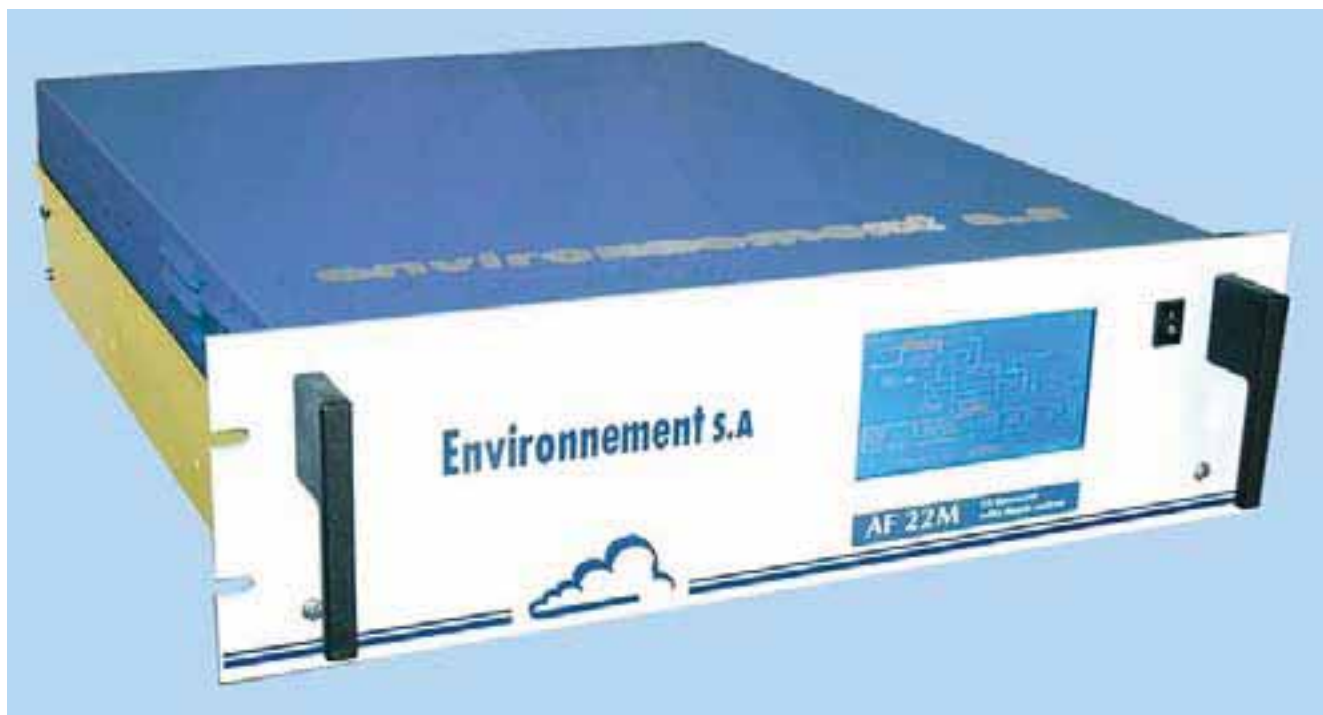


Figure 1-1 - AF22M Presentation

1.1 GENERAL INFORMATION

1.1.1 PRESENTATION

The AF22M is a continuous sulfur dioxide monitor designed for use at low contents in ambient air.

The detection principle is based on fluorescence in ultraviolet.

The monitor provides many advantages by the use of recent advanced electronic and optical technologies and requires very limited maintenance.

The sample is taken with a Teflon tube (outside diameter 6 mm) connected to the back of the unit.

The measurement is indicated by a graphic display on the front panel.

1.1.2 DESCRIPTION

1.1.2.1 Front panel

The front panel includes:

a general switch

a backlit liquid crystal display

- 16 lines 40 columns (240 x 128 pixels)
- the display provides the measurement values according to the selected unit, the information required for programming and testing the unit.

a keyboard with 6 touch-sensitive keys

The control and check functions of the unit are controlled through the keyboard.

- the function of each key varies with the different screens or menus.

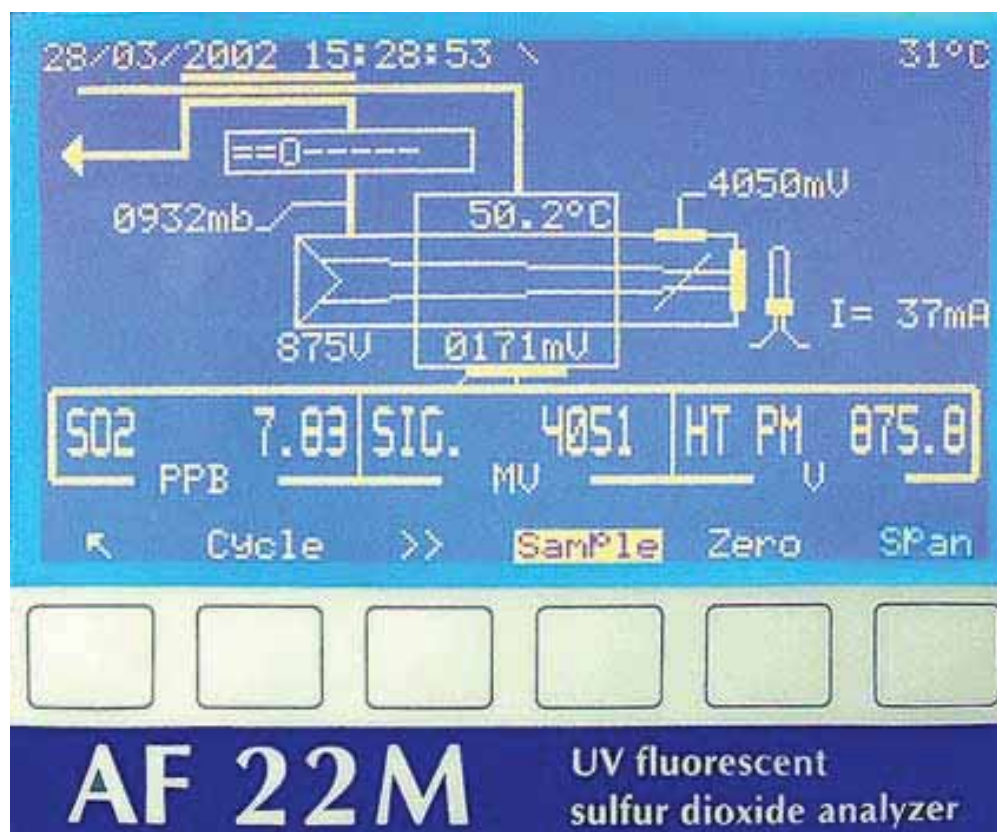


Figure 1-2 - Keyboard and display

1.1.2.2 Rear panel

The rear panel of the AF22M contains the electrical connectors and gas inlets/outlets.

Gas inlets/outlets (right hand-side)

- The inlet of sample to be analyzed is composed of a PVDF 4/6 mm fitting associated with a dust filter holder equipped with Teflon filtering membrane (1).
- The "span gas" inlet (4) consists of a 4/6 mm PVDF fitting to connect an outside span gas delivered at atmospheric pressure, or span gas excess pipe when optional permeation bench is available.



In case of permeation bench option, this inlet becomes an outlet of the permeation bench (certification de calibration certification by built-in permeation bench).

- The "pump outlet" (3) to exhaust the analyzed sample consists of a 4/6 mm PVDF fitting.
- The optional "zero air inlet" (2) allows to connect an external zero air generator.

NOTE : This inlet is sealed when optional built-in zero filter is present.

Optional : - optional permeation bench zero air inlet (10).

Electrical Equipment Connections (left side)

- The main power supply assembly consists of a 3-contact socket (5) for standard power cable connection, and general fuse: 0,6 A / 220 V or 1,2 A / 115 V (6)
- 1 standard 25-pin plug (7), for serial links COM1 (RS 232C – RS 422) and COM 2 (RS232C).
- 1 standard 37 pin (8) (see table 3-1 – DB37 and DB connectors links)

Ventilation

Ventilation is achieved by a fan (9).



Figure 1-3 – Rear panel

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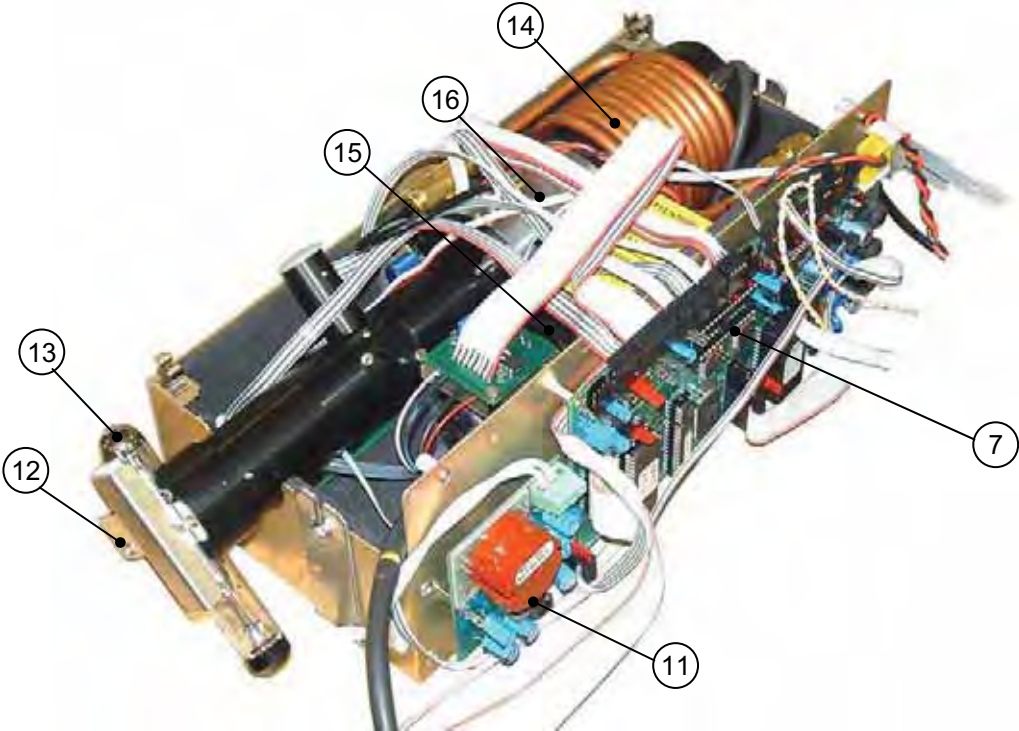
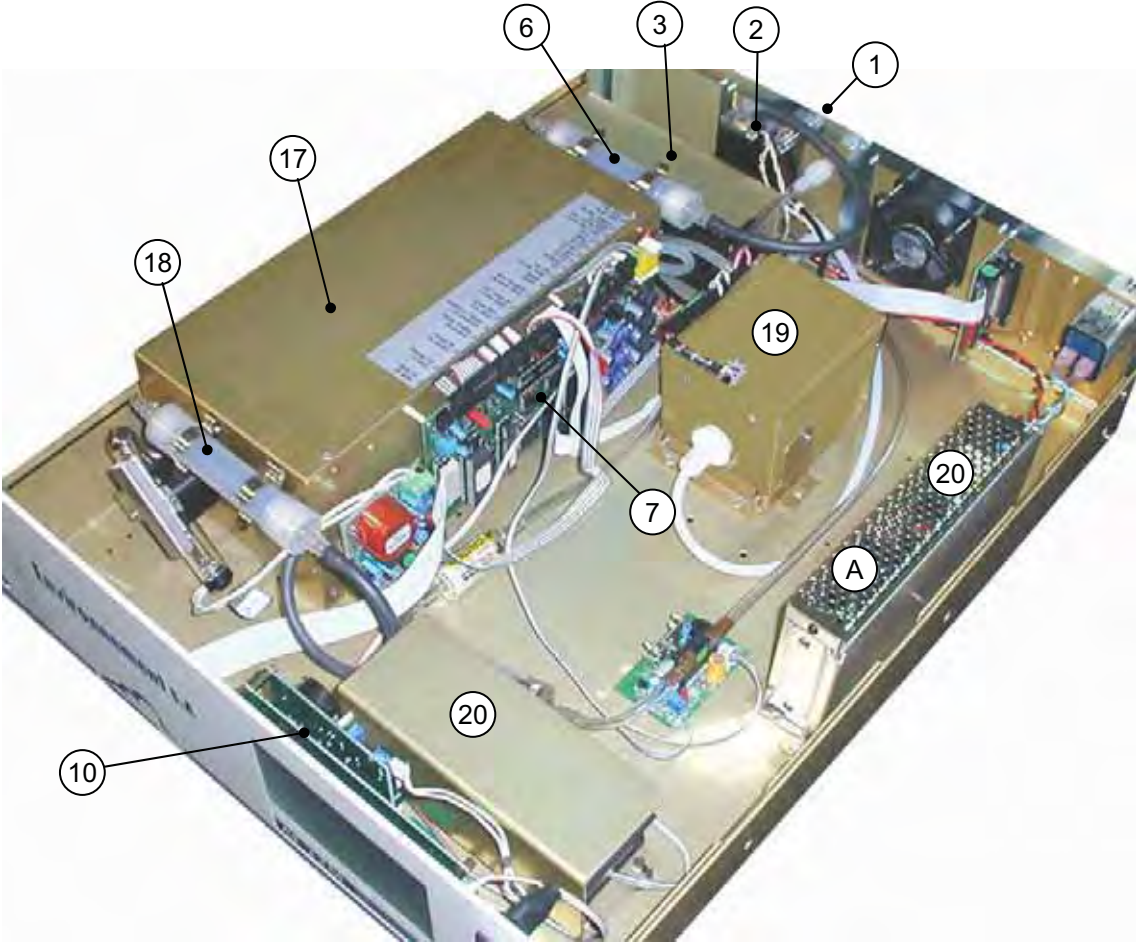


Figure 1-4 - Components location

1.1.2.3 Components locations

The components inside the unit are accessed by simply unscrewing the screws at rear and lateral sides of the unit and removing the upper cover.

Mechanical components

This includes the following equipment:

- the solenoid valves and filter assembly (1 and 2),
- the internal "Zero Air" filter (activated charcoal) (6),
- the carbon kicker filter (activated charcoal) (18),
- the temperature controlled compartment (17),
- the photo multiplier tube (15),
- the UV lamp (13).

After passing through the inlet dust filter (1), the sample to be analyzed is sent to a block of two 3-way solenoid valves (2).

These solenoid valves are used to select:

- either in "MEASUREMENT" mode, the gas sampling to be analyzed,
- or in "ZERO" mode, the filtered ambient air through the activated charcoal filter (6),
- or in "SPAN" mode, a span gas originating from:

- the permeation oven (19) if there is one (optional),

The permeation oven consists of an aluminum chamber inside which the permeation tube is placed. This chamber is regulated at a temperature of 50 °C and ventilated by an additional constant flow rate pump (20).

- or an outside calibration device.

The sample to be analyzed is drawn through a carbon kicker (14) aromatic hydrocarbon elimination device to the reaction chamber (16) in which the fluorescence takes place. Fluorescence is detected by a photo multiplier tube (15).

The temperature of the optical block is regulated at 43 °C by the microprocessor associated with a probe and a heating resistance.

Outside the compartment, the "zinc ray" type lamp (13) generating the UV radiation is powered by a stabilized voltage supply (11). The UV beam emitted is interrupted during the beginning of "ZERO REF" by a shutter (12) in order to measure the PM tube black current.

The analyzed sample passes through a flow regulating restrictor and through the external tube of the carbon kicker to eliminate hydrocarbon molecules. It is then expelled by a membrane pump (3).

The supply bloc 24V DC (20) enables to supply the whole necessary voltages.

If the user wants to modify the original supply voltage (230 V AC to 155 V AC), 2 options are necessary :



- **switch the A switch to the cut-out power supply**
- **modify value of general fuse :**
 - from 0,6 A passes to 1,2 A for a normal analyzer,
 - from 1,6 A passes to 3,2 A if the CH2S option is available.

Electronic part

- The signal output by the photo multiplier tube is amplified and converted into a digital signal on the Module board (7). This board also:
 - supplies DC voltages at + 15 V, – 15 V, + 5 V,
 - detects any alarms, transmits these alarms to the microprocessor board and signals them on the corresponding output connector,
 - performs digital/analog conversion of measurement values output by the microprocessor board.
- The microprocessor (7), calculates and stores measurement values, and manages alarms and automatic cycles.
- The keyboard interface board (10) controls the dialogue between the microprocessor board and the keyboard and the display unit.
- The UV lamp power supply (11) provides the stabilized voltage necessary for the zinc ray lamp. The current in the zinc ray lamp is regulated.

1.1.3 OPERATING MODES**1.1.3.1 Standard**

- Programmable measurement range up to 10 ppm, with a minimum detectable limit of 1 ppb (typical value is below 0,5 ppb).
- Remote-controlled or programmable automatic calibration and zero sequence.
- Automatic control of parameters influencing metrology and tests of correct functioning.
- Measurement values indicated in ppb or μm^3 .
- Memorization of average measurements with a programmable period (capacity 5700 averages).

1.1.3.2 Optional

The monitor can be equipped with the following options:

- Permeation oven.
- ESTEL board(s) :
 - Analog outputs of the SO_2 concentration and two of the 16 multiplexer channels.
 - Remote signaling of "measurement", "zero", "calibration" and "alarm" functions.

1.1.4 ASSOCIATED EQUIPMENT

- Analog recorders and data loggers.
- Numerical data acquisition system
- Serial printer for continuous printout of display measurements (programmable period) and configuration.

1.2 CHARACTERISTICS

1.2.1 TECHNICAL CHARACTERISTICS

Measurement range (programmable)	:	user programmable up to 10.00 ppm *
Units	:	ppm or mg/m ³ (programmable)
Noise (σ)	:	0.0005 ppm (RT 120)
Minimum detectable limit (2σ)	:	0.001 ppm (RT 120)
Response time (0-90 %)	:	20-120" fixed or automatic (programmable)
Zero drift	:	< 1 ppb./24 hours in operation
Span drift	:	< 1% / 24 hours
Linearity	:	$\pm 1\%$
H ₂ O influence	:	Null
Temperature influence	:	0.3 ppb/°C
Sample flow rate	:	about 415 cc/min (internal pump)
Display	:	LCD 240x128 text and graphic modes
Control keyboard	:	6 context dependent keys
Output signals	:	3 analog outputs 0-1V, 0-10V, 0-20 mA or 4-20 mA
Power supply	:	230V-50Hz (115V-60Hz) + ground
Consumption	:	280 VA starting up, 110 VA normal operation
Working temperature	:	+ 10 °C to + 35 °C
Memorization of measurement values	:	Capacity: 5700 last averages of the 3 displayed parameters
Measurement values printing	:	By serial printer connected to COM2
Alarms checks	:	– Permanent – Detection and indication of functioning anomalies: optical block temperature, sample flow rate, UV energy, PM tube high voltage, SO₂ measurement over-threshold, over-range, calibration fault ...
Tests and diagnostics for maintenance	:	Selection at keyboard and display of all parameters.
Backup saving time for data memorized in RAM and of the real-time clock.	:	> 6 months by incorporated battery.

* **0.500 ppm is the only range covered under EPA equivalency designation.**

1.2.2 OPERATING CHARACTERISTICS

Not applicable.

1.2.3 STORAGE CHARACTERISTICS

– Temperature: – 10 ° to 60 °C.

1.2.4 INSTALLATION CHARACTERISTICS**1.2.4.1 Links between units**

The AF22M monitor uses the external links and power supplies illustrated below:

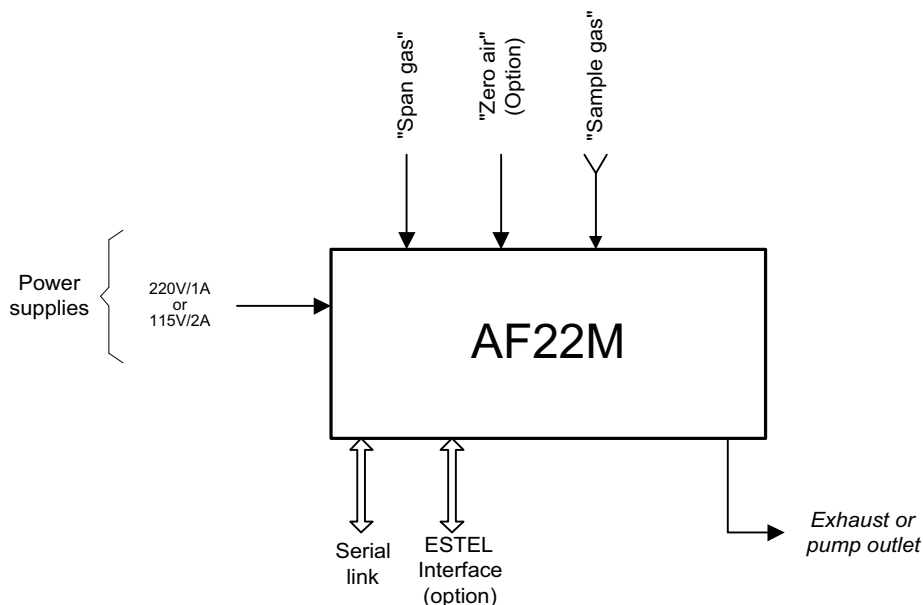


Figure 1-5 - Links between units

1.2.4.2 Dimensions and weight

The analyzer is a standard 19-inch, 3-unit rack high.

Length : 591 mm

Width : 483 mm

Height : 133 mm

Weight : 9 kg

1.2.4.3 Handling and storage

The AF22M monitor must be handled with care to avoid damage to the various connectors and fittings on the rear panel.

Ensure the fluid inlets and outlets on the unit are protected with caps whenever storing the monitor.

The unit is stored in a foam-packed case provided for this purpose.

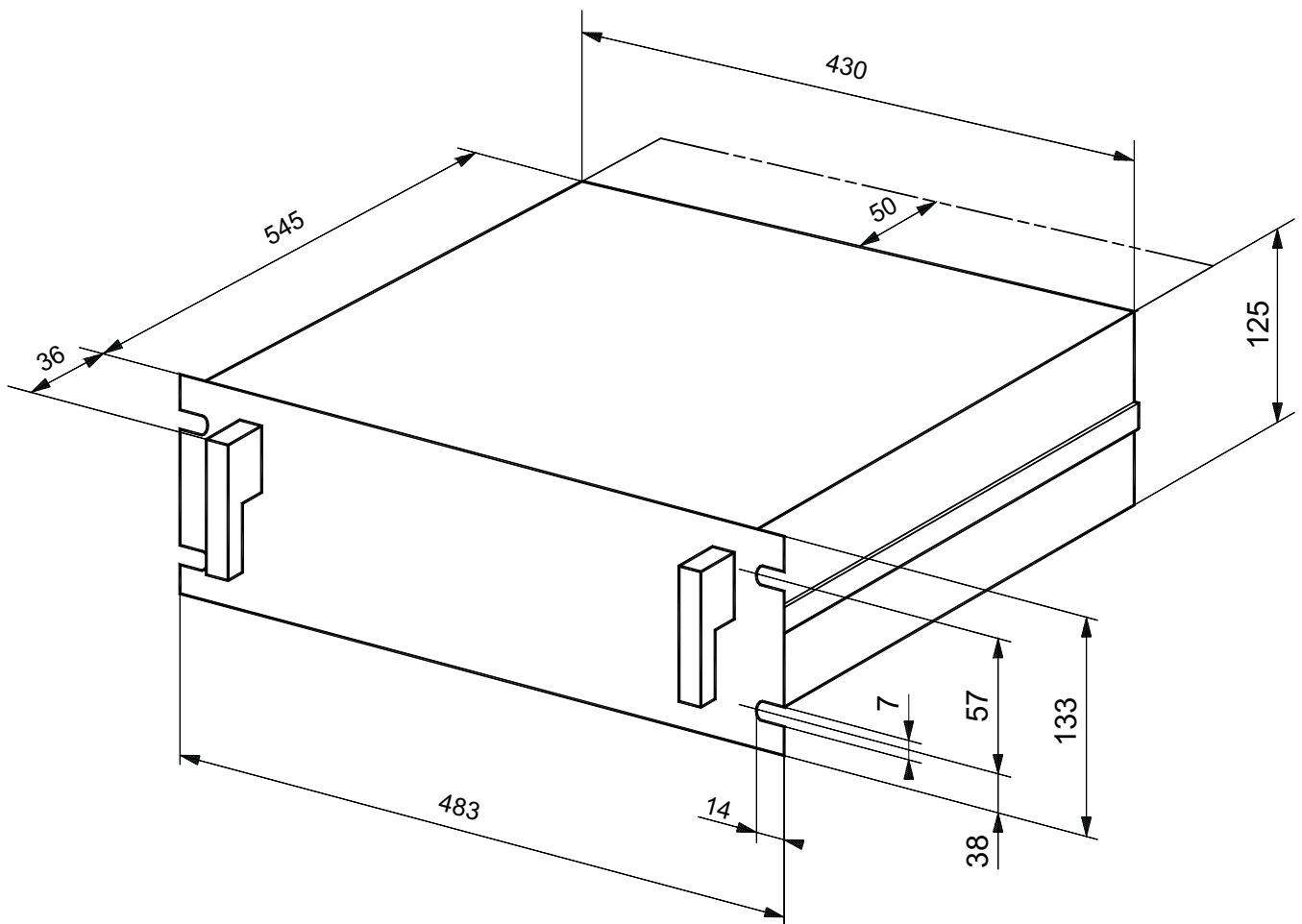


Figure 1-6 – Outline dimensions

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CHAPTER 2

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2

PRINCIPLE OF OPERATION

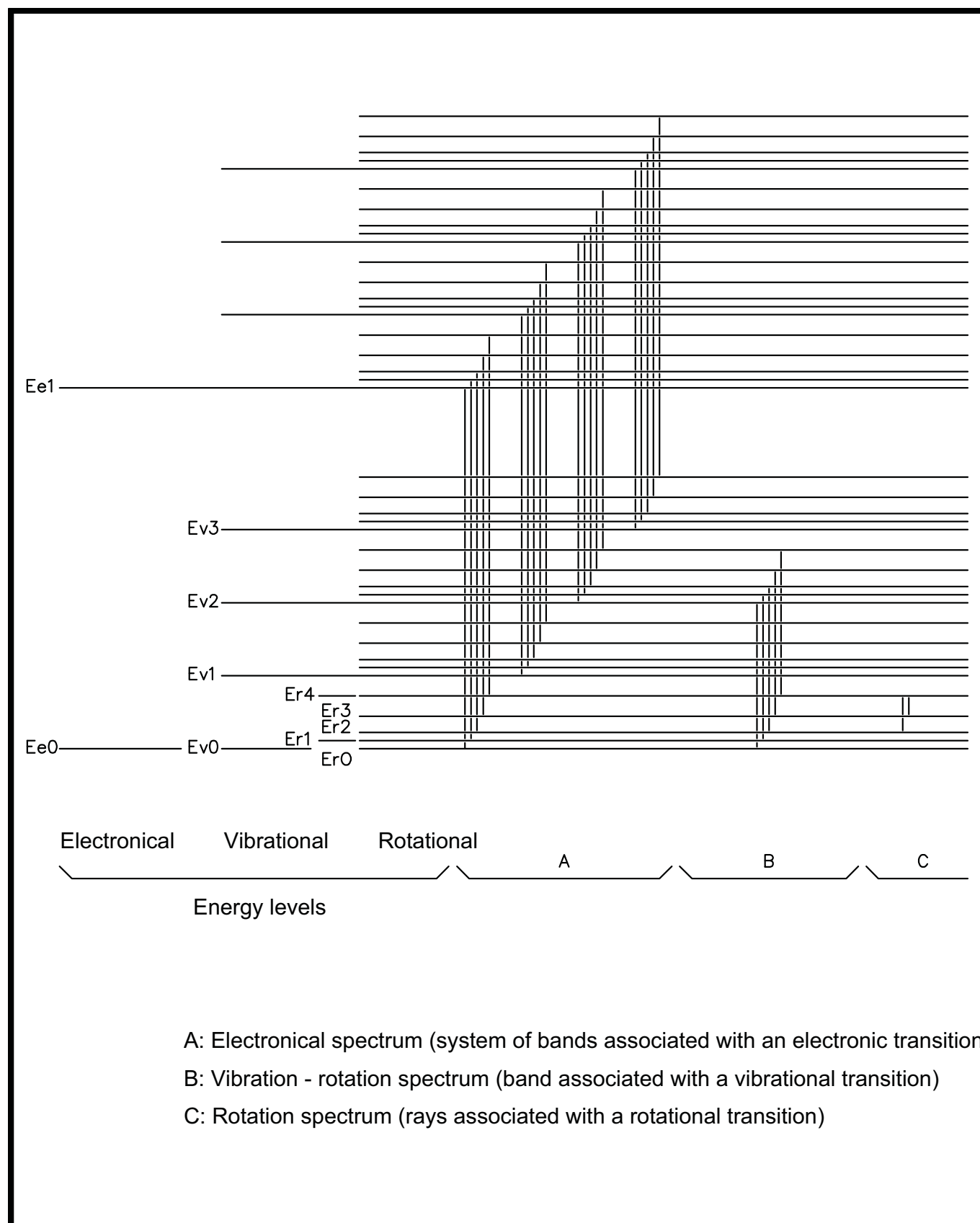


Figure 2-1 - Diagram showing molecule energy levels

2.1 THEORETICAL BASIS (FIG. 2.1)

Energy diagram:

Many developments in quantum mechanics at the beginning of this century enabled theoretical physicists to conceptualize the processes involved in energy exchanges between a gaseous molecule such as sulfur dioxide and its environment.

The diagram shown makes it easier to obtain a global understanding of the various phenomena due to absorption of a radiation by the molecule.

Two essential comments clarify this diagram:

Energy levels are quantified and distributed in accordance with a structure that is different for molecule electronic, vibrational and rotational levels,

At the considered wavelengths, the "electronic" scale of energies only includes the energy levels of molecule valency electrons.

Also, a transition from one electronic level to another by absorption of a photon is always accompanied by lower level vibrational and rotational transitions.

The mathematical formulation of molecular physics associates a spatial and temporal function called a wave function to each state of the molecule, fully characterizing this state. A transition then occurs by a new spatial distribution of the wave function.

Absorption and emission:

In its fundamental state E_{e0} , the SO_2 molecule can only absorb sufficient energy photons to access the first excited state on the electronic scale E_{e1} .

The energy of a photon is given by Einstein's relation:

$$E = h \nu = h \frac{c}{\lambda}$$

where λ is the radiation wavelength output from a low-pressure zinc vapor lamp, namely $\lambda = 213.9 \text{ nm}$, and h and c represent Plank's constant and the speed of light in a vacuum respectively.

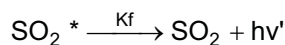
The transition is denoted in the following form:



As we have already said, the molecule always reaches a vibrational and rotational sub-level of E_{e1} , namely higher than this sub-level. It then very quickly dissipates its vibration and rotation energy to remain on level E_{e1} a little longer (a few nanoseconds).

Starting from Ee1, it can reach any sub-level of its fundamental state along different paths:

by fluorescence:



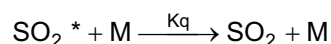
Based on the above,

firstly $\nu' = \frac{c}{\lambda'}$ is less than ν , therefore $\lambda' > \lambda$,

secondly, λ' can be equal to one of several values around an average length, since the sub-level reached can be arbitrary (statistical distribution as a function of the temperature).

The monitor "observes" photons emitted through a filter centered on 350 nm.

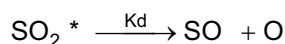
by extinction:



where M represents another gaseous molecule.

The molecule then dissipates its energy mechanically and does not fluoresce. This is referred to as "quenching".

by dissociation:



The energy Ee1 is sufficient to break the SO – O link.

Kf, Kd, Kq denote quantum yields associated with each form of deactivation. They are related to the life of state Ee1, and express probabilities attached to each reaction type. They are determined from the integral, taken over the entire volume, of the product of wave functions of excited and fundamental states.

2.2 PRINCIPLE OF MEASUREMENT

The radiation intensity absorbed by sulfur dioxide in an optical chamber of length L follows the Beer-Lambert law:

$$i_a = i_0 \times (1 - e^{-\alpha L c})$$

where i_0 denotes the intensity at the chamber inlet, α is the characteristic SO₂ absorption coefficient and $c = [\text{SO}_2]$, is the concentration of the gas to be analyzed.

Also, the probability that an excited molecule will fluoresce is given by:

$$\frac{K_f}{K_f + K_q + K_d}$$

The fluorescence intensity received by the PM is then expressed in the following form:

$$I_f = G i_a \frac{K_f}{K_f + K_q + K_d}$$

where G is a constant that depends on the illuminated part of the chamber seen by the PM.

Therefore:

$$i_f = G i_0 \frac{K_f}{K_f + K_q + K_d} \times (1 - e^{-\alpha L c})$$

In our case, $\alpha L c \ll 1$ and $1 - e^{-\alpha L c}$ can be developed to the first order as follows:

$$1 - e^{-\alpha L c} \cong + \alpha L c$$

We then obtain:

$$I_f = \frac{G i_0 K_f \alpha L}{K_f + K_q + K_d} c = \beta \cdot c$$

The radiation captured by the PM is thus directly proportional to the SO₂ concentration. This result is the basis for the measurement technique used with the AF22M monitor.

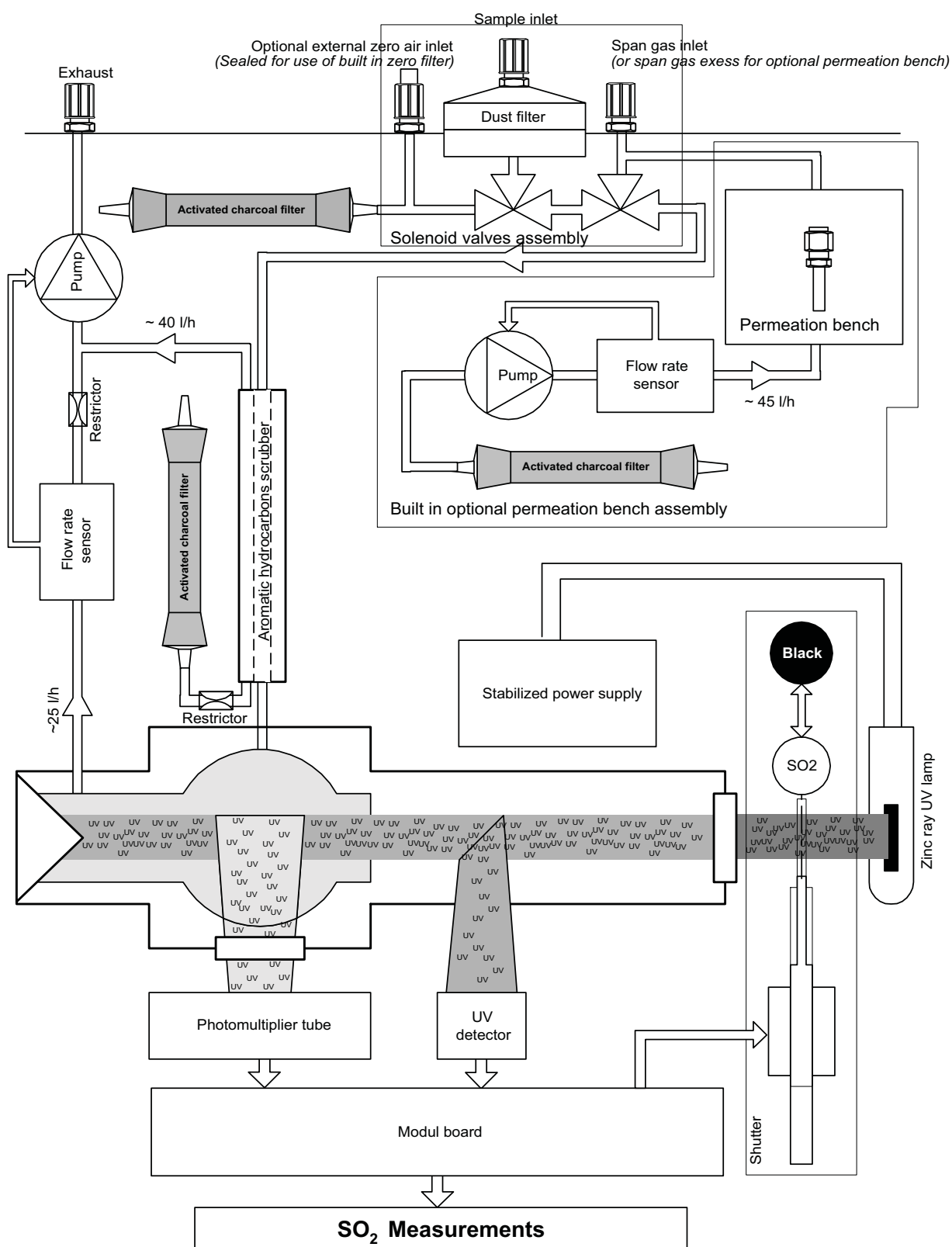


Figure 2-2 – General principle diagram

2.3 TAKING SAMPLES AND ANALYSIS

The sample is taken by a Teflon tube connected to the back of the monitor through a pump placed at the end of the circuit. A Teflon filter provides dust protection.

Filtration of hydrocarbon molecules

The sample to be analyzed is firstly filtered by an elimination device of aromatic hydrocarbon molecule.

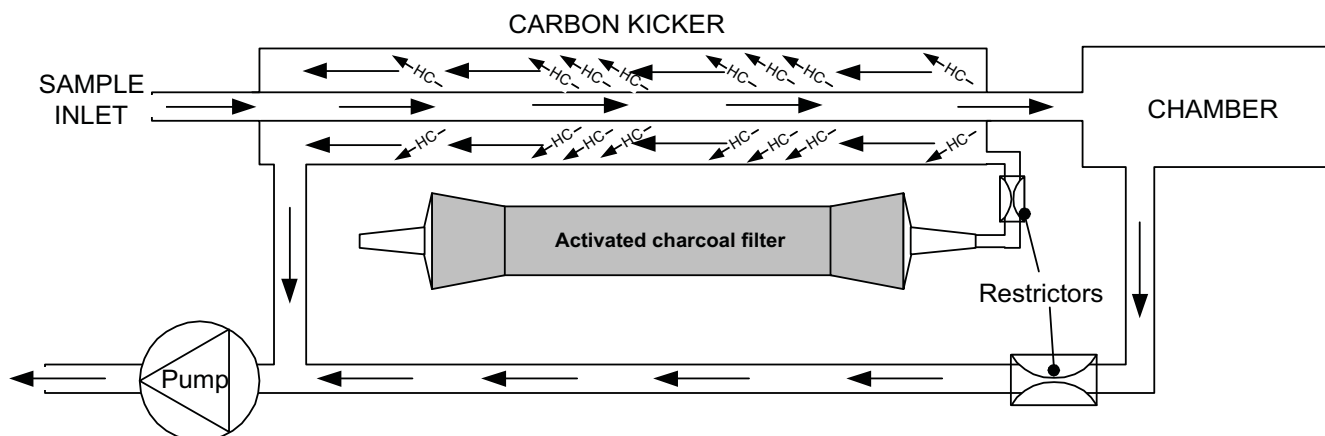


Figure 2-3 – Filtration of hydrocarbon molecules

This device consists of two concentric tubes. The internal tube is made of a special polymer (silicone).

The sample to be analyzed with aromatic HC molecules comes in the internal tube. Aromatic HC molecules are transferred by permeation to the external tube (silicone) with effect that the transfer is done in the direction: more HC molecules in gas to few HC molecules in gas.

The pump creates a vacuum in the external tube, the partial pressure of aromatic compounds decreases due to activated charcoal filtration and molecules are evacuated to the outside of the internal tube.

Analysis

The sample to be analyzed, exempt from HC molecules, is directed to a reaction chamber in which it is irradiated by an ultraviolet radiation centered at 214 nm, the absorption wavelength of SO₂ molecules.

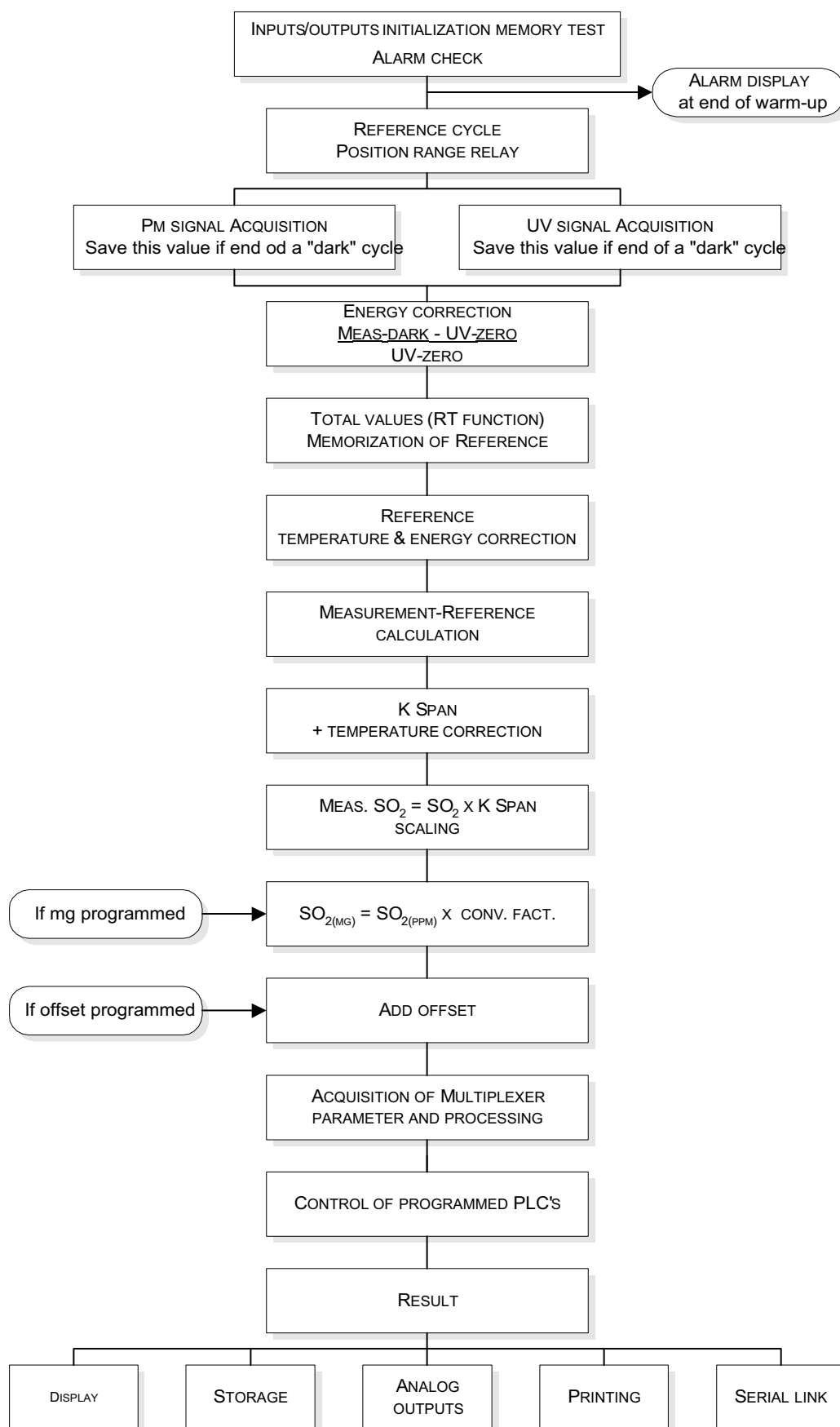
A photodiode measures the ultraviolet radiation generated by the UV lamp, through a mirror. This measurement is used during signal processing in order to compensate for any variation of the UV energy.

Molecules restore a specific fluorescence in the ultraviolet, which is optically filtered between 300 and 400 nm at the outlet in order to eliminate some interfering gases. This fluorescence is visualized by the PM tube placed near the reaction chamber.

At the start of each "zero-ref", a shutter is placed between the UV lamp and the reaction chamber inlet for 40 seconds. This electrical zero corresponds to the PM tube darkness current and the offset voltage of the preamplifier, incorporated into the signal processing, it eliminates the possibility of drifts with temperature and time.

The PM tube signal is amplified and is converted into digital values for processing by a microprocessor that calculates the average of measurement values, checks the alarms and carries out monitor operation diagnosis. These various values and information are displayed on an alphanumeric display unit on the monitor front panel.

2.4 SIMPLIFIED FLOW CHART OF PRINCIPAL PROGRAM



2.5 AUTOMATIC RESPONSE TIME

In order to optimize its metrology, the AF22M monitor is equipped with a software function called "automatic response time" which enables filtration of measurements depending on evolution of concentrations of sulfur dioxide.

2.5.1 SIMPLIFIED PRINCIPLE OF OPERATION

An average of instantaneous readings is carried out corresponding to a minimum response time.

$$[\text{MEAS}]_{\text{AVERAGE}} = \left[\frac{\sum_{1}^{80} [\text{Pm signal}]}{\sum_{1}^{16} [\text{UV detector signal}]} \right] - \text{Reference zero}$$

Then, a weighted average between the filtered values ($[\text{MEAS}]_{\text{FILTERED}}$) and the average measurements ($[\text{MEAS}]_{\text{AVERAGE}}$) is recursively calculated according to the formula:

$$[\text{MEAS}]_{\text{DISPLAYED}} = [\text{MEAS}]_{\text{FILTERED (NEW)}} = X [\text{MEAS}]_{\text{FILTERED (OLD)}} + Y [\text{MEAS}]_{\text{AVERAGE}}$$

$$X + Y = 1$$

When the difference ($[\text{MEAS}]_{\text{FILTERED (OLD)}} - [\text{MEAS}]_{\text{AVERAGE}}$) exceeds a determined threshold, the value of Y is increased, up to a maximum value of 0.98 which corresponds to a fixed response time of TR_{MIN} .

When $[\text{MEAS}]_{\text{FILTERED (OLD)}} - [\text{MEAS}]_{\text{AVERAGE}}$ is below the threshold, Y is progressively decreased.

2.5.2 PROGRAMMING THE RESPONSE TIME

The function of automatic response time may be activated or deactivated in the *configuration* $\Rightarrow$ *measurement mode* menu.

The minimum response time may also be modified in that menu.

See chapter 3 section 3.3.4.2 to obtain more information about programming of those functions.

CHAPTER 3

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3 OPERATING INSTRUCTIONS



Figure 3-1 - Electrical connections

Table 3-1 - DB37 and DB25 connectors links

RS232 / 422 serial links	
COM1	COM2
2 - TX	14 - TX
3 - RX	16 - RX
4 - RTS	7 - GND
7 - GND	
20 - DTR	
21 - TX	
11 - RX	

ESTEL BOARD(S)	
PIN N°	CONNECTION
1	+ ANA OUTPUT 1
2	+ ANA OUTPUT 2
3	+ ANA OUTPUT 3
4	+ ANA OUTPUT 4
5	+ANA INPUT 1
6	+ANA INPUT 2
7	+ANA INPUT 3
8	+ANA INPUT 4
9-28	RELAY 6 CONTACT
10-29	RELAY 5 CONTACT
11-30	RELAY 4 CONTACT
12-31	RELAY 3 CONTACT
13-32	RELAY 2 CONTACT
14-33	RELAY 1 CONTACT
15	REMOTE CONTROL 1
16	REMOTE CONTROL 2

PIN N°	CONNECTION
17	REMOTE CONTROL 3
18	REMOTE CONTROL 4
19	+5VCC
20	ANA OUTPUT GROUND
21	ANA OUTPUT GROUND
22	ANA OUTPUT GROUND
23	ANA OUTPUT GROUND
24	ANA INPUT GROUND
25	ANA INPUT GROUND
26	ANA INPUT GROUND
27	ANA INPUT GROUND
34	REMOTE CONTROL GROUND
35	REMOTE CONTROL GROUND
36	REMOTE CONTROL GROUND
37	REMOTE CONTROL GROUND

NOTE : Output relays contacts are normally open and potential free.

Remote controls are made by closing a potential free dry contact

Analog inputs accept maximum 2.5 VCC.

3.1 INITIAL STARTUP

The monitor is checked and calibrated in the factory before delivery.

3.1.1 PRELIMINARY OPERATIONS

Start-up first consists in carrying out the following preliminary operations:

- Visually examine the interior of the instrument in order to ensure that no element has been damaged during transport.
- **Remove the caps from the "gas" inlets and outlets on the unit** (keep these aside for future storage, see Chapter 1.2.3).

WARNING : Do not remove zero inlet cap if built-in zero filter is present

- Connect the 4/6 Teflon air sampling tube to the "sample inlet", after having checked for the presence of a Teflon filtering membrane in the inlet dust filter (fig.3-2)
- Connect the digital outputs to the DB25 connector (see Table. 3-1).
- Connect the analog inputs / outputs to the DB37 connector(s) (see Table 3-1).
- Connect the mains power supply cable to a socket: 230 V, 50 Hz + ground or 115 V, 60 Hz + ground depending on the voltage specified on order.

Zero air inlet of the permeation bench (option)

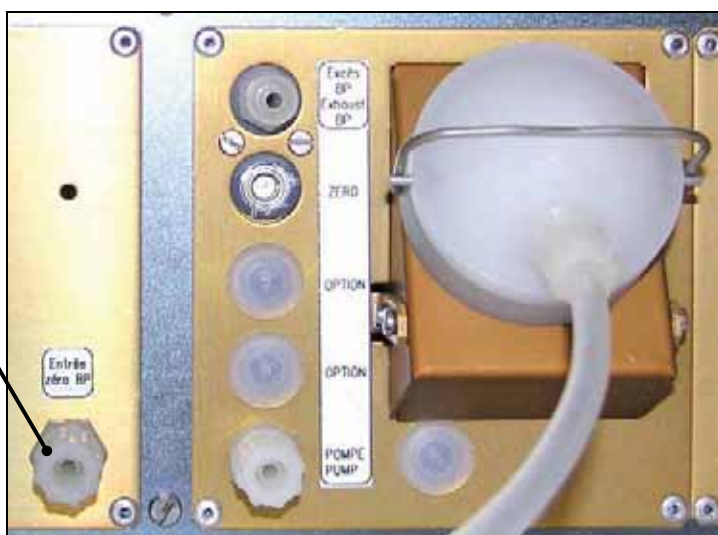


Figure 3-2 - Fluids connection

3.1.2 PERMEATION TUBE INSTALLATION (WHEN OPTION AVAILABLE)

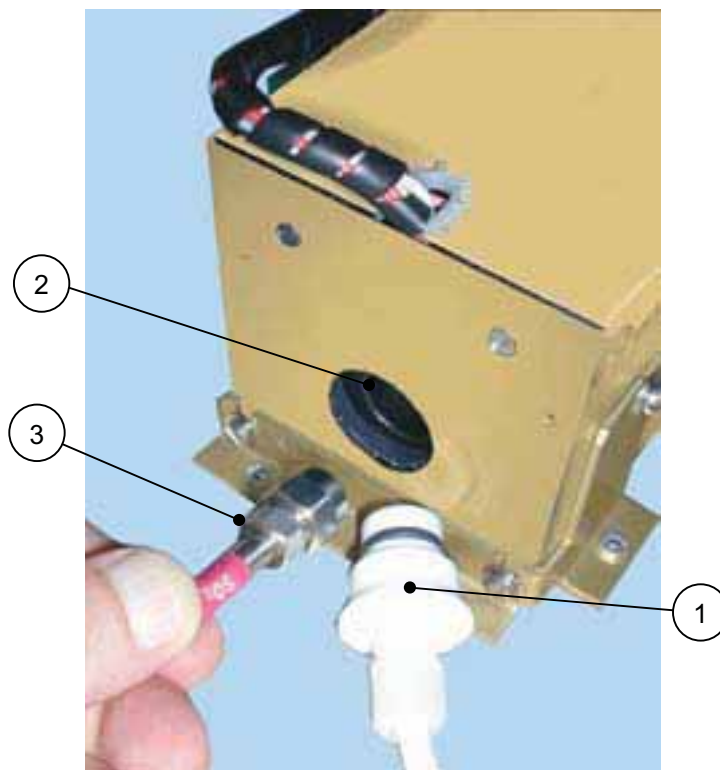


Figure 3-3 – Permeation tube installation

Pull the Teflon tap (1) of permeation oven inlet (2). After putting it out of its housing, insert permeation tube (3) in permeation oven, with porous diaphragm faced to the bottom.

NOTE : *It is essential to NOT OPEN permeation tube and to NOT PERFORATE the porous diaphragm.*

If analyzer has to be kept switched off for a while, it is necessary to remove tube from permeation oven, place it in its delivery box with drying bags, and store it in a cool place.

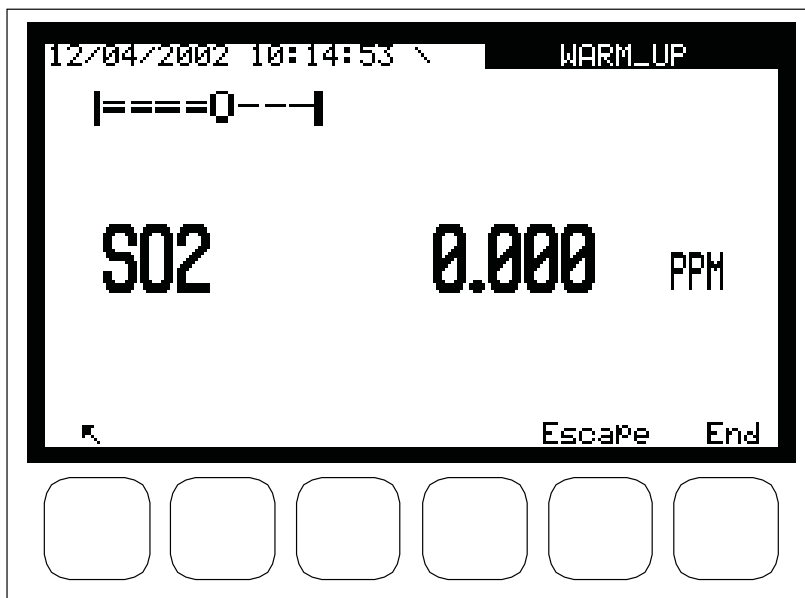
3.1.3 STARTING UP THE UNIT

Press the ON/OFF switch located on the front panel. The monitor goes into the "warm-up" cycle. The duration of this cycle is a function of the time elapsed since the last switch off.

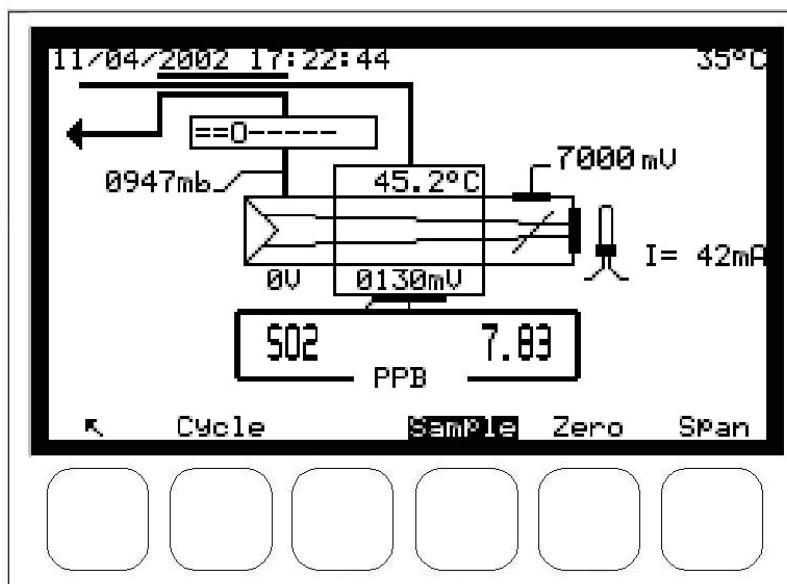
The warm-up cycle is terminated when the two following conditions are satisfied:

- All metrological parameters are within operational limits,
- The analyzer has made 10 consistent measurements within ± 4 ppb.

Display at start-up: the WARM UP message appears in the top right corner.



Display after warm-up: the measurement display mode can be chosen in *CONFIGURATION* $\Rightarrow$ *Measurement mode*. Example given here below: *Synoptic screen*.



After some time (programmable in *CONFIGURATION* $\Rightarrow$ *Measurement mode*) without action on any key, the screen passes in stand-by mode.

Pressing down any key makes it going back to display mode.

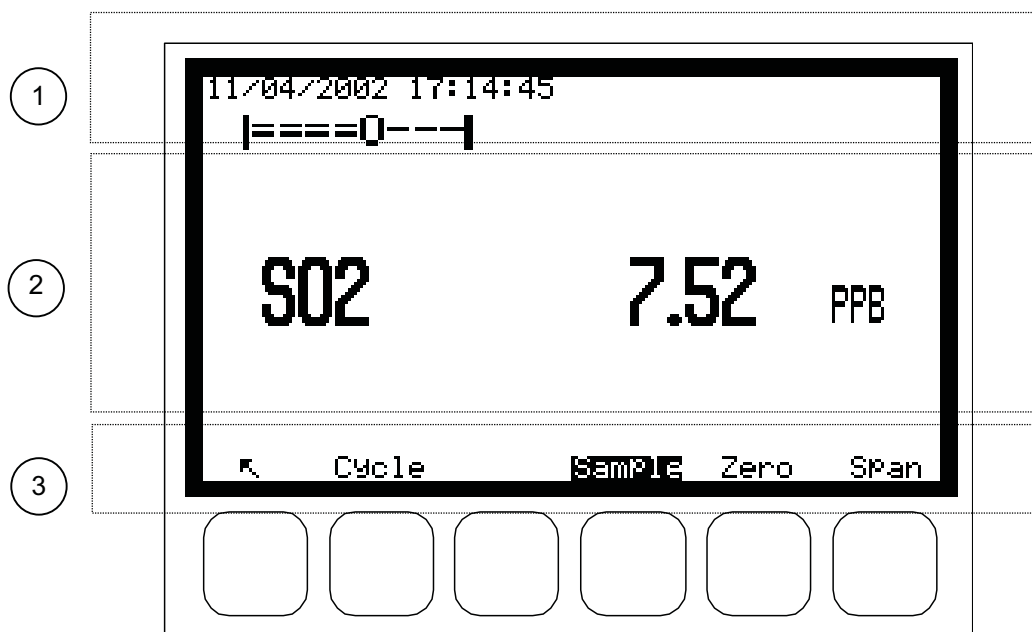
3.2 PROGRAMMING THE AF22M

3.2.1 SELECTION AND MODIFICATION OF THE PROGRAMMABLE PARAMETERS

The keyboard is located under the LCD. The bottom line gives the function of each key for the current screen.

The title of the menus and the selected fields are displayed in reverse video. By default the first line of the menus is selected. In the next paragraphs, the selected fields are symbolized in white on black background.

3.2.1.1 Screen areas definition



1

Information area: displays the date and time in the top left corner. In the top right corner, the WARM UP, REFERENCE or SPAN messages blinks. The ALARM message appears if an operating fault is detected within the instrument operating parameters.

2

Measurement or configuration area: displays the measurement parameters (gas, value, units ...) or the programmable parameters according to the selected menu.

3

Status areas and keys functions: displays the keys functions, the analyzer operating mode and the SO₂ inlet ("sample" in the example above).

NOTE : In the next paragraphs, the keys are symbolized by the icon or function displayed inside a rectangle.

3.2.1.2 Definition of the most used functions of the keyboard

(The availability of these functions is context dependent)

-  Used to display the previous menu or to abort the current operation (parameter programming, etc.)
-  Used to select the required sub-menu and the parameter to be modified. It is also used to increase the digit whose modification is in progress.
-  Used to select the required sub-menu and the parameter to be modified. It is also used to decrease the digit whose modification is in progress.
-  Moves the cursor to the left (only available during numerical parameters modifications).
-  Moves the cursor to the right (only available during numerical parameters modifications).
-  Authorizes the selected parameter modification.
-  Used to valid the selection or the parameter value whose modification is in progress.
-  It is used to print out the current screen.
-  Used to display the next page. When there are several parameters, pressing down this key allows to display the next parameters.

3.2.2 PROGRAMMING THE OPERATING PARAMETERS

3.2.2.1 Programming the digital parameters

Select the parameter with the  or  key in the appropriate menu, press down the  key to access to the modification of the parameter, the 1st digit blinks. Select the digit to be modified with the  or  key then increase it with the  key or decrease it with the  key. The  key validates the modifications of the selected field, the  key cancels the modifications of the selected field.

3.2.2.2 Programming the configurable parameters with toggle list

Select the parameter with the  or  key in the appropriate menu, press down the  key to access to the modification of the parameter, the field blinks. Select with the  or  key the wanted value in the toggle list. The  key validates the modifications of the selected field, the  key cancels the modifications of the selected field.

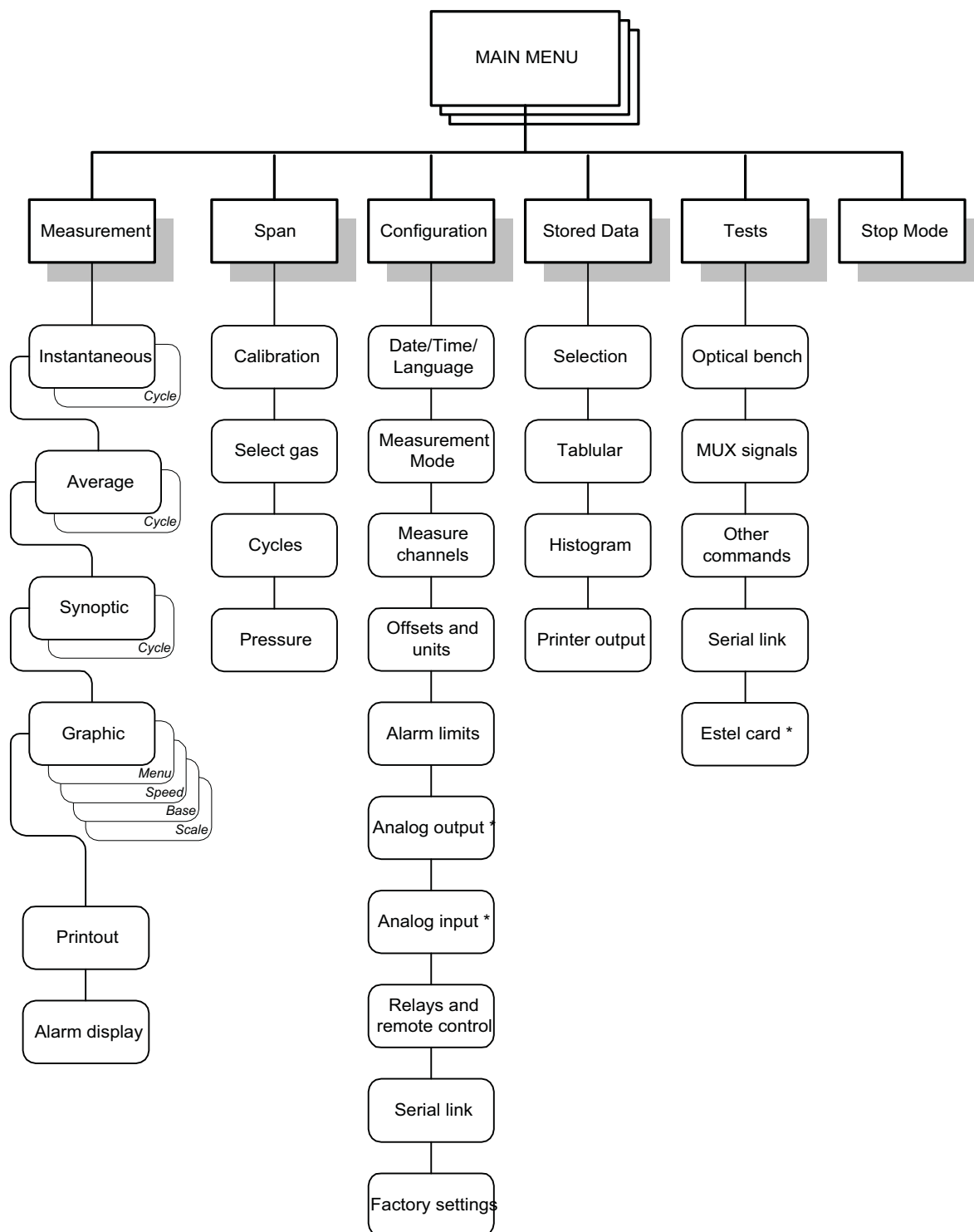


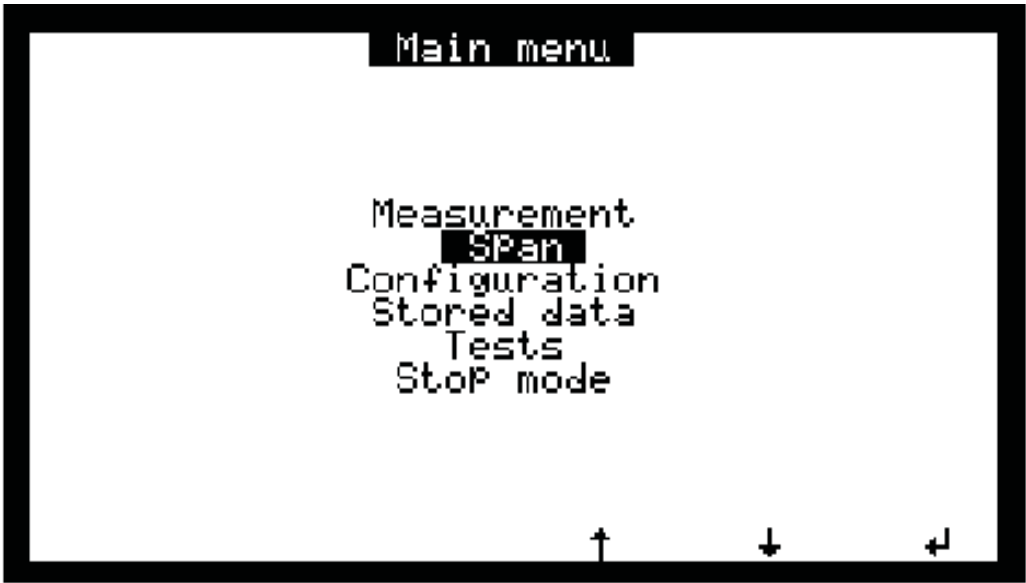
Figure 3-4 – Software overview

*: appears in the menu when option present

3.3 DESCRIPTION OF THE DIFFERENT SCREENS

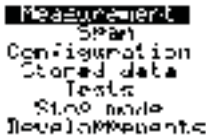
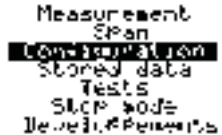
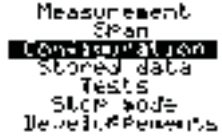
3.3.1 MAIN MENU

This screen is used to choose the menus giving access to the analyzer operating parameters



Select the menu with the  or  key, validates the selection with the  key.

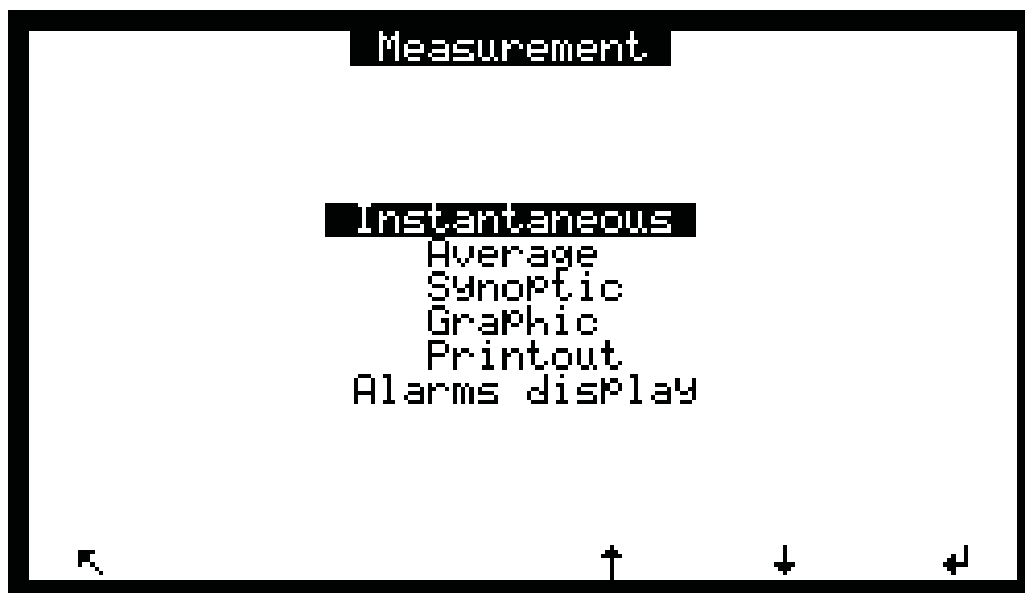
Example:

ACTION	DISPLAY	REMARKS
		- Displays the main menu, the 1 ST item is selected by default.
		- Selects the next item.
		- Selects the next item.
		- Validates the selection (Configuration menu) and displays the sub-menu.
		- Goes back to the previous menu.

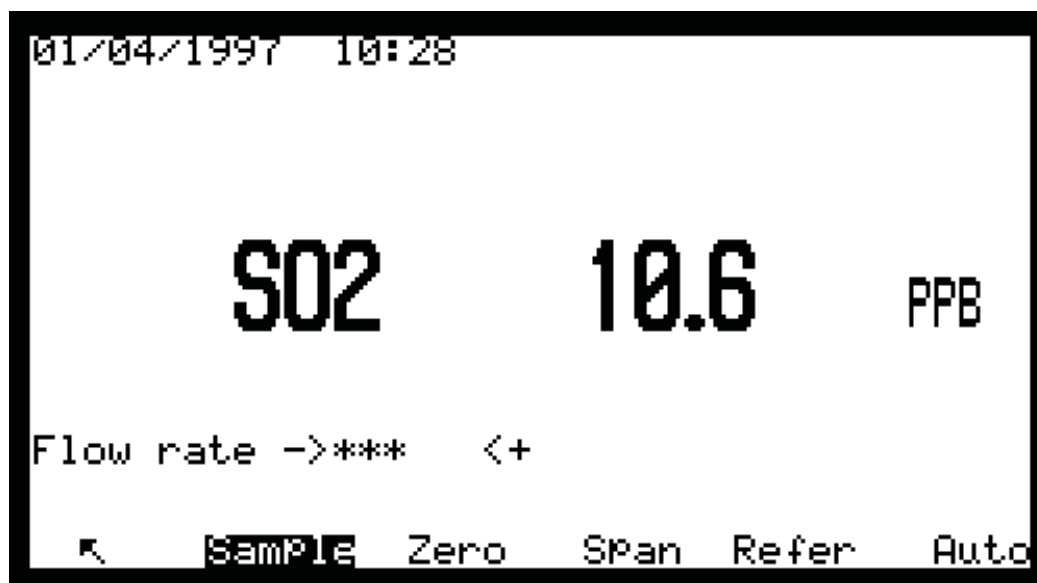
NOTE : To make the reading easier, when a sub menu is quoted in the text, the corresponding menu is reminded before (ex. CONFIGURATION ⇒ Date / time / language).

3.3.2 MEASUREMENT

This screen is used to choose the measurement display mode: instantaneous, average, synoptic (diagram) or graphic, to activate the real-time printout and to display possible alarms.



3.3.2.1 MEASUREMENT ⇒ Instantaneous



Definition of the specific keys to this screen



Selects the sample gas inlet. The gas is continuously sampled through the inlet dust filter. The measurement mode, the unit and the range are those chosen in the *Configuration* menu and in the corresponding sub-menus. This mode can be interrupted at any time by starting an automatic cycle or by manual selection of another mode or gas inlet (zero or span).



Selects the zero gas inlet. Allows manual control of the zero on the built in zero filter or on the optional external zero gas inlet. The analyzer gives its reading on zero gas (possibly increased by a programmed offset).

This operation allows to check the stability and the drift of the zero reading of the analyzer to determine the necessity to launch a reference cycle or to program its repetition period.



Selects the span gas inlet. Allows manual control of the span. The analyzer gives its reading on span gas (possibly increased by a programmed offset).

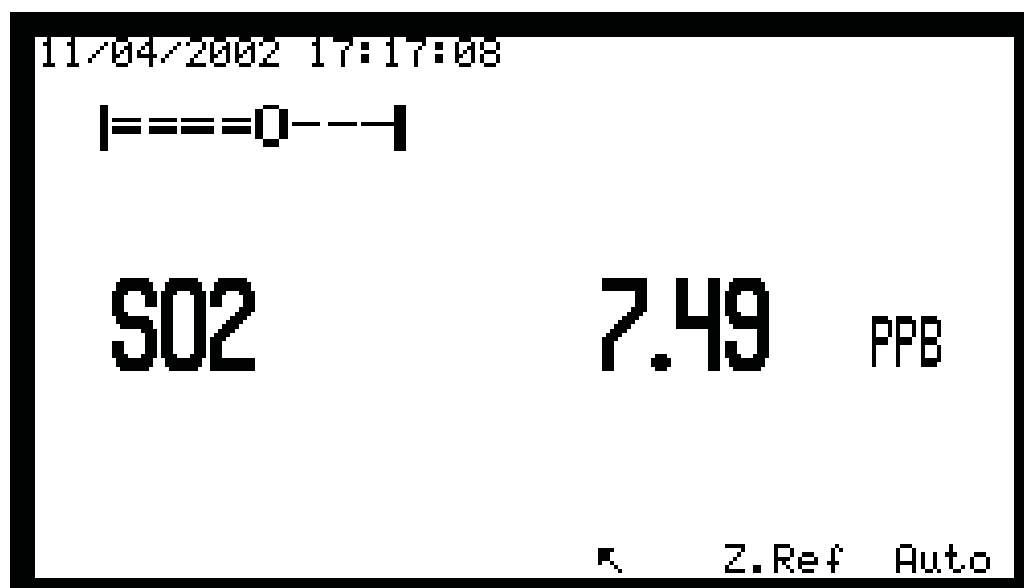
This operation allows to check the stability and the drift of the span reading of the analyzer to determine the necessity to launch an auto-span cycle or to program its repetition period.



Used to display the next page. When there are several parameters, pressing down this key allows to display the next parameters.



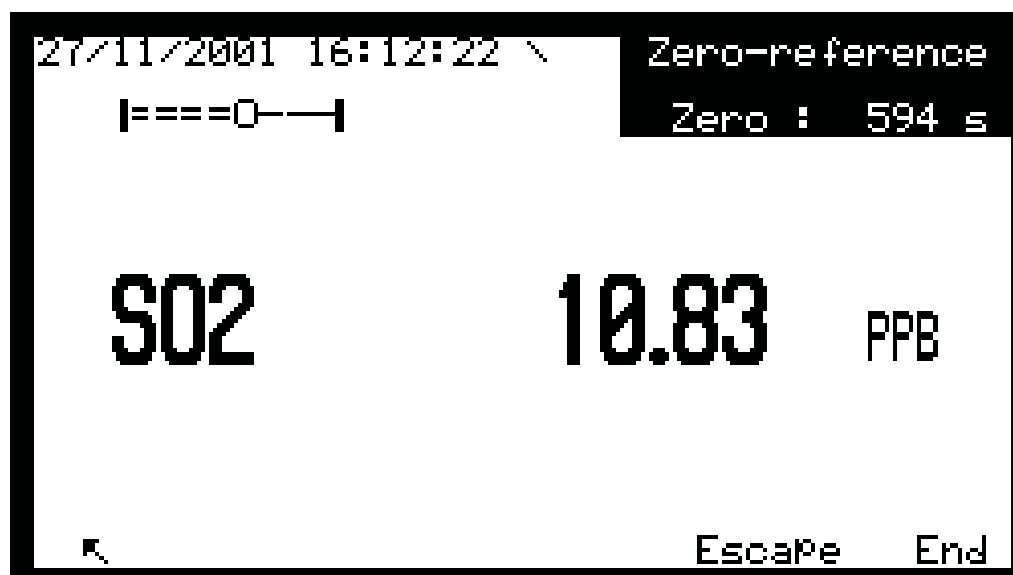
Gives access to the screen allowing manual launching of cycles.



Definition of the specific keys to this screen



Allows to manually launch an automatic corrective cycle in case of electrical zero and zero gas difference. At the top right corner, zero reference and the gas inlet on which the cycle is conducted are displayed.



Allows to skip the current cycle without computing the new Zero-reference.



Allows to manually shorten the cycle duration.

Be careful : the new zero-reference displayed value is computed, this could occur a wrong zero reference value.



Allows to launch manually an automatic span cycle. The analyzer adjusts automatically its span factor K in order to equal its reading value (minus the programmed offset) and span gas concentrations. The span gas concentrations are programmable in the *SPAN* $\Rightarrow$ *Select gas* menu, the concentration programmed for the gas inlet used for calibration is reminded in the top right corner of the screen (CAL=XXXX). The adjustment takes place for the duration programmed in the *SPAN* $\Rightarrow$ *Cycles* menu. The cycle time duration countdown is displayed in the right top corner of the screen, the cycle is achieved when count is 0000 sec. Cycle can be shorten pressing down the  key. The new span factor is then memorized if reading value equals $\pm 5\%$ span concentration.



Allows to skip the current cycle without computing the new K-Span.



Allows to manually shorten the cycle duration.

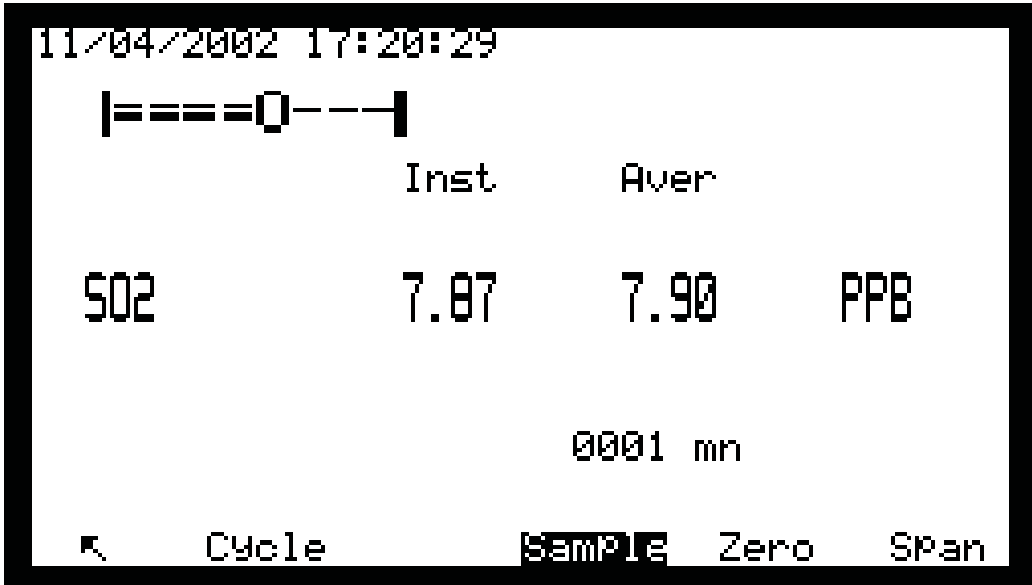
Be careful : the span factor is computed even if the read and the target values are not equal. This could occur a wrong span factor.



This function launches the auto-span using the gas inlet selected before pressing down the  key, select the correct gas inlet before executing an auto span

To go back to normal measurement after manual auto-span, press down the  key to select the sample inlet again.

3.3.2.2 MEASUREMENT ⇒ Average

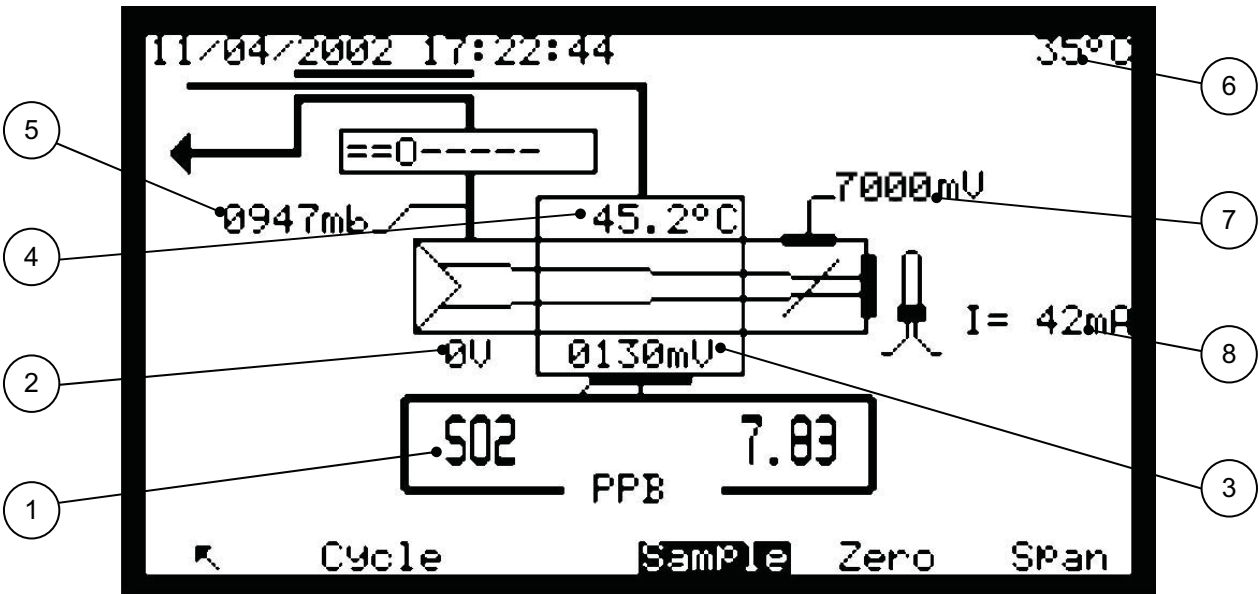


Definition of the specific keys to this screen

Sample Zero Span Cycle have the same function as the screen MEASUREMENT ⇒ Instantaneous.

3.3.2.3 MEASUREMENT ⇒ Synoptic

This screen represents the flow circuit and displays significant operation parameter values: Gas, concentration and unit (1), PM high voltage (2), PM signal (3), optical bench temperature (4), Gas pressure (5), internal temperature (6), UV detector signal (7), voltage proportional to UV lamp current (8).

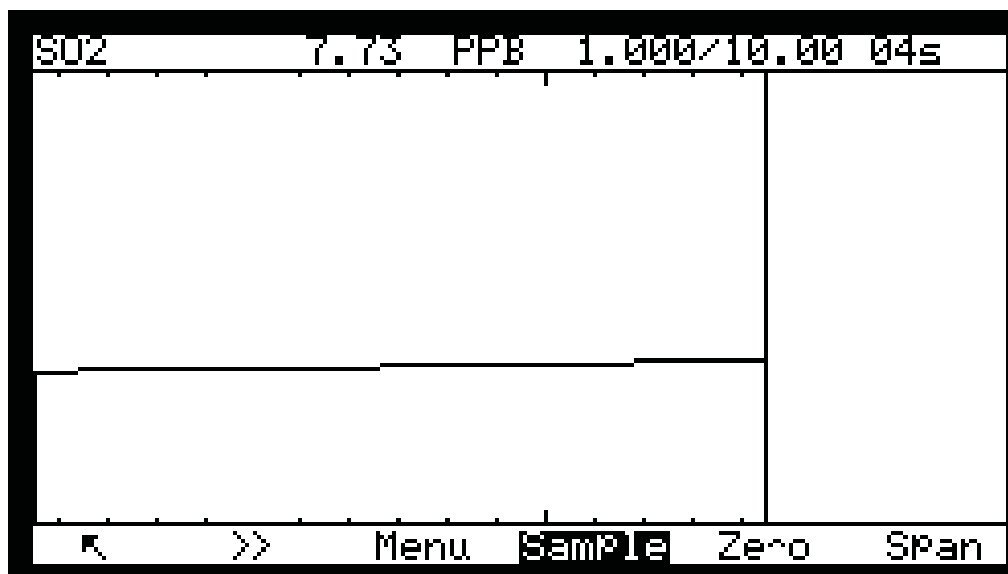


Definition of the specific keys of this screen

Sample Zero Span Cycle Have the same function than the screen MEASUREMENT ⇒ Instantaneous.

3.3.2.4 MEASUREMENT ⇒ Graphic

This screen is used for graphic plotting of the measurement values on sample or zero/span gas inlet. The vertical line shows the current position: The refreshed measurements are given on the left side of this line. The vertical full scale of the graphic is the one programmed for the analog outputs.



Definition of the specific keys to this screen

have the same function as the screen *MEASUREMENT* ⇒ *Instantaneous*.

When sample inlet is already active, pressing down the key refreshes the graph. Press down the

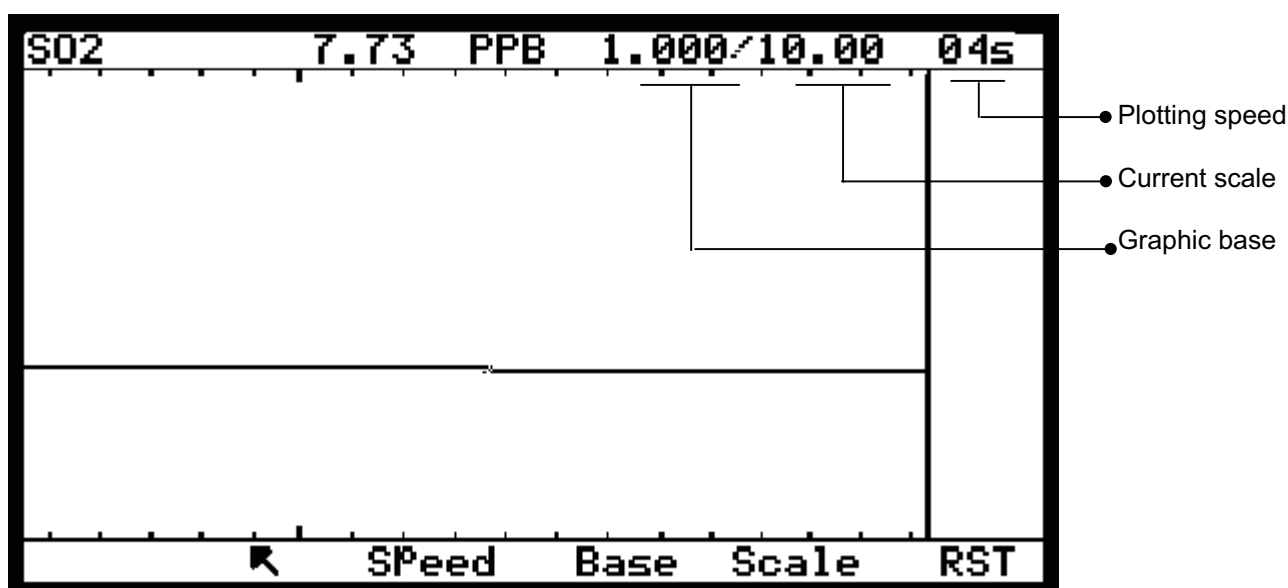
key to select the parameter to display.

3.3.2.4.1 Graphic ⇒ "Menu" screen

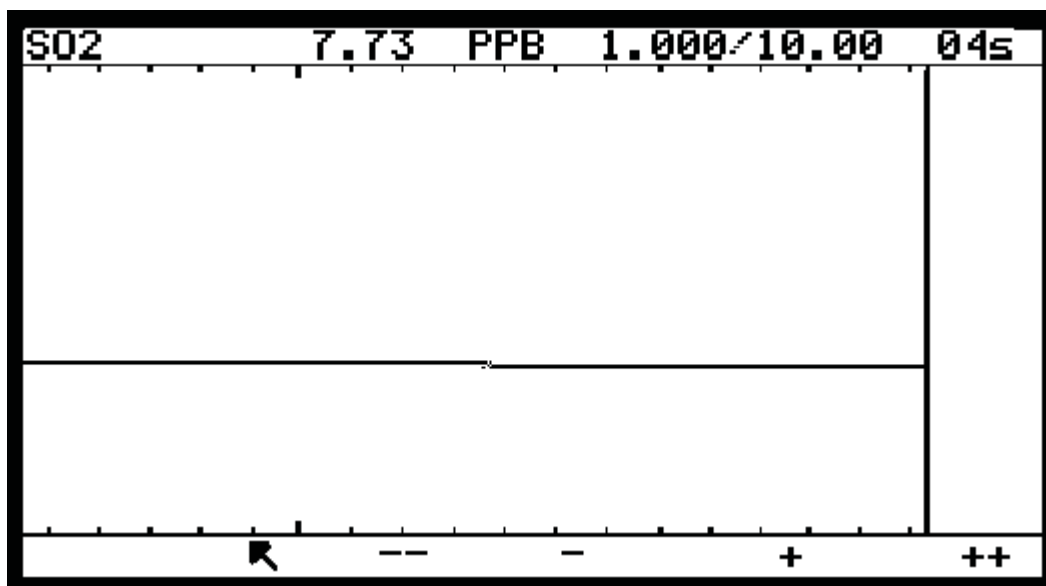
Pressing down the key gives access to the following graphic adjustments:

- Plotting speed
- Base line
- Full scale

The key is used to reset the graph to zero.



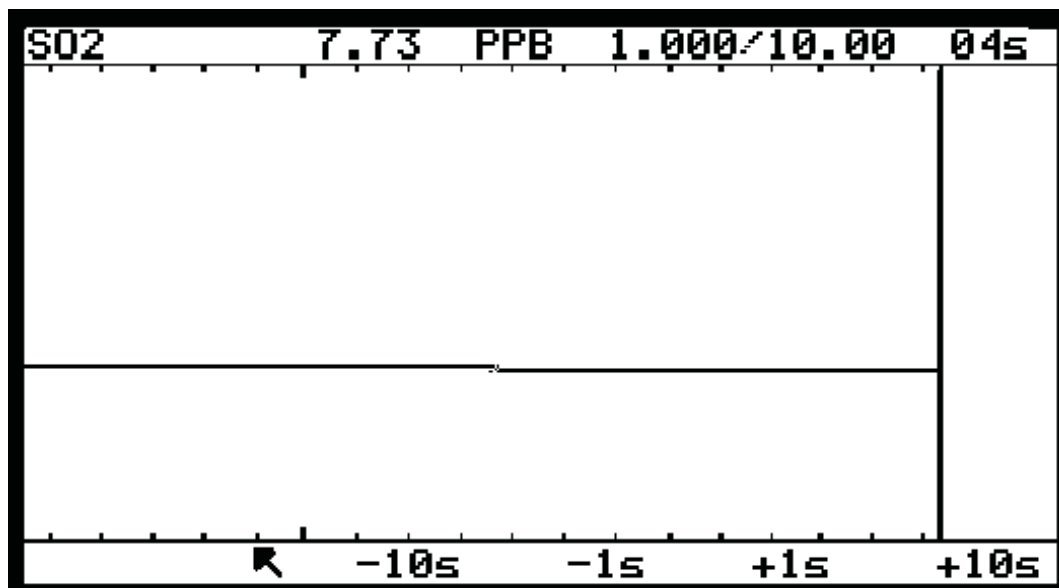
3.3.2.4.2 Graphic ⇒ screen "Base"



Pressing down the Base key allows to adjust the value of the graph base line (the minimum value is zero, the maximum value is just inferior to the full scale).

- Divides 10 times the current base line (when the base line is 5, it resets to zero)
- Selects the inferior base line among 5000, 2000, 1000, 500, 200, 100, 50, 20, 10, 5, 2, 1, 0.
- + Selects the superior base line among 0, 1, 2, 5, 10, 20, 100, 200, 500, 1000, 2000, 5000.
- ++ Multiplies 10 times the current base line.

3.3.2.4.3 Graphic ⇒ screen "Speed"



Pressing down the Speed key allows to adjust the plotting speed on the screen (the minimum value is 1 second, the maximum value is 60 seconds).

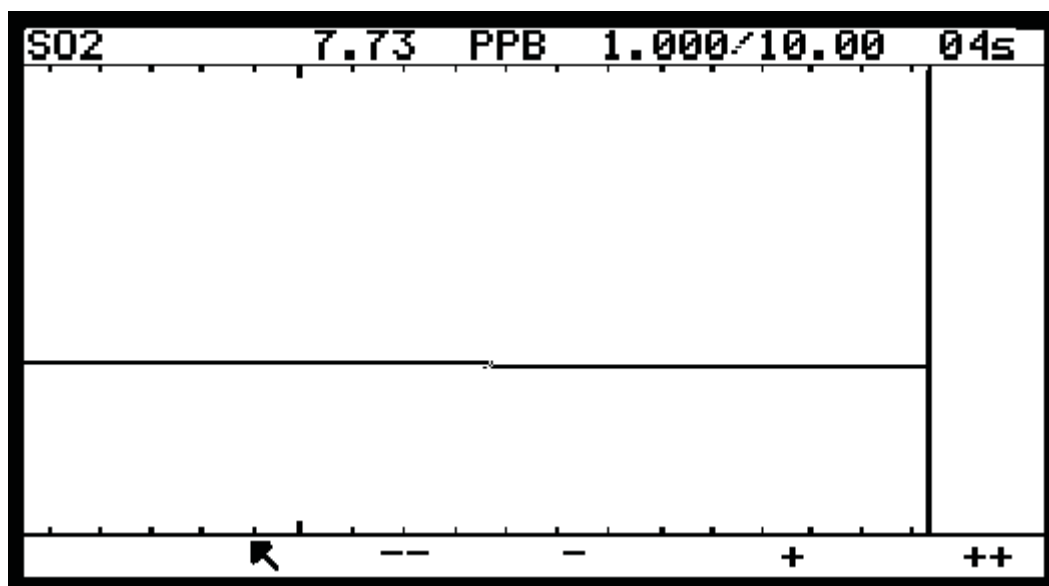
- Decreases 10 seconds to the current plotting speed
- Decreases 1 second to the current plotting speed.
- Increases 1 second to the current plotting speed.
- Increases 10 seconds to the current plotting speed.

The programmed time is the interval duration between 2 points of the graph

E.g.: when the plotting speed is 10 seconds, the graphic screen lasts for $240 \times 10 = 2400$ seconds.

3.3.2.4.4 Graphic $\Rightarrow$ screen "Scale"

Pressing down the key allows to adjust the full scale of the graph (the minimum value is just superior to the base line, the maximum value 10000)

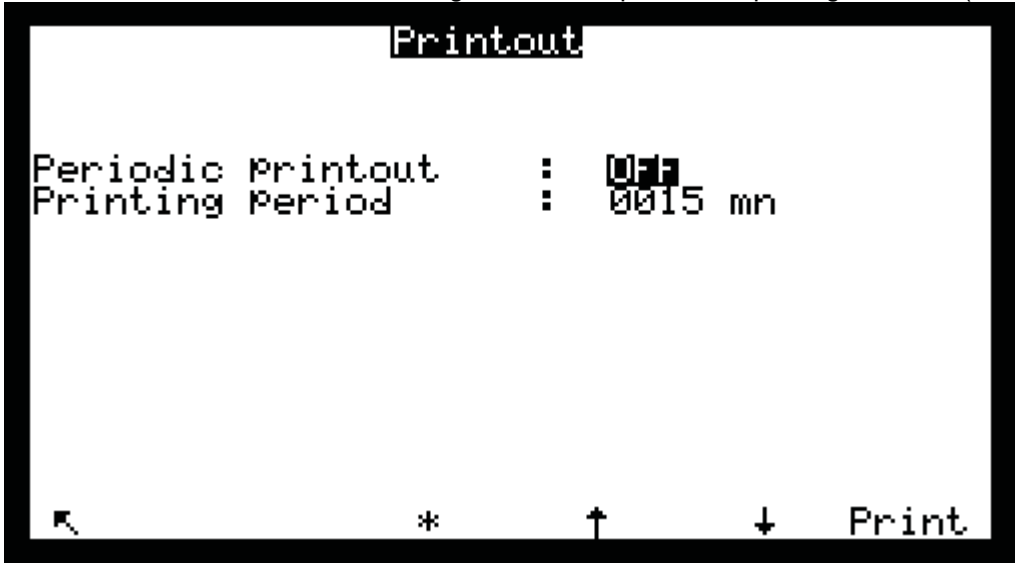


This screen allows to adjust the full scale of the graphic (minimum not exceeding the basis line, maximum value reaching 1000).

- Divides 10 times the current scale (when the scale is 5, it reset to zero.)
- Selects the current scale among 5000, 2000, 1000, 500, 200, 100, 50, 20, 10, 5, 2, 1, 0.
- Selects the current scale among 0, 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000.
- Multiplies 10 times the current scale.

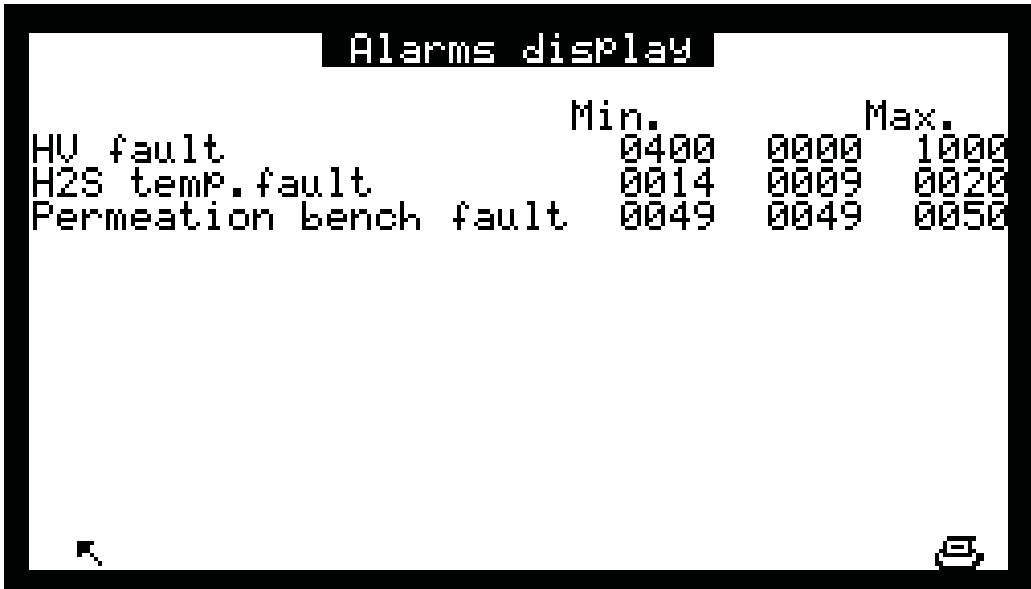
3.3.2.5 MEASUREMENT ⇒ Printout.

This menu is used to start real time printout on a serial printer connected to the COM2 serial port. It is also used to define measurement average calculation period and printing rate from (0001 to 9999mn).



3.3.2.6 MEASUREMENT ⇒ Alarms display

This screen displays the operating faults in case of alarm. Possible corrective actions for these faults are given in chapter 5.



3.3.3 SPAN

This menu gives access to the followings functions:

- Programming the calibration factors K.
- Programming the span gas values.
- Gas inlet selection for span cycles.
- Programming the period and duration of the automatic cycles.
- Pressure sensors calibration

Programmable digital potentiometer (E2Pot) allows to:

- Adjust mercury lamp current and consequently measurements and reference signal amplitude;
- Adjust flow rate control point.



3.3.3.1 SPAN ⇒ Coefficients

This screen is used to modify manually calibration factors. The factors variations after a new auto-cal cycle are displayed in the "Delta %" fields. To reset manually the Delta % in case of calibration alarm due to a wrong use of the auto-cal function, select the Factor field of the gas whose the Delta % is greater than 5.0, press down the  key and the  key. Exit the screen by pressing down the  key and press down the  key to select again the *SPAN ⇒ Calibration* screen in order to refresh the Delta % display.



3.3.3.2 SPAN ⇒ Select Gas.

This screen is used to associate a span gas concentration to each gas inlet. These concentrations are the reference values for manual or automatic auto-calibration cycles.

	Measure	Zero	SPan	
S02	180	0	100	PPB

Navigation icons: cursor, *, ↑, ↓, printer

3.3.3.3 SPAN ⇒ Cycles

This screen is used to program the period and the duration of the automatic cycles, **the durations programmed here are also those of the manually launched cycles.**

The possible automatic cycles are:

ZERO : zero air check

SPAN : span gas check.

Z.Ref : automatic zero adjust correction

AUTO : automatic span factor correction

	Zero	Span	Z.Ref.	Auto.
Function modes				
Cyclical:	OFF	OFF	ON	OFF
Remote:	OFF	OFF	OFF	OFF
Settings				
Inlet:	Zero	Span	Zero	Span
Period:	0024 h	0024 h	0024 h	0024 h
Timing:	0600 s	0600 s	0600 s	0600 s
Starting time:		00:00		

Navigation icons: ← * ↑ ↓ →

The "Remote" fields are used to configure the cycles remote controls (ESTEL board option), ZERO, ZERO REFERENCE, AUTO or SPAN. The status programmed in the "Cyclical" fields (ON = active, OFF = inactive) governs the analyzer reaction when a dry contact is closed on the remote controls inputs (see Table 3.1).

The "Inlet" fields allow to select gas inlets used during the automatic sequences. The reference concentrations for automatic calibration are those programmed in the previous menu.

The "Starting time" field is used to program the hour when the cycles are launched. If a 24 h ZERO cycle, a 24 h AUTO cycle, and a 24 h SPAN cycle are programmed, the sequence is launched at starting time according to the following priority: Z.Ref., ZERO, AUTO, then SPAN.

To inhibit an automatic cycle, program 0000h in the "Period" field. To inhibit both an automatic and a manual cycle, program 0000s in the "Timing" field.

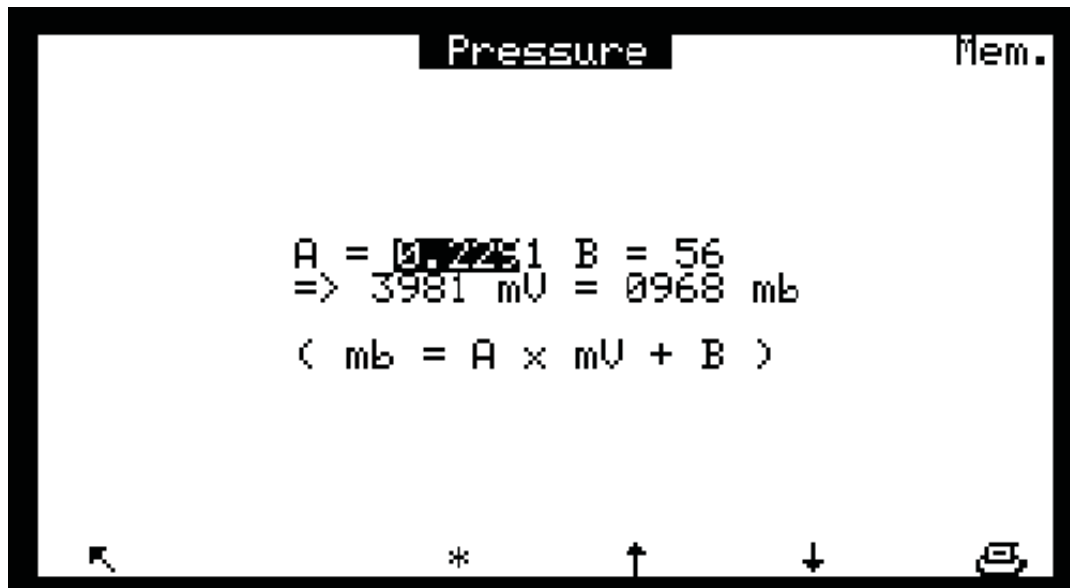
3.3.3.4 SPAN $\Rightarrow$ Pressure

This screen is used to program the calibration curve of the pressure sensors.

Pressure sensors calibration:

Connect a reference pressure sensor in parallel to the pressure sensor to be calibrated.

Enter the slope value (A) and its intercept (B) in the pressure calibration window fields.



3.3.3.5 SPAN $\Rightarrow$ E2Pot

This screen allows to adjust, in a digital way, electric gains on measurement signals.

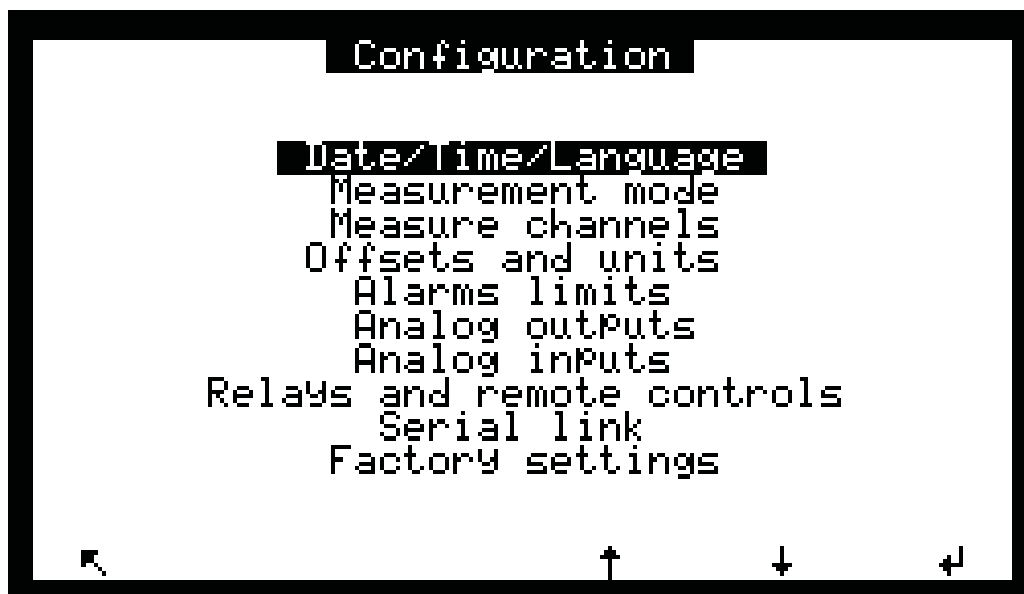
- ☐ + Pressing down this key increase the signal
- ☐ - Pressing down this key decrease the signal



3.3.4 CONFIGURATION

This menu gives access to the following functions:

- Response time programming.
- Dilution function programming.
- Analog outputs configuration.
- Unit change and offset adjustment
- Alarm thresholds, activation and assignment of the alarm relays.
- Serial link programming.
- Reset of the main programmable parameters.



3.3.4.1 CONFIGURATION ⇒ Date/time/language

This screen is used to set the internal clock of the analyzer, as well as to choose the displayed language among French, English, German, Italian and Spanish. It also shows the software version number to remind in case of software dysfunction.



3.3.4.2 CONFIGURATION ⇒ Measurement mode

This screen is used to program the response time from about 20 sec. to 120 sec. The first digit of the programmed number activates (1) or de-activates (0) the automatic response time function, the second item divides the basic electronic integration time (120 sec.)

Examples: Response time = 13 gives an automatic response time with a minimum of 40 sec.
Response time = 03 gives an integration time fixed to 40 sec.

The advised value is; Response time =11

(see chapter 2, automatic response time)

This screen gives also access to the dilution function: to measure very high concentrations (i.e. founded in industrial environment), it is necessary to bring them to values corresponding to the range of analyzer by inserting a dilution system in the sampling line to get:

$$C_{\text{analyser inlet}} = \frac{C_{\text{sample}}}{K_{\text{dilution}}}$$

The real concentration display is obtained by applying a K Dilution multiplying factor to the measured concentration.

"Latch DAC" field: when this field is ON, it latches the analog outputs to the last measurement values during the zero or span cycles in order not to perturb eventual data loggers.

"Maintenance" field: when this field is on, it allows to trigger one of the alarm relays (see section 3.3.4.5 and table 3.1). The maintenance mode condition is reminded on the *MEASUREMENT* screens.

"Starting screen" field allows to choose the screen displayed after warm-up when starting on the analyzer. 4 choices are offered: *Instantaneous*, *Synoptic*, *Average*, *Graphic*, corresponding to the menu MEASUREMENT.

Screen std -by delay" field allows to program the time delay after which, without any action on keyboard, the screen passes in stand-by mode.

"Zero filter life time" field allows to program a day down-counter which triggers, when zero, a filter alarm message. The setting value depends on the analyzer using conditions. The factory setting value, 365 days, corresponds to the advised maintenance frequency.

```

Measurement mode Mem.
Measure
Response time      : 11
Dilution
Probe              : OFF
Factor            : 0001
Other
Latch DAC         : OFF
Maintenance       : OFF
Starting screen    : Instantan.
Screen std-by delay : 8h
Gain              : X10
Zero filter life time: 0174 d
  
```

3.3.4.3 **CONFIGURATION ⇒ Measure channels**

This screen is used to select the parameter, the display format and the unit for each measure channel. The measure channels programming allows to display (*MEASUREMENT ⇒ Instantaneous* screen or *MEASUREMENT ⇒ Average* screen) and to store (*STORED DATA* menu) other parameters than the one displayed by default (SO₂). It allows to store MUX channels and analog inputs (ESTEL option).

Nb	Channels	Formats	Units
1	SO2	Auto.	PPB

Meteo Speed: 00, Direction: 00

- The 8 "Channels" fields are used to choose the parameter among: SO₂, H₂S (if H2S rack option available), TRS (if TRS rack option available), GND, Int.t°, Opt.t°, Aux.t°, Flow r., Pressure, +15V, -15V, I.Pbse, I.IR, Signal.
- The "Formats" fields are used to choose the display format among 4 possibilities (X.XXX, XX.XX, XXXX.X, XXXX). "Auto" manages the comma in order to display the best resolution at any time.

The "Units" fields refer to the units programmed in screen *CONFIGURATION ⇒ Offsets and units*, or *CONFIGURATION ⇒ Analog inputs*.

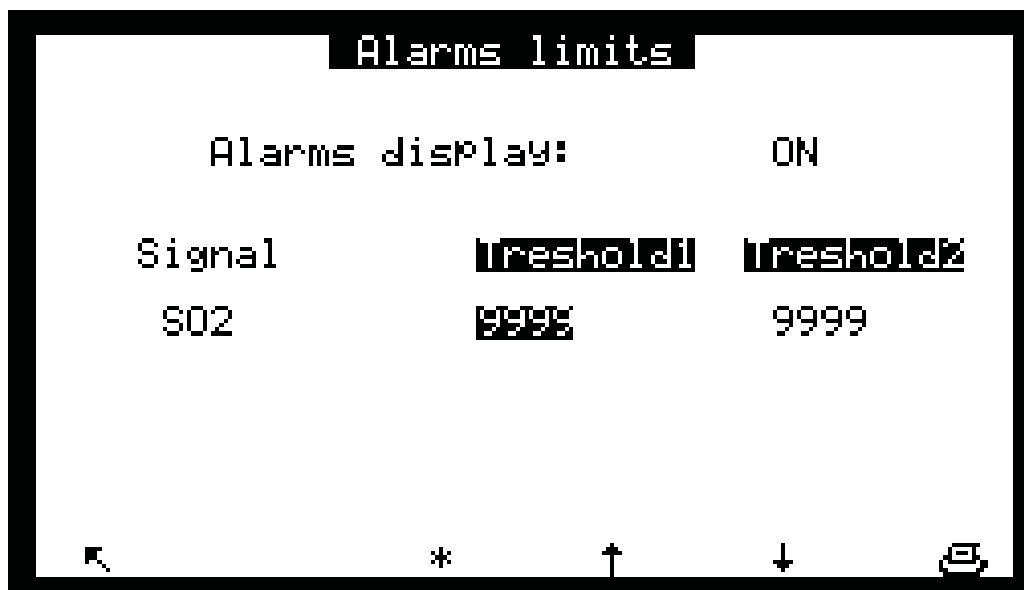
3.3.4.4 **CONFIGURATION ⇒ Offsets and units**

This screen is used to program the offset. This value is added to the measurements. It is also used to program the conversion factors from ppm to mg/m³, when the mg/m³ unit is selected. The conversion factor of 2,660 is applied for normal conditions of temperature and pressure, as follow : (20°C and 101,3 kPa).

Signal	Offset	Unit	Convers.
SO2	010.1	PPB	2,660

3.3.4.5 CONFIGURATION ⇒ Alarm limits

2 limits are programmable for the programmed parameter: Threshold 1 and Threshold 2, allowing to activate relays and alarm messages. When the "Alarms display" field is "OFF", displays and alarm relays are inhibited.



3.3.4.6 CONFIGURATION ⇒ Analog outputs

This screen allows to choose the analog outputs parameters (only when ESTEL board option is available) among:

- The SO2 concentration, the H2S concentration (optional), the TRS concentration (optional).
- From MX01 to MX16, the 16 channels multiplexer
- The external inputs.

The chosen parameters correspond to the analog outputs.

This screen is used to program the ranges for each displayed parameter. The ranges correspond to the analog output full scale.

Scale 1 corresponds to the analyzer standard range. The analyzer switches to scale 2 when scale 1 is exceeded. It switches again to scale 1 when measurement decreases below 85 % of the full scale 1.

This screen is also used to choose the parameters units among ppb, mg/m³, mV, °C or hPa.



3.3.4.7 CONFIGURATION ⇒ Analog inputs

This screen is used to program the analog inputs characteristics.

- "ESTEL card" field allows to select the board to be programmed: each ESTEL board has 4 analog inputs.
- "Name" fields allow to enter 8 alphanumeric digits.
- "Unit" fields allow to choose a unit among : none, ppt, ppb, ppm, $\mu\text{g}/\text{m}^3$, mg/m^3 , gr/m^3 , $\mu\text{g}/\text{Nm}^3$, mg/Nm^3 , gr/Nm^3 , $\mu\text{g}/\text{Sm}^3$, mg/Sm^3 , gr/Sm^3 , %, μgr , mgr, gr, mV, U, °C, °K, hPa, mb, b, l, NI, SI, m^3 , l/min, NI/min, SI/min, m^3/h , Nm^3/h , Sm^3/h , m/s or km/h, in a scrolling menu.
- "aX + b" fields allow to enter the calibration curve for each parameter.
- "Meteo" fields allow to assign the channel where meteorological parameters are connected in order to apply a trigonometric treatment to those data.

Nb	Name	Unit	Ax + B
1	1-1 Ana.	mV	1 0
2	1-2 Ana.	mV	1 0
3	1-3 Ana.	mV	1 0
4	1-4 Ana.	mV	1 0

3.3.4.8 CONFIGURATION ⇒ Relays and remote control

Nb	Relays	Type	Remote controls
1	General Al.	N.C.	Zero
2	General Al.	N.C.	Span
3	General Al.	N.C.	Disable
4	General Al.	N.C.	Disable
5	General Al.	N.C.	
6	General Al.	N.C.	Mode: State

This screen allows to configure the function of each input / output of the Estel board(s).

- "Estel card Nb" field allows to choose what board to configure.
- "Relays" fields allow to control the relays according to the following situations :

Disable	⇒ Relay not assigned
General alarm	⇒ Any operating fault triggers the relay
Ch.1 > Thrs.1	⇒ Limit 1 channel 1 exceedance triggers the relay
Ch.1 > Thrs.2	⇒ Limit 2 channel 1 exceedance triggers the relay
Ch.2 > Thrs.1	⇒ Limit 1 channel 2 exceedance triggers the relay
Ch.2 > Thrs.2	⇒ Limit 2 channel 2 exceedance triggers the relay
Ch.3 > Thrs.1	⇒ Limit 1 channel 3 exceedance triggers the relay
Ch.3 > Thrs.2	⇒ Limit 3 channel 3 exceedance triggers the relay
Over range	⇒ Range 2 exceedance triggers the relay
Flow rate	⇒ Abnormal flow rate triggers the relay
Temperature	⇒ Abnormal temperature in the analyzer triggers the relay
Pressure	⇒ Barometric pressure in chamber
Null gas	⇒ On zero, relay is triggered
Span	⇒ On span, relay is triggered
Ref-Zero	⇒ On Ref-Zero, relay is triggered
Auto Span	⇒ On Auto Span, relay is triggered
Warm-up	⇒ On Warm-up, relay is triggered
Stand-by	⇒ On Stand-by, relay is triggered
Maintenance	⇒ Relay triggered when the analyzer is in maintenance mode
- "Type" fields allow to control (NC) or not (NO) the relays when alarm OFF.
- "Mode" field allows to configure working remote controls mode.

Two different modes are possible:

"State" mode: control is activated as long as remote control is active. When remote control falls down, control is no more active.

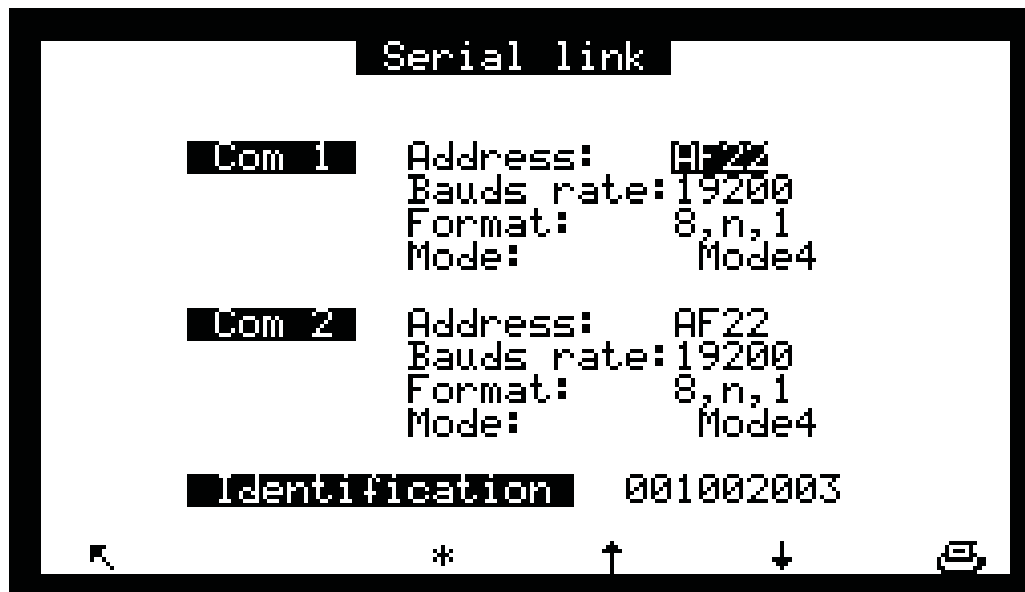
"Rise" mode: control is activated when remote control activation is detected. When it is down, control remains active. It is necessary to re-activate the same remote control to make down the previous remote control.

3.3.4.9 CONFIGURATION ⇒ Serial link

This screen is used to configure the Serial links (COM 1 and 2).

The baud rate, format and communication mode of the 2 channels are programmable among:

- Baud rate: 1200, 2400, 4800, 9600, 19200, 38400 (limited to 19200 bds at present time)
- Format : 7n1, 7o1, 7e1, 7n2, 7o2, 7e2, 8n1, 8o1, 8e1, 8n2, 8o2, 8e2
- Communication mode: Mode 4, impress. to send measurements to printer in real time, Jbus, Special1, and Special2.

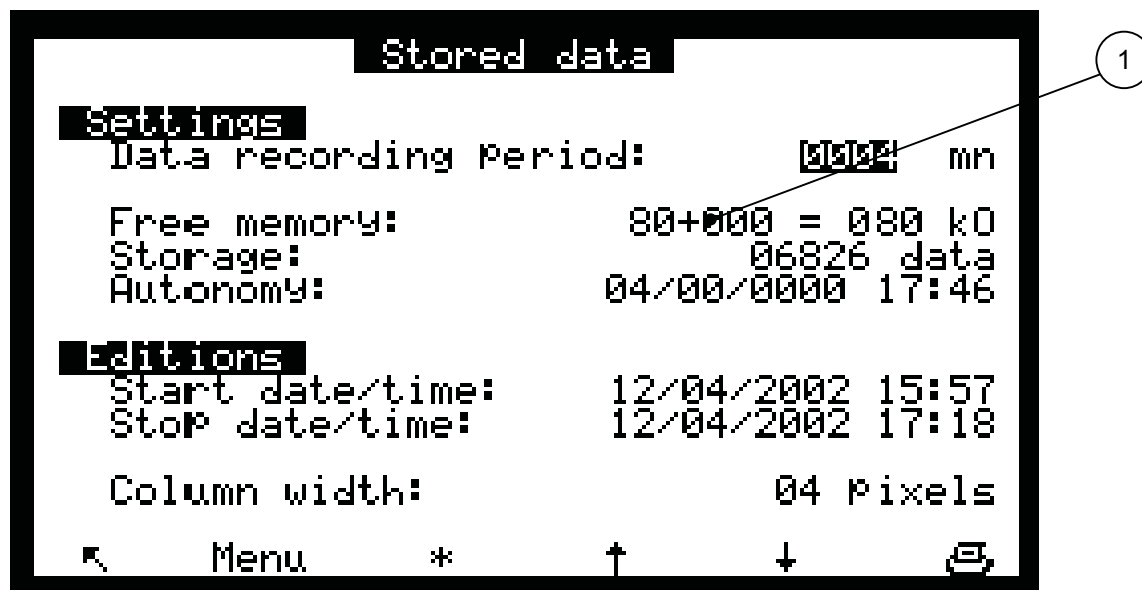
**3.3.4.10 CONFIGURATION ⇒ Factory settings**

When this item is selected, pressing down the  key displays the screen shown here below:



3.3.5 STORED DATA

The access to stored data management is directly done from Main Menu. The stored data consists in the average of analyzer measurements within a defined time interval.



This screen allows to parameter data recording period from 1 to 1440 min (i.e. 24 hours) and informs about memory status:

Free memory: from 80 Ko in standard operation, it can be increased to 464 Ko in adding a 384 Ko memory board (optional). This board is automatically detected when switching on the analyzer and it is indicated on the screen (1).

Storage : it is the possible records number, it depends on free memory

Autonomy: it is the duration (days, months, years, hours, minutes) while memory can store data, considering free space and data recording period. In the here-above example: 4 days, 0 month, 17 hours, 46 minutes.

Data can be edited in the form as table or histogram: this screen allows to program date and hour of edition beginning, date and hour of edition end, histogram column width.

 key gives access to table or histogram data display, printing functions, and memory reset-to-zero.

Stored data edition in the form as tabular

This screen presents stored data list according to parameters defined in the screen before. The running mode (measurement, zero, calibration...), during a memorization period, is coded in the status column. The status codes meaning are:

00	Measurement valid
01	Range 2 over shooting
02	General alarm
04	Calibration fault
08	Zero measurement
10	Span measurement
20	Maintenance
40	Less than 2/3 of valid measurements during the average period
80	Power supply failure
FF	Configuration modification

The displayed status code corresponds to the summation of the status codes (hexadecimal numbers) that occurs during the memorization period.

Example: with an average period of 20 min:

5 min zero and 15 min measurement give the 00 status code and the displayed average is the 15 min. measurement average.

11 min. zero and 9 min. measurement give the 08 status code and the displayed average is the 11 min. zero average.

Date/Time	Status	SO2	PPM
12/04/2002 10:00	FF	0.000	
12/04/2002 10:15	FF	0.000	
12/04/2002 10:30	FF	0.000	
12/04/2002 13:30	82	0.000	
12/04/2002 13:45	82	0.383	
12/04/2002 14:00	00	0.791	
12/04/2002 14:15	00	0.869	
12/04/2002 14:30	00	0.928	
12/04/2002 14:45	00	0.961	
12/04/2002 15:00	00	0.981	
12/04/2002 15:15	00	0.988	
12/04/2002 15:30	00	0.993	

Definition of the specific keys to this screen:

Select the previous or the next page.



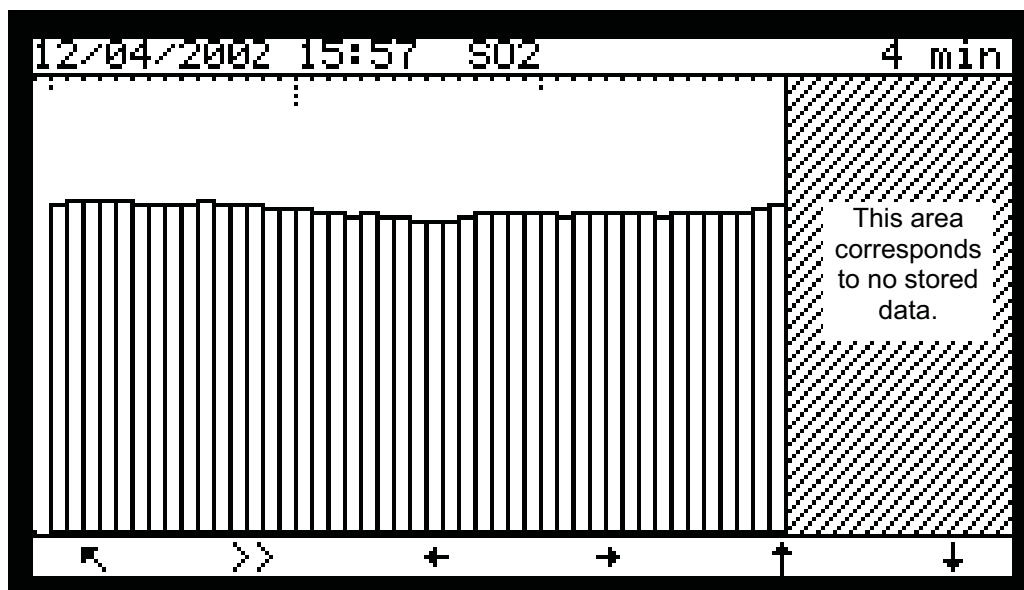
Select stored data beginning or end.



Display the other measure channels if more than 3 channels are programmed in *CONFIGURATION* ⇒ *Measure channels* screen

Stored data edition in the form as histogram

This screen displays records in the form as columns; each column corresponds to the measurements average within the data-recording period as defined in *STORED DATA* screen. Only one channel is displayed at once. The information line gives first record date and hour, the channel name, and, alternatively blinking, full scale with unit, and data recording period.

**Definition of the specific keys to this screen**

-  Return to previous menu.
-  Display previous stored data plotting.
-  Display next stored data plotting.
-  X 2 zoom in
-  1/2 zoom out.
-  Select the next measure channel, when more than one measure channel is programmed.

Stored data printing

To print data, press down the  key found in "Menu" function of "Memorized data" screen. The blinking message "Printing..." indicates printed data output. Data printing can be suspend at any time, pressing down F1 key. When printing is finished, the screen displays the message "Printing finished". When none communication port is programmed on printer output (serial port), the error message "Printing not set" is displayed.



AF22M[3.1]				
11-10-2001				
		S02	EXT1	MX13
HH:MM	status	PPB	hPa	° C
10:15	00	14.7	1001.7	39.4
10:30	00	21.7	1001.4	39.5
10:45	00	21.4	1002.1	39.4
11:00	00	17.9	1002.6	39.4
11:15	00	16.0	1002.4	39.4
11:30	00	14.7	1000.7	39.4
11:45	00	13.9	1001.4	39.4
12:00	00	13.5	1002.1	39.5
12:15	08	1.0	1001.8	39.4
12:30	00	12.9	1001.6	39.3
12:45	00	12.7	1001.1	39.2
13:00	00	12.3	1000.8	39.1
13:15	00	11.8	1001.0	39.0
13:30	00	10.6	1001.3	38.9
13:45	00	9.2	1001.5	38.7
14:00	00	8.5	1001.8	38.7
14:15	00	8.0	1002.1	38.5
14:30	00	7.7	1002.3	38.3
14:45	00	6.8	1002.5	38.2
15:00	00	7.6	1002.6	38.1
15:15	00	7.5	1002.4	38.0
15:45	00	8.1	1002.4	37.9
16:00	00	8.5	1002.3	37.5
16:15	00	8.5	1002.6	37.5
16:30	00	8.7	1002.6	37.7
16:45	00	9.1	1002.5	37.8
17:00	00	9.2	1001.8	37.9
17:15	00	9.2	1002.4	38.0
17:30	00	9.0	1002.2	37.9
17:45	00	8.1	1002.0	37.8

Figure 3-5 - Printout example

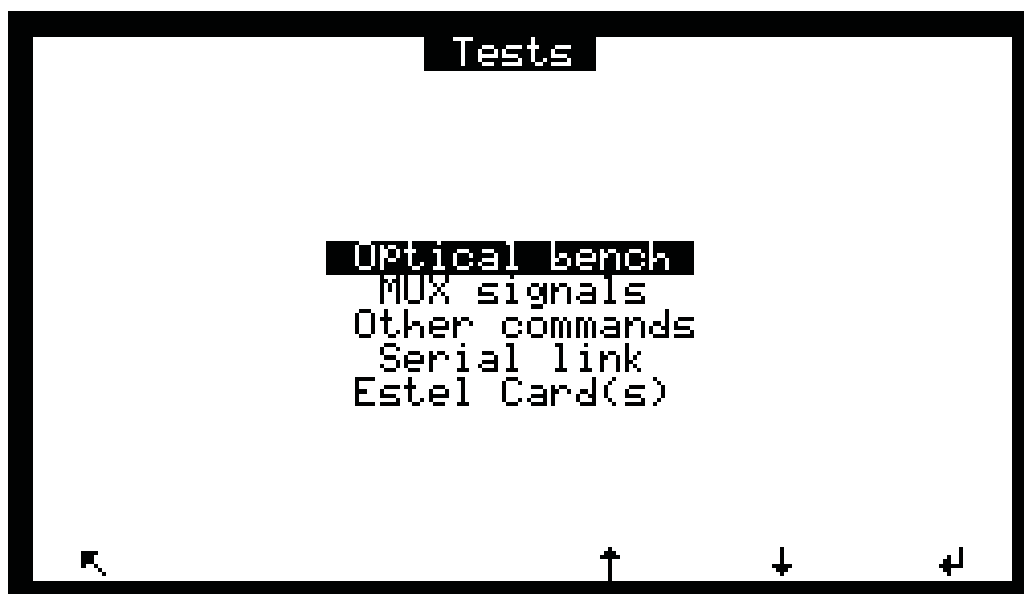
Memory reset to zero

Pressing down the  key allows to empty storage memory. **This action is irreversible:** before to do it, the software asks you to confirm. If your answer is "YES", the software resets end edition dates and hours to the current dates and hours.

3.3.6 TESTS

This screen gives access to the following functions:

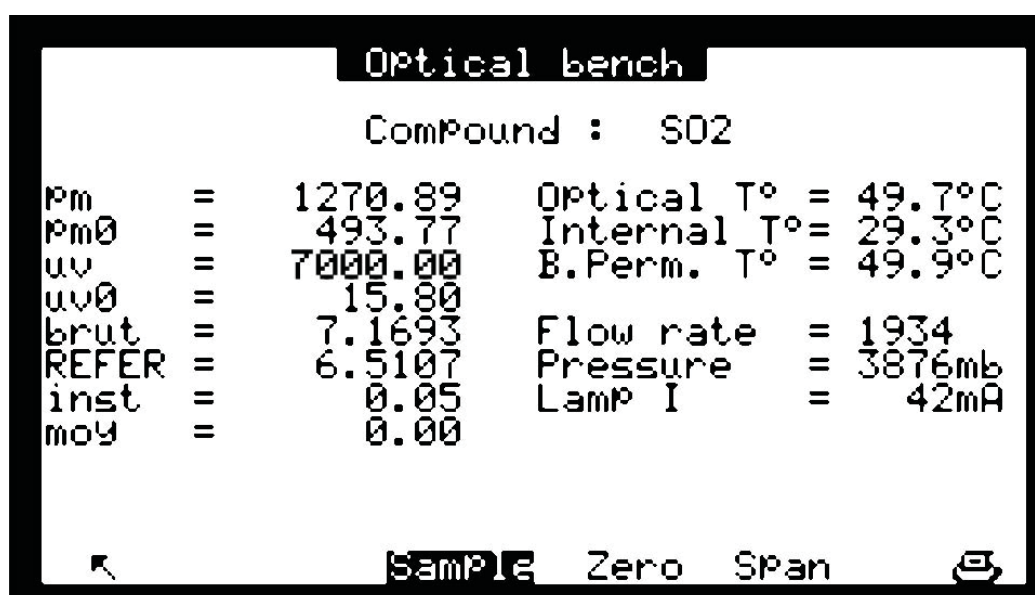
- Optical and flow parameters checking when maintenance operations occurs.
- Serial link checking.
- Checking of the ESTEL board working (when option available)



3.3.6.1 TESTS ⇒ Optical bench

This screen is used to follow-up the measurement parameters periodically or occasionally.

- Pm = instantaneous amplified PM signal
- Pm0 = PM black signal (shutter closed) memorized during the last reference cycle
- UV = Instantaneous amplified UV detector signal
- uv0 = UV detector black signal (shutter closed) memorized during the last reference cycle
- brut = UV energy compensated raw measurement
- REFER = "brut" memorized during the last reference cycle
- inst = instantaneous measurement in ppb
- moy = last memorized measurement average
- Optical T° = optical bench temperature
- Internal T° = internal temperature of the analyzer
- B. Perm. T° = Permeation bench temperature (optional)
- Flow rate = Voltage proportional to flow rate
- Pressure = vacuum pressure in the chamber.
- Lamp I = lamp current



Definition of the specific keys of this screen

Sample Zero Span have the same function as the screen *Measurement* ⇒ *Instantaneous*.

3.3.6.2 TESTS ⇒ MUX signals

This screen is used to check the multiplexer signals.

MUX signals					
GND	0	mV	UV Sig.	7000	mV
Int. T°	361	mV	-----	492	mV
Opt. T°	450	mV	PM sig.	95	mV
PM HV	3167	mV	UV cur.	325	mV
Flow r.	3326	mV	H ₂ S T°	0	mV
Pressure	3981	mV	B.P. T°	0	mV
-15V	1477	mV	O ₂	2	mV
+15V	1499	mV	2.5V ref	2507	mV
SAMPLE Zero Span					

NOTE : The "XXXX mV" displayed value will be checked according to the acceptable limits in next table.

Table 3-2 - MUX signals (Acceptable limits on the multiplexer 1 to 16 channels)

Channel	Display	Parameters	Lower Limit	Normal	Upper limit
1	GND	Analog ground	0 mV	0 mV	50 mV
2	Int. T°	Internal temperature of the analyzer	100 mV	300 mV	600 mV
3	Opt. T°	Temperature of optical chamber	—	450 mV	500 mV
4	PM HV	Voltage proportional to high voltage applied to photo multiplier tube (***)	Embase		
			HAMA SDS		
			2400 mV 3000 mV	2800 mV 3500 mV	3200 mV 4000 mV
5	Flow r.	Flow rate sensor voltage.	1000 mV	3300 mV	5000 mV
6	Pressure	Pressure of measurement chamber (*)	1200 mV	4000 mV	5000 mV
7	-15V	Test point of power supply voltage	1200 mV	1500 mV	1600 mV
8	+15V	Test point of power supply voltage	1200 mV	1500 mV	1600 mV
9	UV Sig	Measurement of UV lamp signal used to compensate for any possible drift.	1000 mV	7000 mV	9000 mV
10	-----	Not used	—	—	—
11	PM sig.	Measurement signal at PM amplifier	0 mV	—	9900 mV
12	UV cur.	Lamp current	250 mV	380 mV	440 mV
13	H ₂ S T°	Thermocouple probe temperature (***)	14 mV	17 mV	22 mV
	TRS T°		0 mV	100 mV	500 mV
14	B.P. T°	Voltage proportional to temperature of built-in permeation oven (**)	2200 mV	2234 mV	2260 mV
15	O ₂	O ₂ sensor (**)	—	—	—
16	2.5 V ref.	Reference voltage of analog / digital converter	2480 mV	2500 mV	2520 mV

(*) These values are given as information and they depends on the A and B linearization coefficients (SPAN ⇒ Pressure screen)

(**) Only when the corresponding options are available in the analyzer.

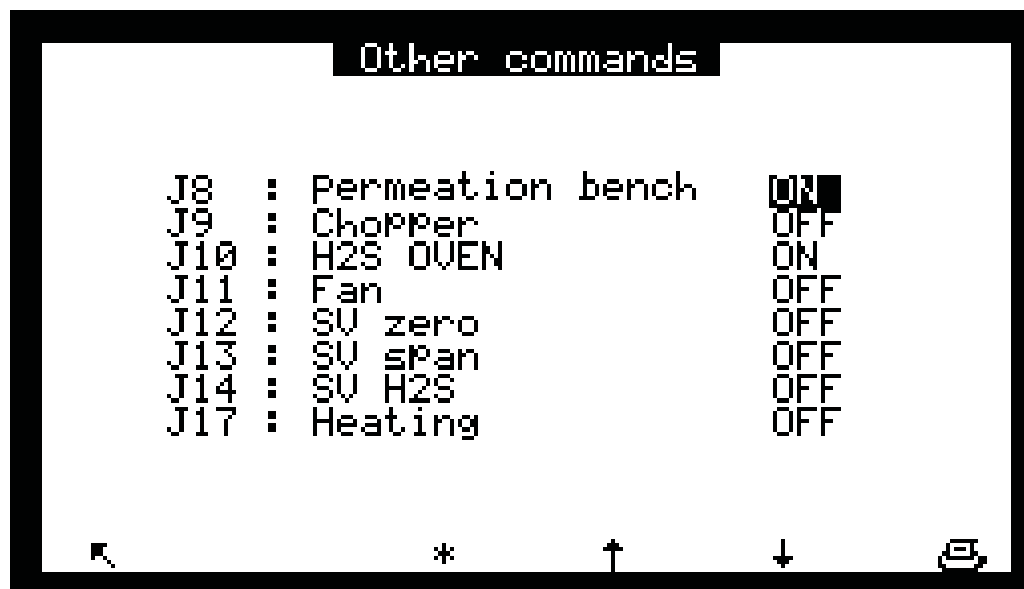
(***) According to the PM embase (HAMA. or SDS) specified in the Check List attached to the analyzer.

3.3.6.3 TESTS ⇒ Other commands



The selection of this menu makes ineffective some commands and regulations. When the instrument is again in measurement mode, some alarms could occur.

This screen is used to check that the MODULE board is working correctly.

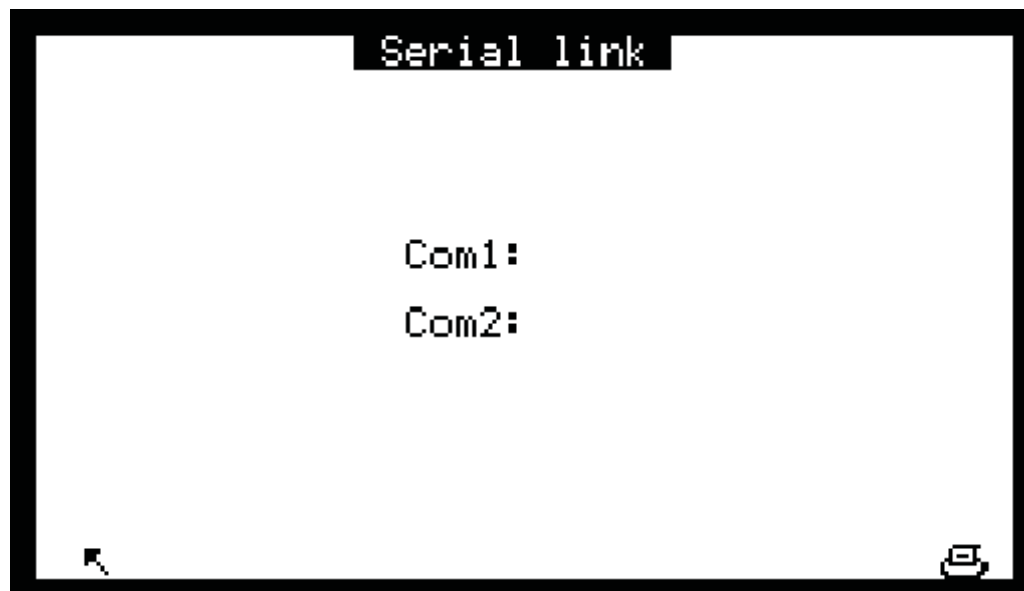


Jxx refers to the connector command, SV refers to solenoid valve.

3.3.6.4 TESTS ⇒ Serial link

This screen allows to check the serial link and shows the inputs / outputs which have to be strapped when the serial link is not connected and you still want to check it are:

2-3: Emission/Reception, 4-6 and 7-8: modem signals.



3.3.6.5 TESTS ⇒ ESTEL card

This screen is only displayed when the option is available.

It is used to set the analog outputs and to monitor the working state of the remote controls and analog inputs.

Nb	DAC	Ax + B	Out	ADC	Rem
1	4000	1 0	ON	0586	OFF
2	4000	1 0	ON	0489	OFF
3	4000	1 0	ON	0835	OFF
4	4000	1 0	ON	0538	OFF
5			ON		OFF
6			ON		OFF

(Points) (mV)

Navigation keys: * 0/OFF

The "Estel card Nb:" field is used to select the board to be tested.

The "DA.C" (digital to analog converter) fields are used to program the number of points generated at analog output.

The "Ax + B" fields are used to program the span factors of each output. These factors are calculated according to the value measured at the output.

The "Out" fields are used to control the relays manually.

The "AD.C" & "Rem" fields are used to read the status of these inputs.

Definition of this screen specific keys:

The sixth key is a toggle key, its function toggles from to .



Gives 0 pt on all analog outputs and opens all relays contacts.



Gives the full scale (4000 pts) on all analog outputs and closes all relays contacts.

3.3.7 STOP MODE

This screen is used to activate the "Stop mode". To return to measurement mode, it is necessary to press down the "Sample" key in any screen of Measurement Menu (see Chapter 3.3.2).



Stop mode is used to turn pumping unit off, all the other regulations keep on operation.

3.4 CALIBRATION

3.4.1 OVERVIEW OF CALIBRATION AND CONCEPTS

To ensure the accuracy of the measurements performed using the AF22M monitor, the unit must be regularly checked, calibrated and adjusted, following the quality assurance plan of the user.

– **Check of zero and span point:**

This operation consists of comparing the monitor response, for zero air and a span point of the range used, to the gas standards used.

This check is used to measure the monitor drift in time without modifying the adjustment coefficient.

This check can be performed using the internal zero air and span.

Frequency: generally 24 hours in automatic cycle mode.

– **2-point calibration:**

This is a procedure for checking and correcting the response of the monitor at zero and at a span point located at approximately 80 % of the full scale of the measurement range used.

Frequency: monthly, or more frequently if the monitor requires it.

– **Span (multi-point calibration):**

This involves a complete check up of the monitor performance characteristics (linearity).

Frequency: quarterly, or following out-of-tolerance calibration check results requiring an intervention on the monitor.

Note about gas generation devices:

For devices providing pressurized gas, it is necessary to provide an excess system to deliver the gas at atmospheric pressure to the monitor inlet. The materials making up this device should be neutral for the gas used. When used in an automatic cycle with a cylinder, provide a shut-off solenoid valve that can be remote-controlled by the monitor (see figure 3.6).

Note about gas cylinders connections:

When the analyzer is equipped with the internal zero filter and/or the internal permeation bench, these respective inlets are no more available (see figure 2-2), therefore during the calibration procedures the calibration gas cylinders will be connected to the sample inlet

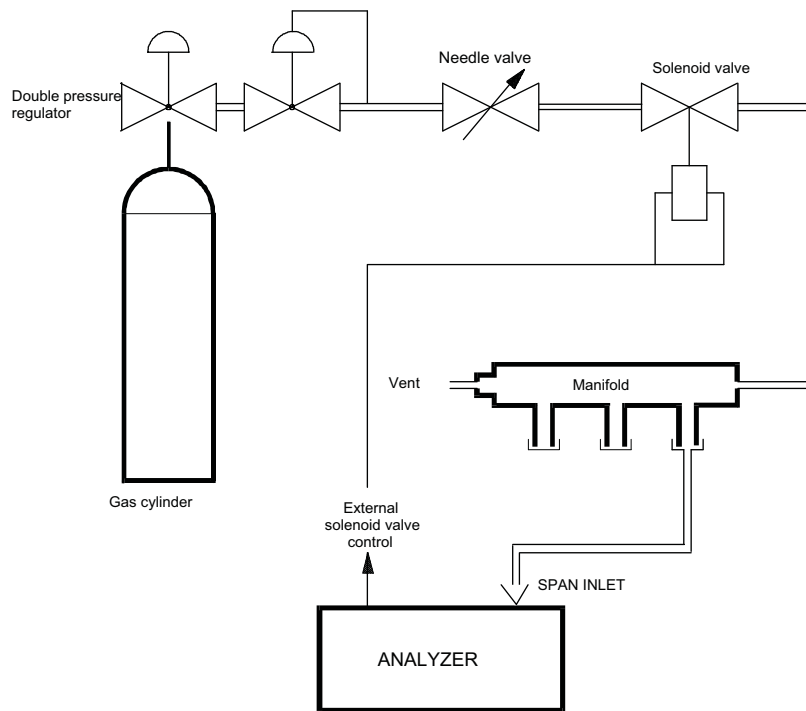


Figure 3-6 - Pressurized gas connection example

3.4.1.1 Zero air generation

- Check: internal zero filter.
- Calibration: cylinder of reconstituted air or zero air generator. The zero air should be free of any constituent likely to be measured by the monitor and should not contain more than 0.0005 ppm of SO₂.

3.4.1.2 Span gas generation (SO₂)

SO₂ calibration gas source standards used for adjustments should be traceable to a National Institute of Standard & Technology (NIST), Standard reference Material (SRM) or Certified Reference Material (CRM) in accordance with EPA's Protocol n° 2.

If a permeation device calibration system is used, be sure to allow 24 hours for warm-up.

Zero air used for the dilution of standard gas concentration and for zeroing the analyzer should be the same and contain less than 0.0005 ppm of SO₂.

All fittings, valves, pneumatic lines and other components that may contact span test gas must be fabricated with highly cleaned inert to SO₂ material (i.e. stainless steel).

– Calibration check:

- Cylinder of SO₂ (1 % accuracy) with a concentration of approximately 80 % of the upper range limit. The cylinder used should be regularly certified in accordance with the user's quality assurance plan.
- Optional built in permeation oven with an SO₂ wafer device. The concentration generated by the permeation oven is recorded on the check sheet (only for span check).
- External permeation device equipped with an SO₂ permeation tube and able to generate a concentration of approximately 80 % of the upper range limit.

– 2-point and multi-point calibration:

- Cylinder of SO₂ (1 % accuracy) associated to a calibrator allowing the generation of 6 concentrations, including zero, up to 80 % of the upper range limit. The cylinder used should be regularly certified in accordance with the user's quality assurance plan.
- External permeation device equipped with an SO₂ permeation tube and able to generate 6 concentrations, including zero, up to 80 % of the upper range limit (refer to section Section10.3 of Appendix A to 40CFR50).

NOTE : The devices used to generate the span gases should deliver 700 cc/min. minimum and 2500 cc/min. maximum. All fittings, valves, pneumatic lines and other components that may contact span gases must be fabricated or coated with cleaned PTFE, glass or stainless steel.

3.4.1.3 Internal solenoid valves

During multipoint calibration the gas sources will be connected to the sample inlet of the monitor. After this calibration operation, the gas sources will be connected to their respective inlets and a check will be made to see that the monitor's response is identical on any inlet used. The various gas inlets can therefore be used for zero and span point checks and for 2-point calibration of the monitor. Otherwise, the solenoid valves must be cleaned or replaced.

3.4.2 CHECK OF ZERO AND SPAN POINT

3.4.2.1 Facilities required

– "Zero" air:

a sufficient quality of "zero" air is obtained using the monitor "ZERO AIR" filter (Purafil/activated vegetable charcoal).

– Span point:

- SO₂ gas cylinder, which concentration is below 1 PPM and connected to the monitor "span gas" inlet.
- Or internal permeation oven with SO₂ tube (connected to monitor "span gas" inlet). The concentration generated by the oven is recorded on the check sheet.
- Or portable calibrator (type VE3M) equipped with an SO₂ tube and connected to the monitor "span gas" inlet.

NOTE : If the analyzer is equipped with an internal permeation oven, the sampling input must be used to connect the reference cylinder or the portable calibrator.

3.4.2.2 Procedure

– Zero check:

Select the monitor "zero air" inlet using the  key or appropriate inlet if required and wait for the measurement to stabilize. The reading should be less than ± 2 ppb. If this is not the case, a zero adjustment is necessary.

– Check of scale point:

Select the gas inlet where the span gas is connected using the  key or the  key, as per the above note, and wait for the measurement to stabilize. The result will be compared to the concentration generated by the device used, taking into account its accuracy.

If the difference between span gas and reading is more than ± 10 %, a span adjustment is necessary.

3.4.2.3 Use of automatic cycles

To program the cycles, see section 3.3.3.3 *Span* ⇔ *Cycles* menu.

– *Zero cycle:*

The "zero" air generator is permanently connected to the monitor "zero air" inlet. The minimum recommended duration of the zero check is 10 minutes.

– *Calibration cycle:*

The scale point generator (the internal permeation oven if option is present) is permanently connected to the monitor "span gas" inlet, it can be the internal permeation oven. The SO₂ concentration must be below the full scale of the range used for the measurement. The minimum recommended duration of the check is 10 minutes.

3.4.3 SPAN ADJUSTMENT

3.4.3.1 Facilities required

– *"Zero" air:*

See section 3.4.1.1.

– *Span point:*

See section 3.4.1.2.

3.4.3.2 Procedure

– *Zero reference:*

- Start by doing a reference cycle (minimum of 600 seconds). Feed zero air to the desired inlet (or span) and press the  key in one of the *Measurement* screen, then the  key of the cycle screen, the cycle duration countdown is displayed in the top right corner of the screen.

– *Span adjustment:*

- *automatic:*

Select the measurement range (*Configuration* ⇔ *Signals/Ranges/Units* menu) corresponding to the concentration (*Span* ⇔ *Select gas* menu) of the standard gas and the inlet used. Press the  key, then the  key of the cycle screen. The analyzer automatically changes its calibration factor as a function of the concentration programmed. It is advisable that the AUTO-CAL lasts 600 seconds, the cycle duration countdown and the concentration of the span gas are displayed in the top right corner.

NOTE : During the cycle, the SO₂ displayed measurement uses the former K span, it is updated when the AUTO-CAL cycle is achieved.

If the new K span is out of the bounds $K_{(former)} \pm 50 \%$, the analyzer gives a calibration alarm, in that case check:

- the calibration pneumatic circuit
- the generated concentration of the gas source used
- the programmed span gas concentration
- the programmed gas inlet

If one of the above points is wrong, correct it and repeat the AUTO-CAL cycle. If nothing is wrong, proceed to a manual adjustment of the span factor as described after.

- manual:

Select the inlet to which the gas is connected by means of the **Sample** or **Span** keys. Wait for the measurement to stabilize. Program the new K (span) in the *Span* $\Rightarrow$ *factors mode* menu.

Calculation of new coefficient:

$$K_{(new)} = K_{(former)} \times \frac{\text{span gas value}}{\text{read value (without offset)}}$$

CAUTION: It is advisable to note $K_{(former)}$ before changing it, since it will be deleted when you enter $K_{(new)}$.

3.4.3.3 Use the AUTO-CAL automatic cycle

To program this cycle, see paragraph 3.3.3.3 *Span* $\Rightarrow$ *Cycles* menu. To configure the concentration, see paragraph 3.3.3.2, *Span* $\Rightarrow$ *Select gas* menu.

The gas generation system is connected permanently to the monitor "span gas" inlet. The recommended time for automatic calibration is 600 seconds.

If the span gas meets the requirements described in § 3.4.2.1, this feature could be covered under US EPA designation.

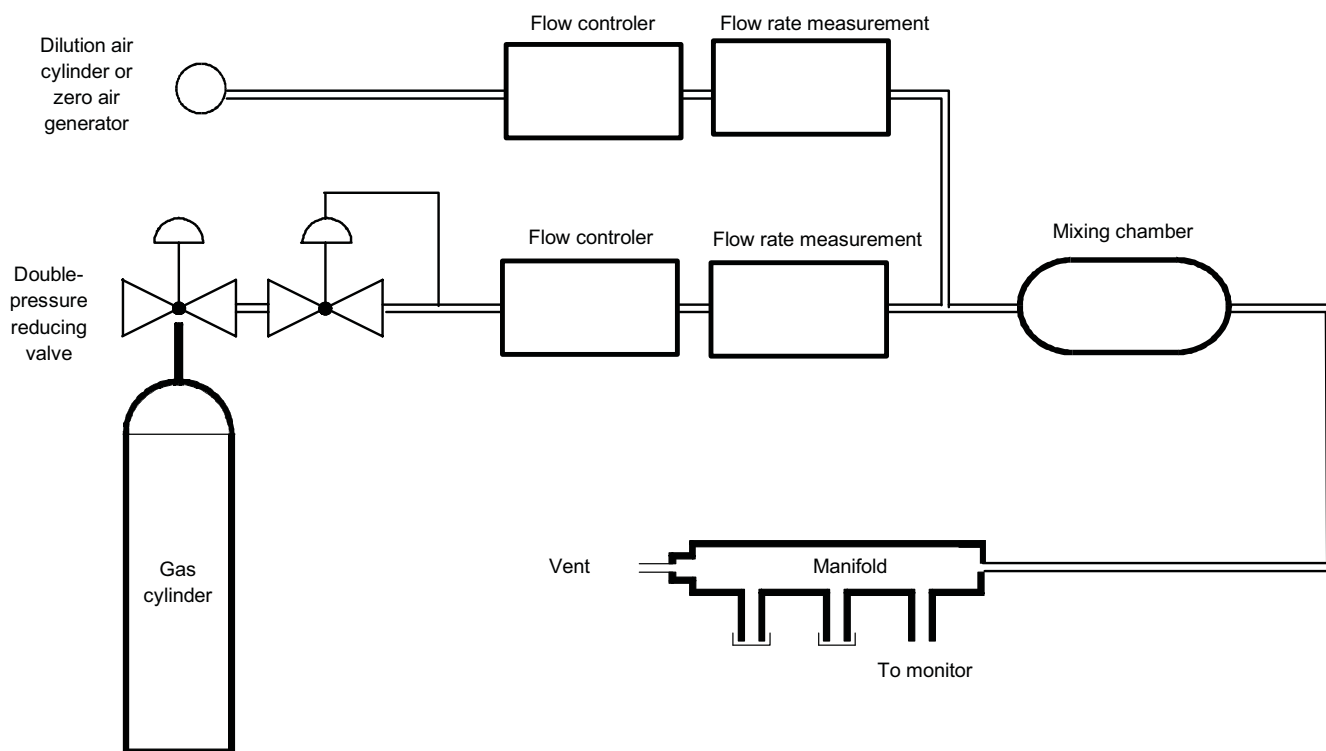


Figure 3-7 - Typical calibrator

3.4.4 MULTIPOINT CALIBRATION

3.4.4.1 Overview

The minimum device will be made up of a diluter, a certified cylinder of SO₂ (1 % accuracy), the calibration of which can be traced to reference materials from the National Institute of Standards (NIST) for calibration in the framework of measurements according to the US EPA, and a zero air generator. Figure 3.7 shows an example of a typical calibrator.

The gases will be applied to the sample inlet of the monitor at atmospheric pressure.

Calibration of the monitor requires generation, in addition to the zero, of 6 gas concentrations (for example, 15 %, 30 %, 45 %, 60 %, 75 % and 90 % of full scale of range used). The dilution air must be the same as the zero air.

It is advisable to connect the analog output of the monitor to a strip chart recorder for acquisition and processing of the data.

3.4.4.2 Equipment Required

– Diluter:

- flow regulators: they must regulate flow rates to the nearest ± 1 %.
- flow meters: they must read and record flow rates to the nearest ± 2 %.
- mixing chamber: its shape and volume should allow a homogeneous mixture of SO₂ and dilution air.

– Manifold:

The manifold must include at least one outlet for the monitor and one outlet for the excess flow meter. The outlet must have a sufficient diameter so as not to create any load losses at the inlet of the monitor. The outlet of the excess flow (at least 20 % of the total flow rate) will be designed so that the pressure in the manifold is quite near the atmospheric pressure (no overpressure), but so that the ambient air cannot be back scattered.

– Dilution air (zero air):

Zero air generator or cylinder of reconstituted air free of any contaminants likely to be measured by monitor and of which the SO₂ concentration does not exceed 0.0005 ppm.

– Span gas (SO₂):

Certified cylinder of SO₂ in air at 1 % of which the concentration will allow generation of 6 ranges between 15 % and 90 % of the full scale of the measurement range used.

3.4.4.3 Procedure

- Switch on monitor at least 6 hours before making the calibration.
- Configure monitor as follows (advised configuration):

Span $\Rightarrow$ *Cycles* menu

REFER. time 0600

AUTO-CAL time 0600

(deactivate all cycles or check that no one of them will be triggered during calibration).

Configuration $\Rightarrow$ *Measurement mode* menu

Resp. time 01

Configuration $\Rightarrow$ *Ranges / Units* menu

Range 1 - Range 2.....range currently used in measurement

Configuration $\Rightarrow$ *Offset / Conversion* menu

SO₂ offset 10.00 ppb (at least or 5% of full scale of the range used).

- Connect analog output (see table 3-1) of monitor to recorder. Use response from recorder to calibrate device as described below.
- Connect sample inlet of monitor to dilution system manifold.
- Adjust dilution system so that it generates a total zero air flow rate that is more than 20 % greater than the sample flow rate of the monitor.
- Generate zero air. Wait for reading to stabilize (recommended duration: 600 sec). Note value of Z_{SO_2} , expressed in percentage of maximum recorder response (for example, 400 mV / 10 V = 4 %).
- Follow steps in section 3.4.3.2. ("span point correction" section) to adjust calibration coefficient.

Use formula below to calculate value of span generated:

$$[SO_2]_{\text{generated}} = \frac{[SO_2]_{\text{Cylinder}} \times F_{SO_2}}{F_D + F_{SO_2}}$$

where:

$[SO_2]_{\text{Generated}}$ is the concentration of SO_2 in the gas generated at the outlet of the manifold,

$[SO_2]_{\text{Cylinder}}$ is the concentration of SO_2 in the certified cylinder,

F_{SO_2} is the flow rate of SO_2 in NI/min,

F_D is the dilution air flow rate in NI/min.

Use formula below to calculate the reading value in ppm from the value in % full scale to be taken into account:

$$[SO_2]_{\text{read}} = \left(\frac{S_{\text{record}} - Z_{SO_2}}{100} \right) \times ECH$$

where:

S_{record} is the value noted on the recorder, expressed in percentage of the full scale of the recorder,

ECH is the full scale of the monitor,

Z_{SO_2} was measured previously.

- Then generate the other five SO_2 concentrations between 15 and 90 % of the full scale, varying the flow rates F_{SO_2} and/or F_D .
- Plot the $[SO_2]_{\text{read}}$ values as a function of the $[SO_2]_{\text{generated}}$ values, including the zero air point. Check linearity.
- Plot or calculate, using the formula for least error squares below, the straight line that will constitute the straight calibration line for the monitor.

$$[SO_2]_{\text{read}} = a \cdot [SO_2]_{\text{generated}} + b$$

where:

a ; is the linear regression coefficient (slope), and is calculated as follows:

$$a = \frac{n \cdot \sum [SO_2]_{\text{generated}} \cdot [SO_2]_{\text{read}} - \sum [SO_2]_{\text{generated}} \cdot \sum [SO_2]_{\text{read}}}{n \cdot \sum [SO_2]_{\text{generated}}^2 - \left(\sum [SO_2]_{\text{generated}} \right)^2}$$

b ; is the constant linear regression term (intercept) ; it is calculated as follows:

$$b = \frac{\sum [SO_2]_{\text{read}} - a \cdot \sum [SO_2]_{\text{generated}}}{n}$$

n is the number of items of data.

3.4.5 INTERNAL PERMEATION OVEN (OPTIONAL)

The scale point check can be performed using a permeation oven equipped with a SO₂ source.

– **Principle:**

The SO₂ used is maintained in a state of liquid/vapor phase equilibrium in a closed cylindrical tube fitted with a polymer diaphragm. Due to the differential partial pressure of the gas on either side of the diaphragm and the diaphragm "permeability", gas is diffused toward the outside of the tube. The weight of the gas diffused by unit of time, called "permeation rate", depends on several parameters: type of gas, thickness, surface and nature of diaphragm, partial gas pressures on either side of diaphragm, temperature.

– **Setup:**

The permeation tube is placed in a block which is thermostat-controlled to 50 °C (± 0,1 °C) and continuously ventilated with a scavenging gas whose flow is regulated to about 45 l/h. The rinsing gas is filtered by an active charcoal filter located at the rear of the analyzer and directly connected to the inlet of the permeation oven. The outlet of the permeation oven (PTFE plug) is connected to the "span gas" inlet of the analyzer.

When this inlet is selected, the monitor samples a part of the generated gas and thus measures the concentration of this gas.

– **Procedure:**

- The internal oven is selected using the "span" key, or automatically on a calibration cycle. Once stabilized, the SO₂ measured by the monitor must be compared to the concentration recorded on the check sheet.
- Since SO₂ permeation is very sensitive to the quality of the rinsing air, the condition of the activated vegetable charcoal filter should be checked periodically.
- The oven temperature is set using the *Tests* ⇔ *Mux signals* function on channel No. 14 (see Table 3.3).
- The system operating temperature range is 10 to 35 °C (ambient temperature).
- When setting into service or following an extended shutdown period, the stabilization time is about 24 hours.
- The permeation source supplied with the monitor has an autonomy of about 16 months.
If the monitor is not used, the source should be removed from the oven, and placed in its original package with the desiccant and absorbent bags, then stored in a cool, ventilated location.

- To maintain the best precision, the following characteristics should be checked regularly:
 - flow rate of oven diluent air,
 - permeation tube rate (theoretically constant).

A differential weighing using an accurate balance should be performed to the 10 th, or better to the 100 th of mg.

Calculation of permeation (P) and (CG) concentration rates generated by oven:

m0 = Initial weight of tube (ng)

m1 = Final weight of tube (ng)

m0 – m1 = Weight of SO₂ diffused (ng)

t = Time between two weighings (min)

P = (m0 – m1) / t = Tube permeation rate (ng/min.)

F = Real total air flow rate passing through the oven (CC/min.)

Km = Molar coefficient, for SO₂ = 0.382

CG = Km x P / F = Concentration of span gas generated (ppm)

NOTE : These calculations must be repeated each time a permeation tube is changed, or, more simply, the concentration generated by the new tube can be determined by reading the concentration on the oven just after calibration of the analyzer (see section on calibration).

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**CHAPTER 4
PREVENTIVE MAINTENANCE**

4.1	SAFETY INSTRUCTIONS	4-2
4.2	MAINTENANCE CALENDAR	4-3
4.3	MAINTENANCE OPERATION SHEETS	4-4
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4 PREVENTIVE MAINTENANCE

4.1 SAFETY INSTRUCTIONS

The safety instructions must be observed at any time by the user.

Whenever possible, cut off the power supplies when performing any work on the monitor.

Only qualified personnel should intervene on the monitor.

As concerns safety, the manufacturer shall not be responsible for any consequences resulting from:

use of the monitor by non-qualified people,

use of the monitor under conditions other than those specified in this document,

modification of the monitor by the user,

no-maintenance of the monitor.

A systematic periodic inspection is required.

4.2 MAINTENANCE CALENDAR

By its design, the AF22M requires very limited maintenance. However, to ensure the monitor's performance characteristics over time, the unit must be serviced regularly. The periodicities indicated below are given as an example and can vary according to operating conditions.

Operation	Periodicity	Sheet No.
– Sample inlet PTFE filter	15 days	4.3.1
– Check of electrical parameters	1 month	4.3.2
– Check of zero and calibration	1 month	4.3.3
– Replacement of activated charcoal (internal zero filter) and kicker zero filter cartridge	One year	4.3.4
– Inspection of pump valves and diaphragms	Six months	4.3.5
– Replacement of UV lamp	2 years	4.3.6

Annual check

The monitor must be returned to laboratory for complete cleaning (measurement chambers, restrictors, flow circuit, etc.) and check of all metrological parameters.

4.3 MAINTENANCE OPERATION SHEETS
MAINTENANCE SHEET

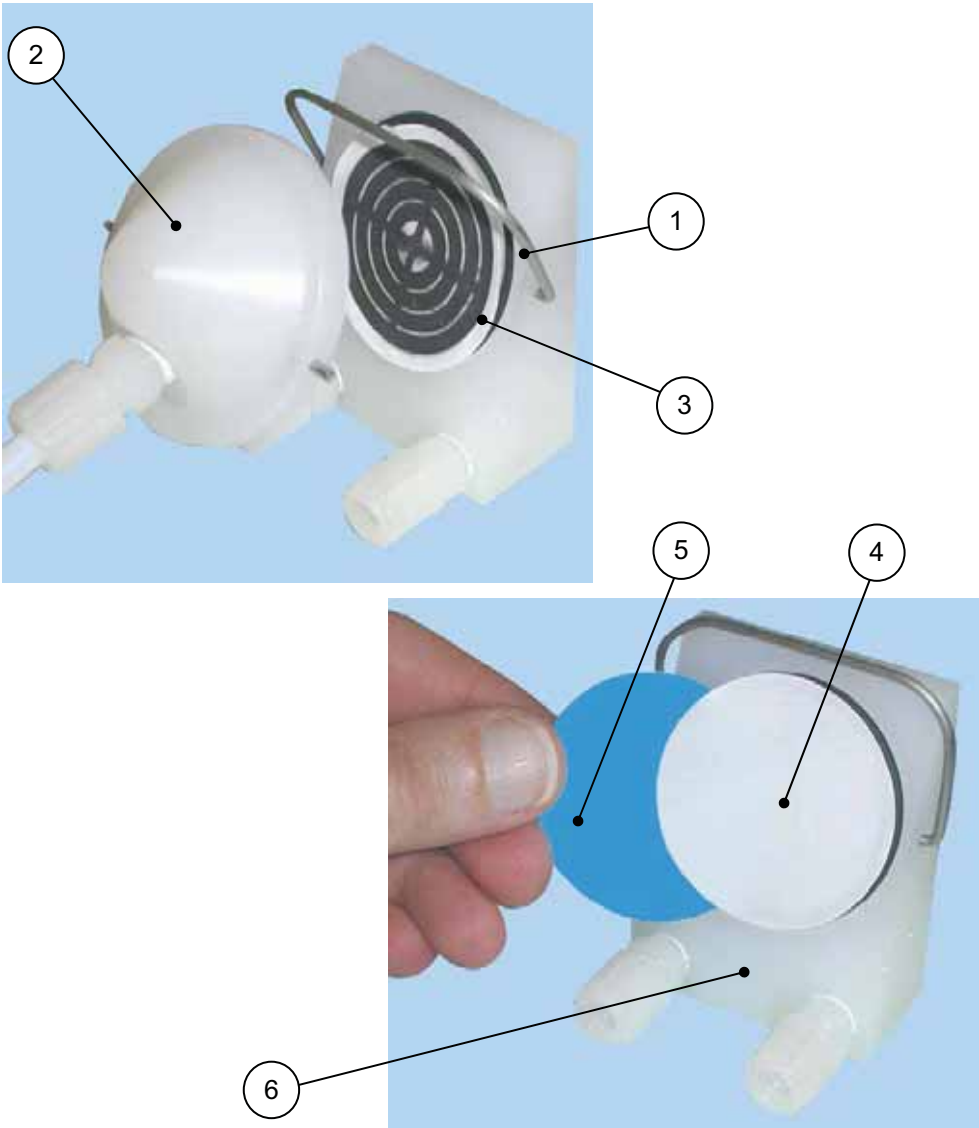
MONITOR Serial No.:		OPERATION SHEET: 4.3.1	
Scope: Replacement of filters		PAGE: 1/1	Periodicity: 15 days
Sample inlet PTFE filter: "MITEX" Teflon filter - 5 µm porosity - 47 mm dia. Ref.: F05-11-842 a) Unlock the filter cover (2) by rotating the locking spring (1) b) Remove the dirty filter (3) c) Place the new filter (4) by slipping it slightly from its paper protection(5) onto the filter holder (6).			Date
			
Tools required			
• none			

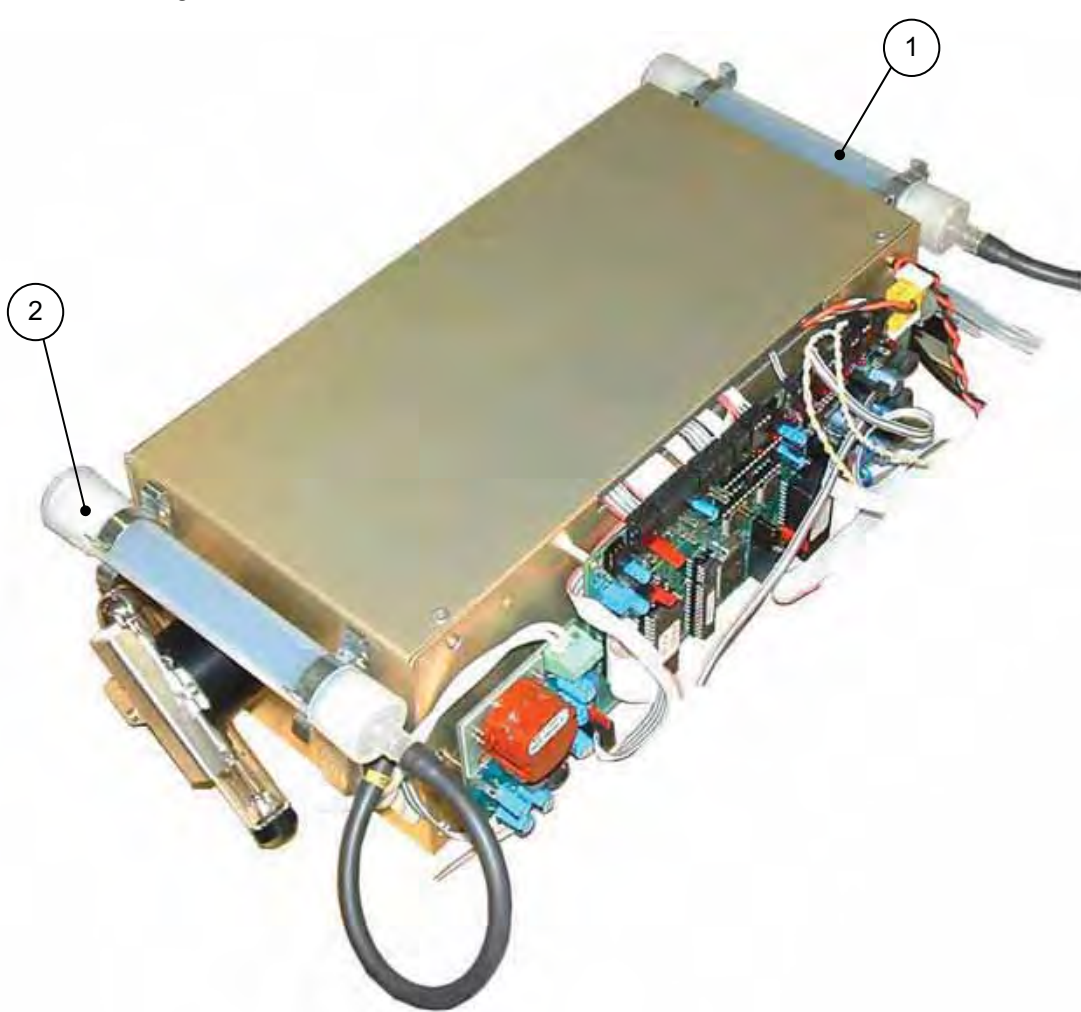
Figure 4-1 – Inlet dust filter

MONITOR Serial No.:						OPERATION SHEET: 4.3.2				
Scope: Check of electrical parameters						PAGE: 1/1		Periodicity: 1 month		
- Electrical values are checked using the Tests ⇒ Mux signals (see Chapter 3, paragraph 3.3.6.2). - Compare the voltage values at each input of the multiplexer to the values given for the same inputs on the test sheet provided with the monitor.										
Dates	Mux Reading									
	2	6	7	8	9	10	11	12	14	16
- Tools required • None.										

MAINTENANCE SHEET

MONITOR Serial No.:	OPERATION SHEET: 4.3.3		
Scope: Checks of zero and calibration	PAGE: 1/1	Periodicity: 1 month	
– Check of zero (see section 3.4.2) Zero checking operations can be carried out: • Either using a pressure-packed, synthetic air cylinder guaranteed to be free of any trace of SO₂, and injected in the analyzer sample inlet at atmospheric pressure. • or using a zero air provided by the pure-air generator, for example of the AADCO type, and injected under the same conditions as the previous example. • or using the internal zero filter, by actuating the  key, which controls the internal solenoid valve. – Manual calibration of monitor (see section 3.4.3). It is highly advisable to carry out this operation on site, under regular operating conditions. Once the calibration is made, the connections of the analyzer must not be handled. This operation requires the use of a SO₂ in synthetic air gas cylinder for calibration (this cylinder should be fitted with a double pressure relief valve) or the use of a calibration device (portable calibrator). The gas used for calibration should be delivered at atmospheric pressure (for example, through a manifold at the "span gas" inlet of the analyzer). – Tools required • None	Dates	ZERO reading	Span reading

MAINTENANCE SHEET

MONITOR Serial No.:		OPERATION SHEET: 4.3.4	
Scope: Replacement activated charcoal (internal zero filter) and kicker zero filter cartridge.		PAGE: 1/1	Periodicity: Six months
– Disconnect and remove used cartridge. – Connect the tube to the fitting provided for this purpose on the Zero solenoid valve. – Position cartridge.  Figure 4-2 – Zero filter cartridges position NOTE : The built-in zero filter (1) lifetime is 1 year. The hydrocarbon elimination device scavenging air zero filter (2) is 6 months (double flow rate than filter (1) and continuous working)			Date
– Tools required • None			

MAINTENANCE SHEET

MONITOR serial No.:		OPERATION SHEET: 4.3.5	
Scope: Check of pump valves and diaphragm		PAGE: 1/1	Frequency: Yearly
– Check condition of diaphragm and pump valves. Replace them if necessary. <u>Replacement of diaphragm and valves</u> – The diaphragms and valves are parts that wear out; these parts are replaced very easily. <u>Replacement of diaphragm and pump valves</u> – Power instrument off, unplug the power cord a) Disconnect supply connector. b) Disconnect gas flow connections. c) Loose the 3 screws (1) fastening pump assembly to frame in order to release the 3 pump holder slots. d) Unscrew the 2 screws (3) fastening pump on its holder. e) Draw a permanent mark (13) on the pump body (8) for reassembling purpose. f) Remove the nut (5) fastening pump body (8) on the vibrating blade (6). g) Unscrew the 2 screws (7) fastening pump body to pump housing (9). h) Remove pump body. i) Separate the various parts.. j) Clean diaphragm (10), the two valves (11) and the gasket (12). k) Dust off or clean the different parts with alcohol (dismount valves if only necessary) l) Check diaphragm, change it if necessary. m) Reassemble, repeating steps above in reverse order.		Type of operation	Date
– Tools required • Dia. 4 mm cross-tip screwdriver, • Dia. 5 mm flat screwdriver, • Combination pliers • Alcohol solution			
<u>Type of operation:</u> D: Replacement of diaphragm V: Replacement of valves			

MAINTENANCE SHEET

MONITOR serial No.:	OPERATION SHEET: 4.3.5	
Scope: Check of pump valves and diaphragm	PAGE: 2/2	Frequency: Yearly

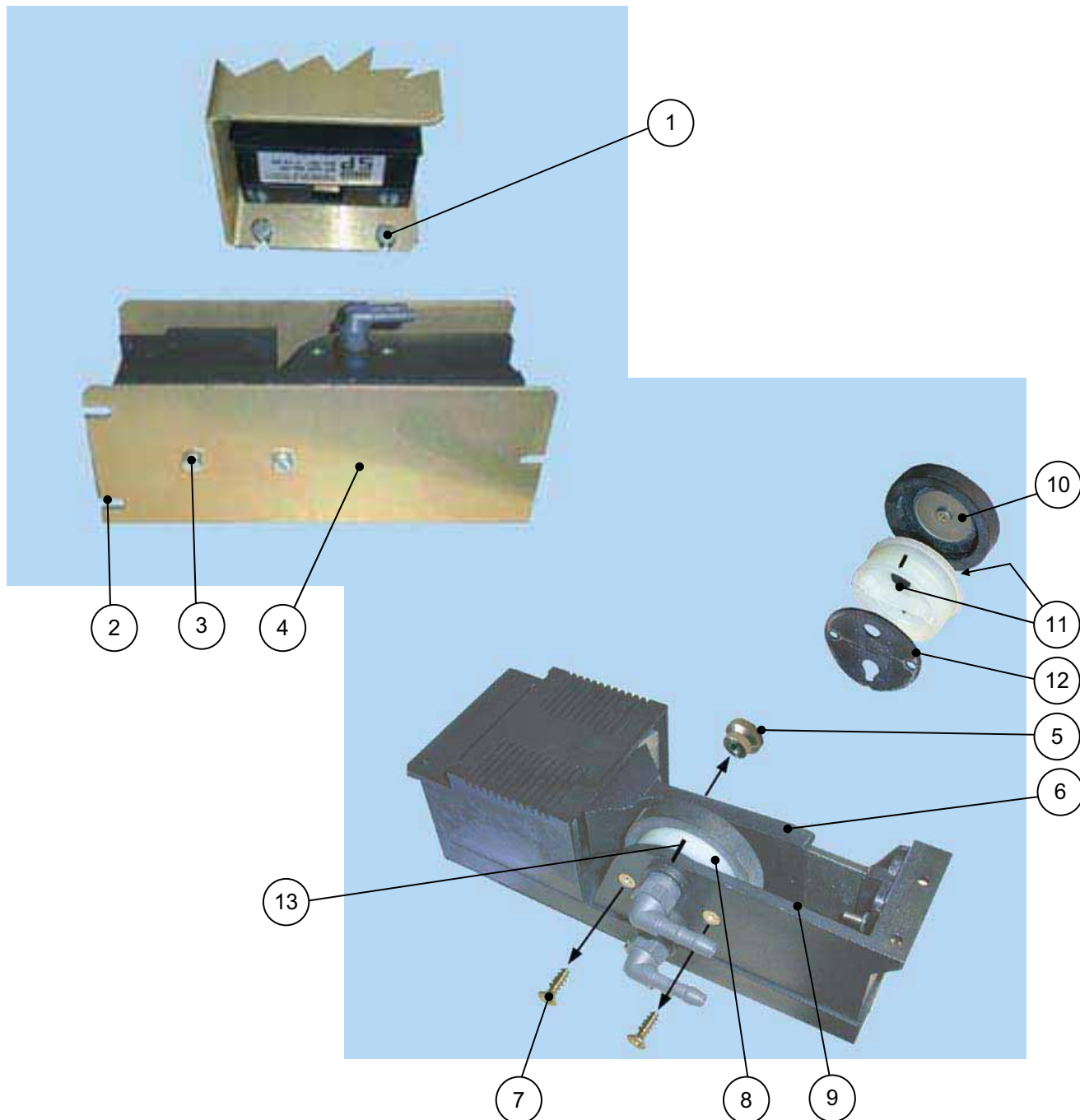


Figure 4-3 – Exploded view of the pump.

MAINTENANCE SHEET

MONITOR Serial No.:	OPERATION SHEET: 4.3.6	
Scope: Replacement of UV lamp	PAGE: 1/1	Periodicity: 2 years

WARNING: Do not touch the glass part of the lamp.

- a) Switch off the monitor.
- b) Disconnect the 2 supply wires of the lamp, loosening them from the UV supply board.
- c) Unscrew the 2 screws (1) to remove the lamp / shutter assembly.
- d) Loose off 3 mm the 2 screws (2) to release the UV lamp.
- e) Replace the UV lamp.
- f) Align mechanically the new lamp, taking into account the MAX OUTPUT PORT mark (3). This window must face the hole (4).
- g) Remount the lamp assembly.
- h) Select the Tests ⇒ MUX signals screen:
 - Check the voltage and current of the UV lamp. I-UV supply: 350-450 mV adjusted on UV lamp supply board
 - Adjust the position of the lamp by rotating and translating it so as to have the maximum UV signal (channel 9 of MUX) voltage. Adjust to approximately 4300 mV.
- i) Tighten the 2 screws (2) when maximum signal is obtained to fix the UV lamp.
- j) After stabilization of the UV power, run a reference cycle, and then calibrate the analyzer.

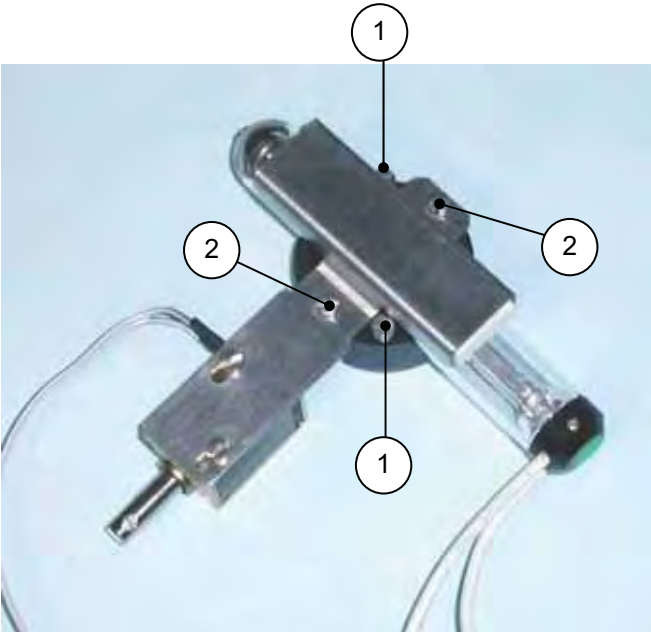


Figure 4-4 – Lamp / shutter assembly (front face)

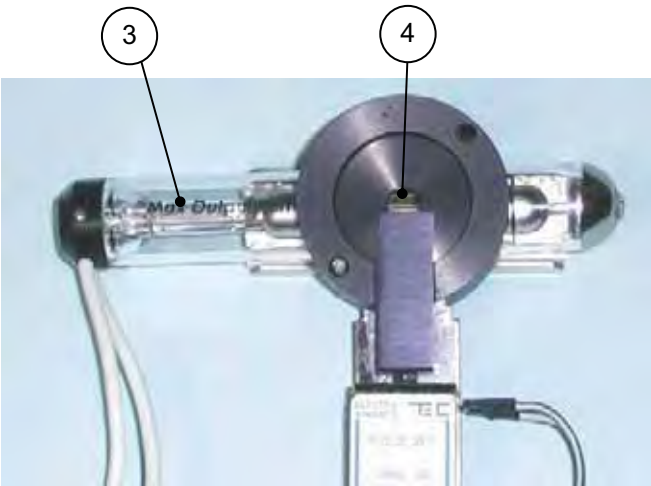


Figure 4-5 – Lamp / shutter assembly (rear face)

- Tools required
 - 3.5 x 75 mm - 2.5 x 50 mm screwdrivers

4.4 AF22M MAINTENANCE KIT FOR 1 YEAR

Kit reference : AF22-K

This kit consists in:

1 box of 25 filtering membranes for sample inlet particulate filter

SAV-K-000042-A

3 filtering cartridges

F05-0128-1

1 maintenance kit for internal pump

V02-K-0041-A

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CHAPTER 5

CORRECTIVE MAINTENANCE

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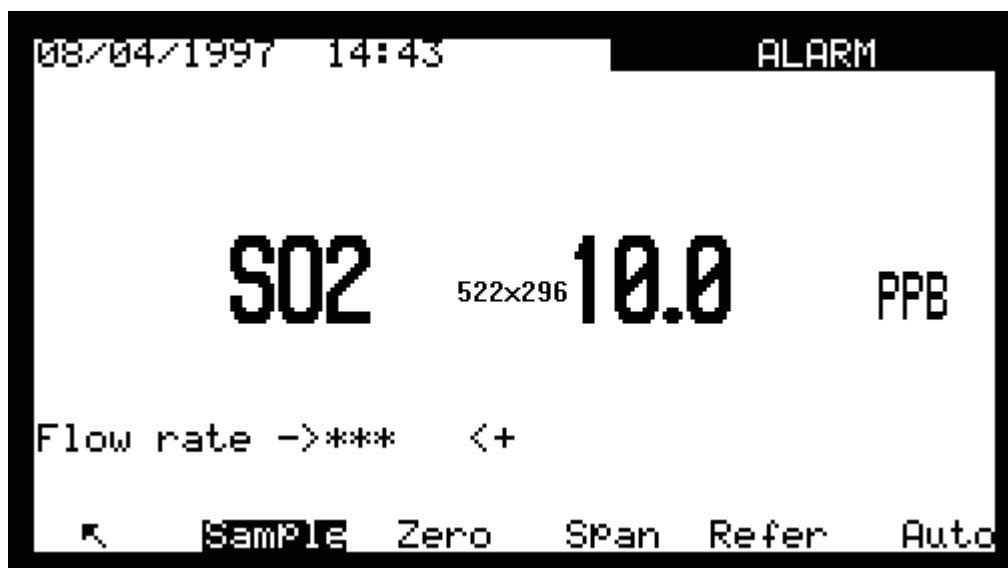
5. CORRECTIVE MAINTENANCE

Corrective maintenance of the monitor should only be performed by qualified people using the information provided in this document.

The monitor automatically and continuously self-tests its main components. Any malfunction detected is indicated by a clear message on the display and a buzzer.

Table 5.1 summarizes the main faults indicated by the unit with corresponding corrective possible actions.

In case of operating fault, the ALARM message blinks in the top right corner.



To check which operating fault is present select the menu *Measurement* ⇒ *Alarms display*.

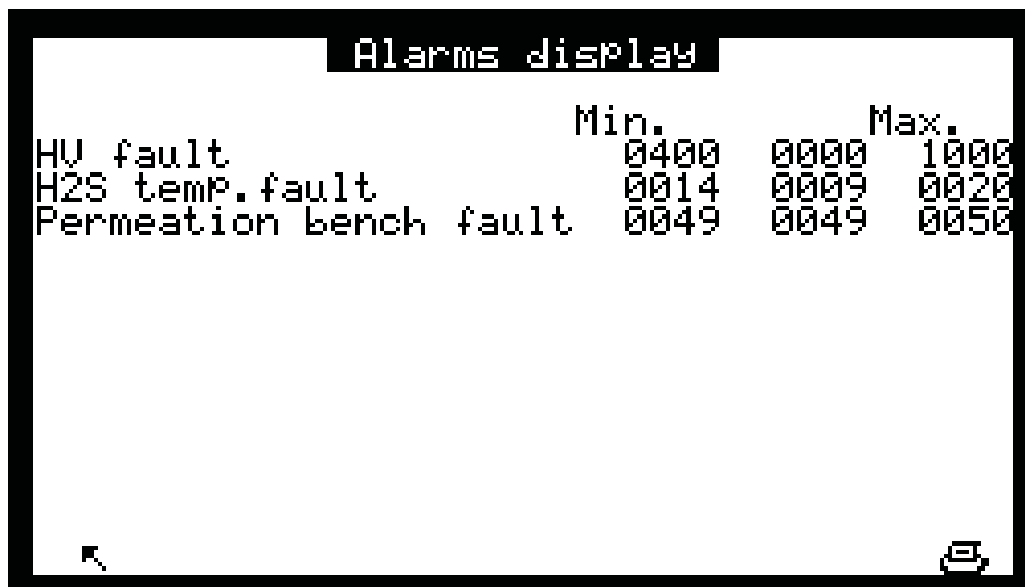


Table 5-1– List of faults and corrective actions

ALARM MESSAGE	CAUSE	POSSIBLE ACTION
CHOPPER FAULT	– The shutter does not cut off the UV beam.	– Check that connector J9 is indeed inserted. – Check that the shutter is not blocked by its supply cable. – Check the solenoid control. • To do this: start a Zero Ref. cycle. • Put a multimeter between the two terminals of the connector. For the first 40 seconds of the cycle, the multimeter should give a voltage reading of 24 V. • If it does not, disconnect J9. Check the electrical continuity of the coil of the solenoid (between end terminals of J1). Replace solenoid if necessary. Otherwise, check mechanical positioning of the shutter. If you still do not locate the problem, contact us. • If this is not the case: check that the 5 V is available in the control transistor emitter, located on the motherboard, throughout the duration of the control. If the voltage is indeed available, check the availability of the D.C. power supply. If you still do not locate the problem, contact us.
OVERRANGE	– The measurement value exceeds the range 2.	– Change ranges.
Optical T°C FAULT	– The temperature of the optical bench does not fall within the boundary values for correct operation (38 °C < T < 48 °C) – The temperature probe is defective. – The Module board is defective.	– Check that connector J17 is correctly inserted. Run a warm up of the monitor by stopping it for a few seconds, then switching it back on. Check the availability of the 24 V voltage at both end terminals on J17. – Contact us. – Replace board or contact us.
Flow rate Fault	– Incorrect flow rate in measurement chamber; the pump is not working. – Leak in fluid circuit.	– Check connection. If everything is OK, overhaul pump completely. – Check all connections. – Place a flow meter at the outlet of the monitor. Plug the sample or calibration inlet, according to the case. The ball in the flow meter should drop to zero. If it does not, there is a leak.

ALARM MESSAGE	CAUSE	POSSIBLE ACTION
Calibration fault	– Span gas source – Following a manual reset of the span K factor	– Check the availability of the span gas at atmospheric pressure on the selected inlet. – Run a manual calibration.
UV Power FAULT	– The control voltage of the UV power does not fall in the boundary values for correct operation $2,500 \text{ mV} < \text{UV voltage} < 4,500 \text{ mV}$. – The voltage function of lamp current does not fall in the boundary values $80 < \text{UV cur} < 440$	– Check that connector J7 is correctly inserted. Check the stability of the regulated voltage between the right-hand end point of connector J7 and the ground. If one of these voltages is off, contact us. - Set voltage between 350 mV and 450 mV using potentiometer P (MUX 12) I. UV. • move the lamp to adjust maximum UV energy. If it is still not possible: • increase the UV lamp supply current using SPAN $\Rightarrow$ E2Pot screen, I (mA) lamp function. If it is still impossible to make the setting, or if the voltage is regulated but the failure persists, contact us. – After stabilization of the UV power, run a Zero Ref. cycle, then calibrate the analyzer.
UV SOURCE FAULT	– The UV signal does not fall in the boundary values for correct operation $1000 \text{ mV} < \text{UV signal} < 9800 \text{ mV}$.	– Refer to possible actions described for UV power fault – Adjust the SPAN $\Rightarrow$ E2Pot, UV signal function.
Pressure Fault	– The signal given by the pressure sensor (optional) doesn't fall in the limits $500 < \text{Pressure} < 1050$	– Check that sample is indeed at atmospheric pressure. – Check that there is no clogging. – Disconnect the tube on pressure sensor. Check that measurement corresponds to the ambient barometric pressure.
Permeation T° Fault	– Ambient working temperature out of bounds – Permeation oven temperature regulation fault	– +15V permeation oven pcb's fault.
ZERO filter life time	– The zero filter life time down counter has reached 0.	– Replace the filter (see maintenance sheet 4.3.4) – Reset the zero filter life time down counter (<i>Configuration $\Rightarrow$ Measurement mode</i> screen).

Table 5-1– List of faults and corrective actions

SYMPTOM (no fault indication)	POSSIBLE CAUSES	ACTIONS
Monitor does not react when switched on.	– Mains fault – Faulty mains cable – Connector incorrectly plugged in – Mains fuse blown	– Check for presence of mains power supply. – Test continuity of mains cable. – Check fuse in connector assembly.
Monitor remains in preheating status	– Module board faulty. – Reset circuit blocked – Micro 5 V faulty.	– Check that display flashes. – If not flashing, check if the microprocessor board is in place. – Replace if necessary. – If flashing, wait for 15 min. and possible fault indication.



Switching power supply is protected from short circuits. In such case, it is necessary to disconnect / re-connect the power cord to reset

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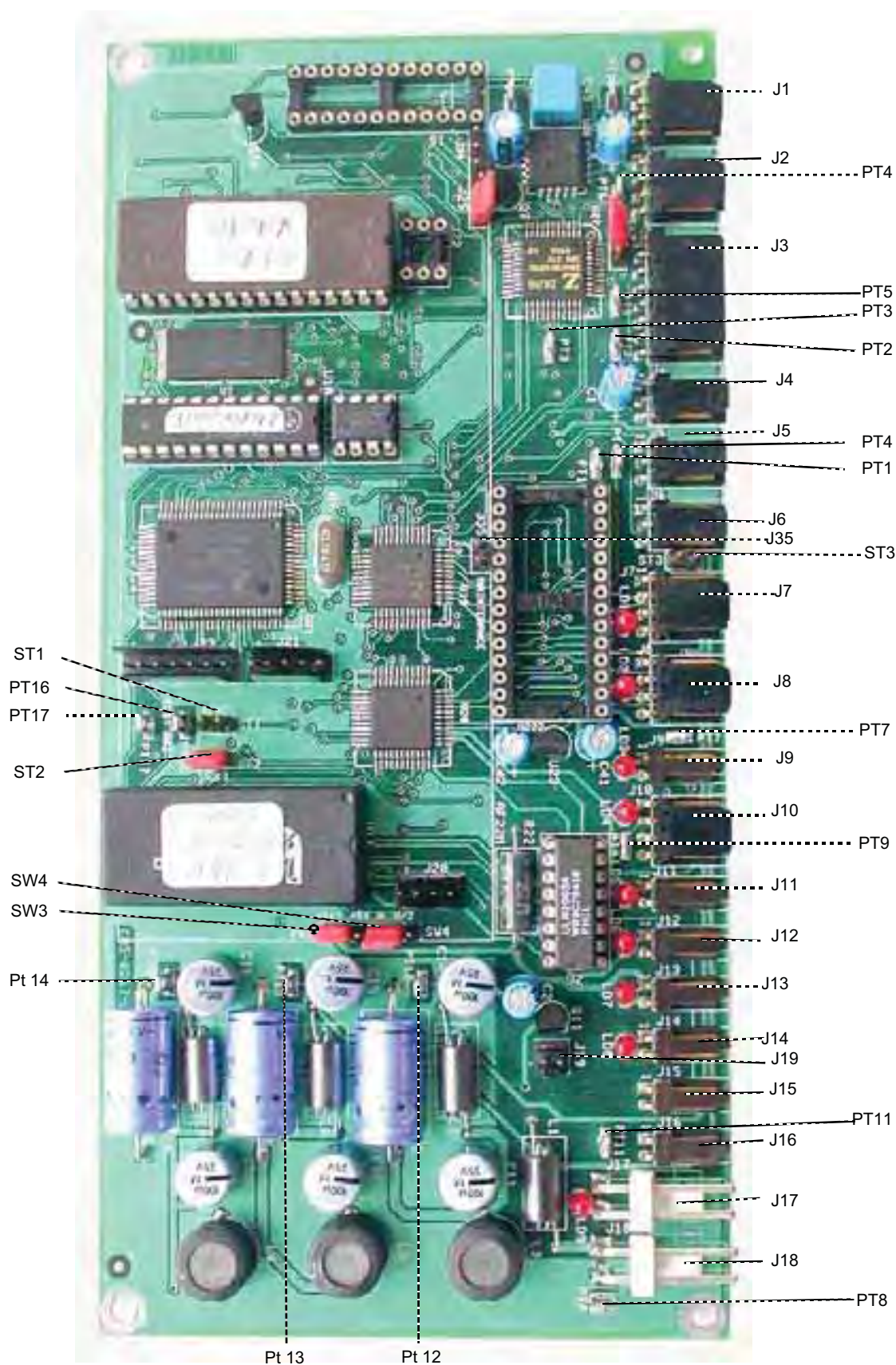


Figure 5-1 – AF22M Module board

Table 5-2 – AF22M Module board configuration

Test points.	Type of signal	Connector	Connection
PT1	Flow rate	J1	O ₂ option
PT2	Chamber pressure	J2	UV detector
PT3	PM signal	J3	PM preamplifier
PT4	Gas temperature	J4	Gas temperature
PT5	A/D RUN	J5	Chamber temperature
PT6	A/D STATUS	J6	Flow rate sensor
PT7	Permeation bench	J7	UV lamp supply
PT8	0 V (GND)	J8	Permeation bench
PT9	H2S converter	J9	Chopper
PT10	MUX output	J10	H2S/TRS converter
PT11	+24VCC	J11	Fan
PT12	+5VCC	J12	Zero SV
PT13	+15 VCC	J13	Span SV
PT14	-15VCC	J14	H2S SV
PT15	Not used	J15	Pump 2 (permeation)
PT16	I off+	J16	Flow rate
PT17	I off -	J17	Chamber heating
		J18	+24VCC supply
		J19	+24VCC on/off switch
		J20	Estel board I2C bus
		J21	Synchro bus
		J22	RAM extention option
		J23	RS4i board I2C bus
		J24	Not used
		J25	Not used
		J26	Not used
		J27	Not used
		J28	Not used
		J29	Not used
		J30	Not used
		J31	Not used
		J32	Not used
		J33	Not used
		J34	Stand by LED option
		J35	Maintenance switch option

Jumpers		Nature of operation
SW1 EPROM Configuration		Selects 27C20 (2 Mbytes)
		Selects 27C40 (4 Mbytes)
SW3 Power supply		5V I2C Bus
		24V I2C Bus
SW4 Internal clock		=CLK/2
		= CLK (default position)
ST1		(default position)
		Resets microprocessor
ST2 Watch dog		Inactive
		Active (default position)

Table 5-3 – Board RS4i Configuration

Jumpers references	Symbols	Nature of operation
SW1, SW2		Channel 1 on RS422 standard
SW3		Channel 1 on RS232 standard
ST1		Load RX bus RS422 active
		Load RX bus RS422 inactive
ST2		Load TX bus RS422 active
		Load TX bus RS422 inactive
ST3		Not used

NOTE : Channel 2 is on the RS232 standard.

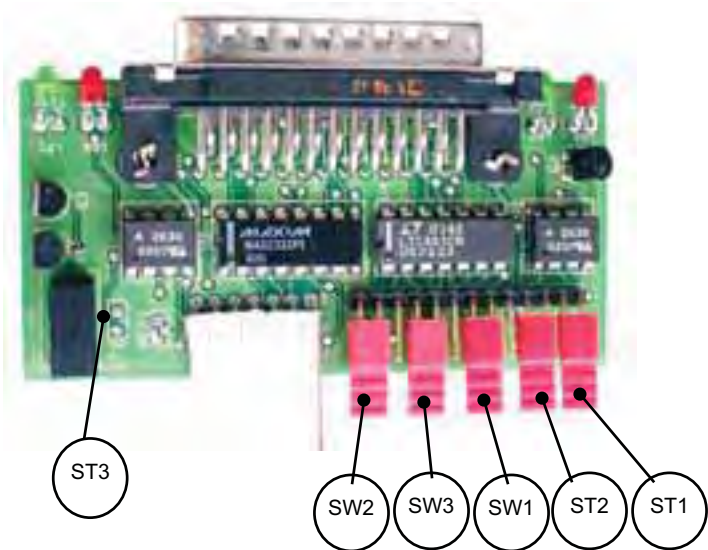


Figure 5-2 – Card RS4i Configuration

Table 5-4 – Keyboard Interface configuration

Jumpers references	Symbols	Nature of operation
ST1		Keyboard inhibited
		Keyboard active
P1	LCD contrast adjustment potentiometer	

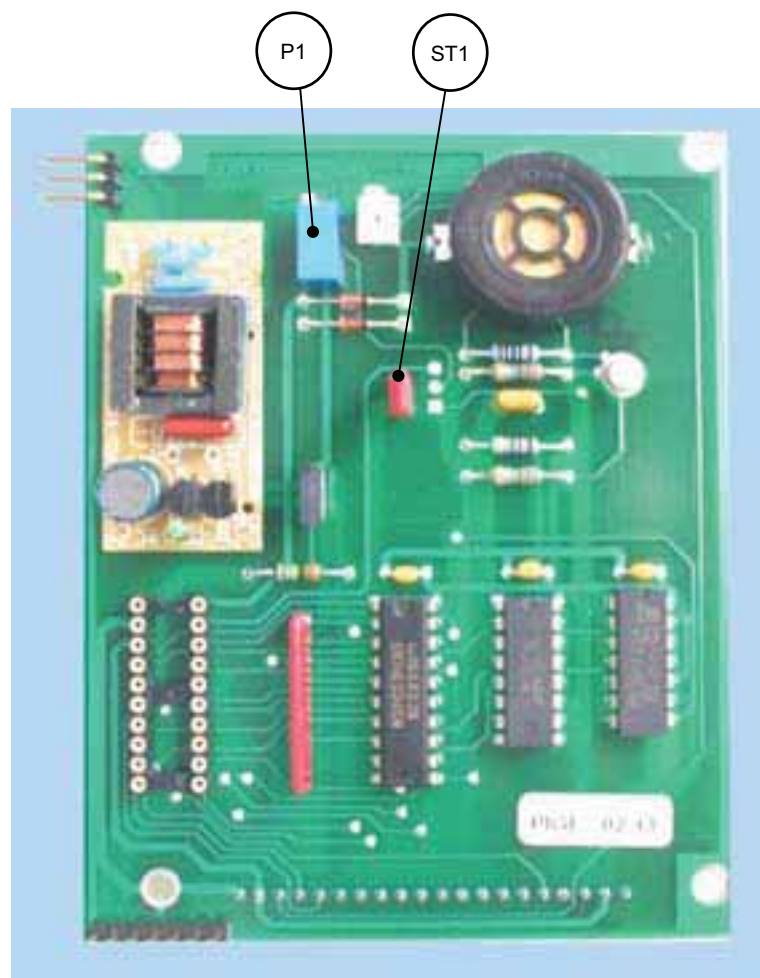


Figure 5-3 – Keyboard Interface board

Table 5-5 – UV lamp board configuration

References	Nature of operation
P1	Lamp current pre-adjustment
PT1	GND
PT2	LM2596 output (SV)
PT3	Frequency (25 kHz)
PT4	V. f(I)
PT5	I lamp adjust
PT6	I lamp

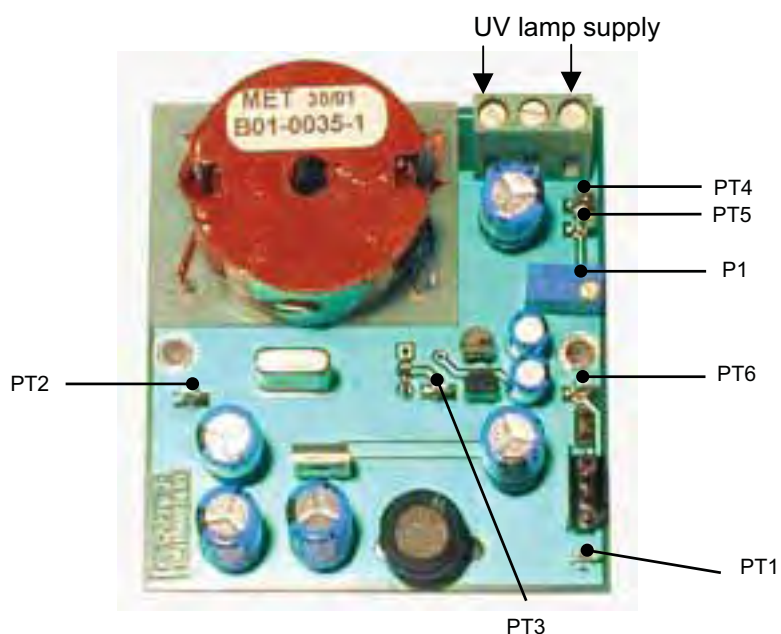


Figure 5-4 – UV lamp board

Table 5-6 – Configuration of UV lamp supply board

Jumpers			Nature of operations
SW1		3 2 1	AF22M Measure lamp (2-3 are linked)
ST1			ON
ST2			OFF



Figure 5-5 – UV lamp supply board

Table 5-6 – Flow rate control board configuration and setting

Designation		Nature of operations
ST1		Ground referenced to Module Board
		Direct ground (default position)
P1		Flow rate adjustment
P2		Pump frequency adjustment
PT0		Ground
PT1		Flow rate signal
PT2		Pump supply voltage

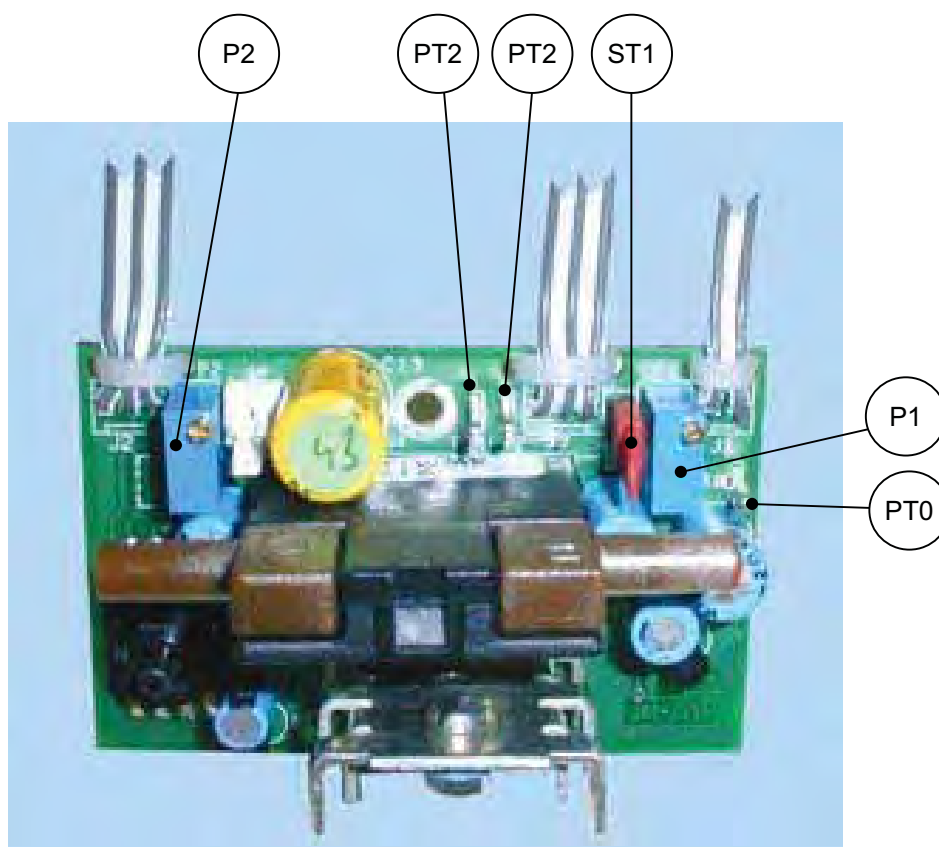


Figure 5-5 – Flow rate control board

CHAPTER 6

APPENDIX

ESTEL board

SOREL board

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ESTEL Board

INPUTS / OUTPUTS BOARD

OPTION OF 2M ANALYZERS

- June 2009 -

WARNING

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ESTEL BOARD

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Up-to-date:

Pages	Up-to-date	Pages	Up-to-date	Pages	Up-to-date
1	05-2004	9	06-2009	17	06-2009
2	05-2004	10	06-2009	18	06-2009
3	05-2004	11	06-2009		
4	05-2004	12	06-2009		
5	05-2004	13	06-2009		
6	05-2004	14	06-2009		
7	05-2004	15	06-2009		
8	06-2009	16	06-2009		

1. ESTEL BOARD

ESTEL board is a universal board of logic and analog inputs/outputs for the 2M series analyzers. It is optional: it is possible to install up to 2 ESTEL boards in an analyzer.

1.1 FUNCTION AND USE

The ESTEL board has 4 functions:

- 4 Analog inputs,
- 4 Analog outputs,
- 6 Relays,
- 4 Remote controls.

The ESTEL board enables dialog with the measurement module and relieves it of the Inputs/Outputs functions. It enables remote control and/or remote signalling of certain functions as: "measurement", "zero", "calibration", "alarm".

The analog inputs are used to connect independent monitors in order to follow-up, for example, weather parameters.

The analog outputs enable to send numeric parameters (gas concentration to be analyzed, MUX channels) to analog independent peripherals in order, for example, to store and process several data of several months.

Equipped with an ESTEL board, the analyzer can work as an autonomous unit of analysis.

1.2 TECHNICAL CHARACTERISTICS

Management by specialized micro controller:

- 4 analog inputs of 12 bits, 0-2,5 volts full scale,
- 4 not-insulated analog outputs, configurable into: 0-1 volts, 0-10 volts, 0-20 mA, 4-20 mA (maximum load of 1000 Ohm).
- 4 insulated by optocoupler logic inputs,
- 6 potential-free contacts for remote signalling,
- only one power supply of 8 to 24 volts,
- i2C communication visualization using a LED.

Electric connection:

- 4-point connector for link with Module boards of 2M series,
- Inputs / Outputs centralized on only one female connector SUB D 37 points. This connector is screwed on the rear panel of the analyzer.
- Option of external connection, see paragraph 1.6

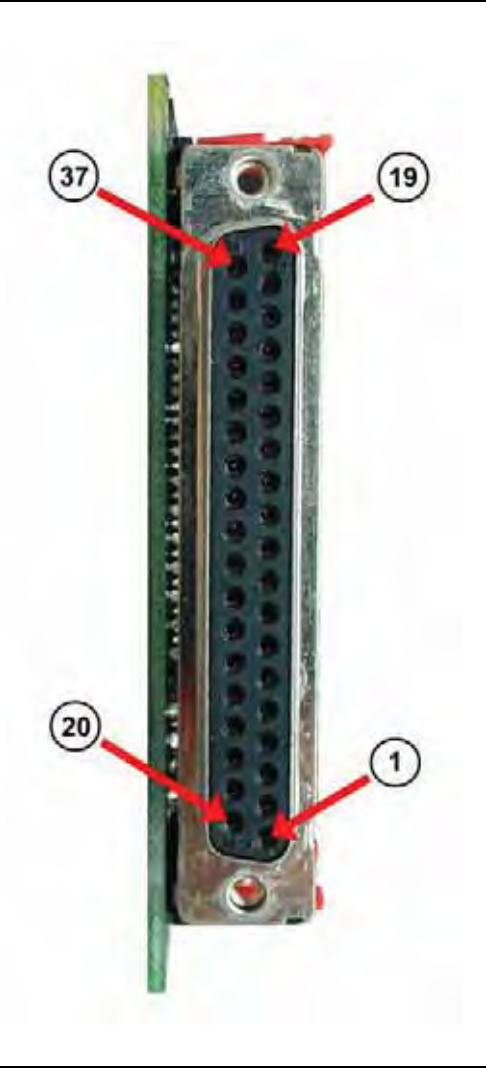
Voltage and current on relays:

- Maximum voltage by relay contact : 50 volts
- Maximum current by relay contact : 1 Ampere at 24 V D.C. (resistive load)

Remote controls:

- By dry contact between (1-4) Remote control and ground Remote control.

1.3 CONFIGURATION

PIN N°	CONNECTION		PIN N°	CONNECTION
1 + 20 GND	Analog output 1		14-33	Relay contact 1
2 + 21 GND	Analog output 2		13-32	Relay contact 2
3 + 22 GND	Analog output 3		12-31	Relay contact 3
4 + 23 GND	Analog output 4		11-30	Relay contact 4
5 + 24 GND	Analog input 1		10-29	Relay contact 5
6 + 25 GND	Analog input 2		9-28	Relay contact 6
7 + 26 GND	Analog input 3			
8 + 27 GND	Analog input 4		15 + 34 GNDI	Remote control 1
			16 + 35 GNDI	Remote control 2
			17 + 36 GNDI	Remote control 3
			18 + 37 GNDI	Remote control 4
			19	5 VCC or + 24 VCC

(*) according to SW5 jumper position
GND: ground
GNDI: insulated ground

Table 1 - Configuration of ESTEL board_index A

Jumpers mark	Symbols	Nature of operations
ST1, ST2, ST8		ESTEL selection, board N° 1
		ESTEL selection, board N° 2
		ESTEL selection, board N° 3
		ESTEL selection, board N° 4
ST3		0 V to ground (default)
		Floating 0 V
DAC1 DAC2 DAC3 DAC4		0-1 V, idem for the 4 DAC.
		0-10 V, idem for the 4 DAC.
		0-20 mA, idem for the 4 DAC.
		4-20 mA, idem for the 4 DAC.
		4-20 mA, idem for the 4 DAC.
P1, P2, P3, P4		4 mA adjustment in 4-20 mA mode

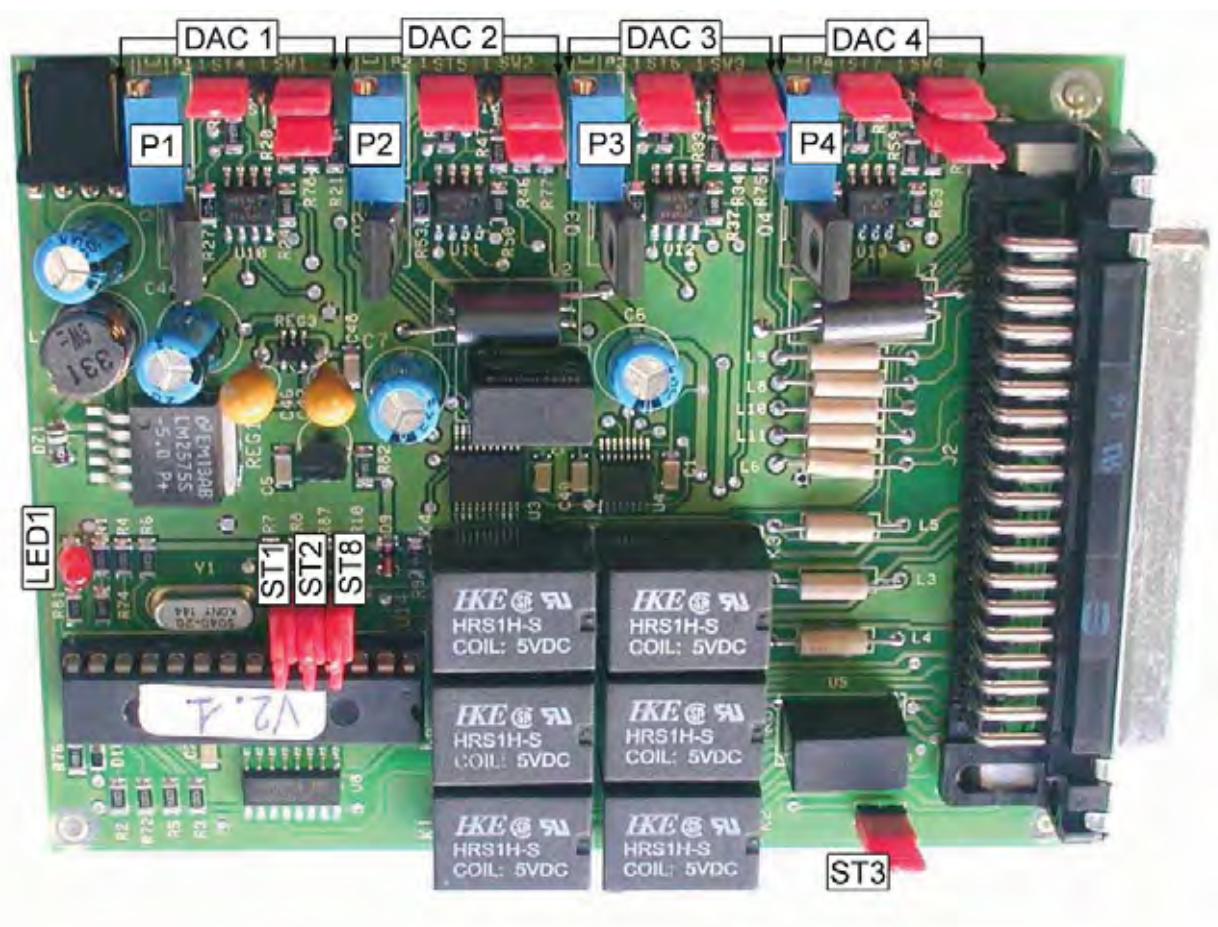


Figure 1 - ESTEL board_index A

Table 2 - Configuration of ESTEL board_index B

Jumpers mark	Symbols	Nature of operations
ST7, ST6, ST5		ESTEL selection, if 1 board
		ESTEL selection, if 2 boards
		ESTEL selection, if 3 boards
		ESTEL selection, if 4 boards
ST8		0 V to ground (default)
		Floating 0V
DAC 1		0-1 V (or optional 2,5 V and 10 V) idem for the 4 DAC
DAC 2		0-10 V, idem for the 4 DAC
DAC 3		0-20 mA, idem for the 4 DAC
DAC 4		4-20 mA, idem for the 4 DAC
SW5		Output 5 V
		Output 24 V

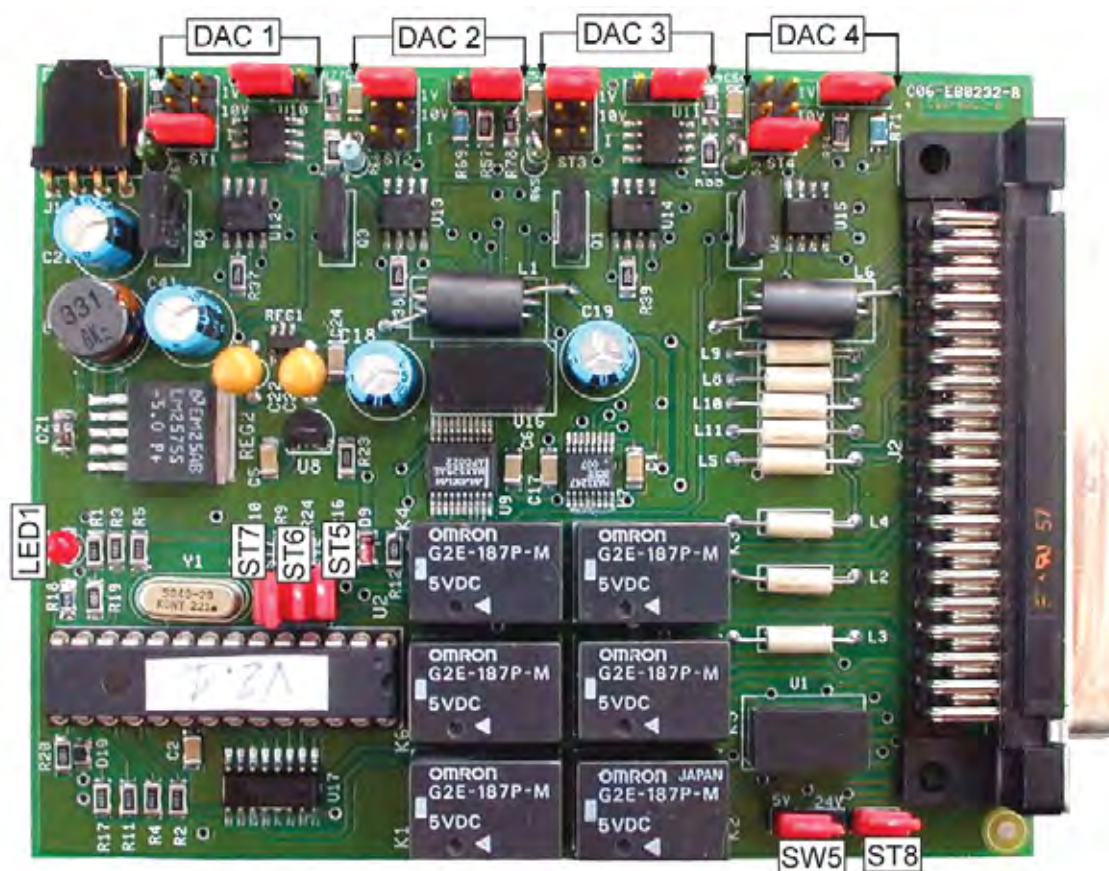


Figure 2 – ESTEL board-index B

Specific configuration of output 0-5 volts instead of 0-10 volts

There are 4 possible configurations for 0-5 volts output:

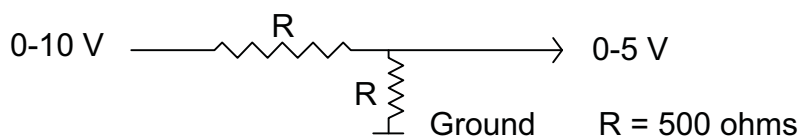
- Board configured into 0-10 volts with addition of a by-2 divider bridge :

The user (customer) carries out himself the operation at input of his acquisition system.

Operating mode:

Connect each analog output, previously configured into 0-10 volts, to ground through 2 resistances of equal value within 500 and 1000 ohms.

Take off the divided-by-2 signal at terminals of the resistance that is connected to the ground.



- Board configured into 0-10 Volts with adjustment of half gain :

In menu *Tests* ⇒ *ESTEL boards*, adjust the A and B coefficients of each channel in order to obtain 0-5V at analog output for 0-4000 pts resolution of analog-to-digital converter.

- Board configured into 0-20 mA

The user (customer) carries out himself the operation at output of his acquisition system.

Operating mode:

Connect each analog output, previously configured into 0-20 mA, to ground using a 250 Ohms resistance, tolerance 1 %.

Voltage, thus generated, is equal to $U_{mV} = 250 \times I_{mA}$, that is to say 5 V for $I = 20$ mA.

Note: place resistance the nearest possible of receiver equipment.

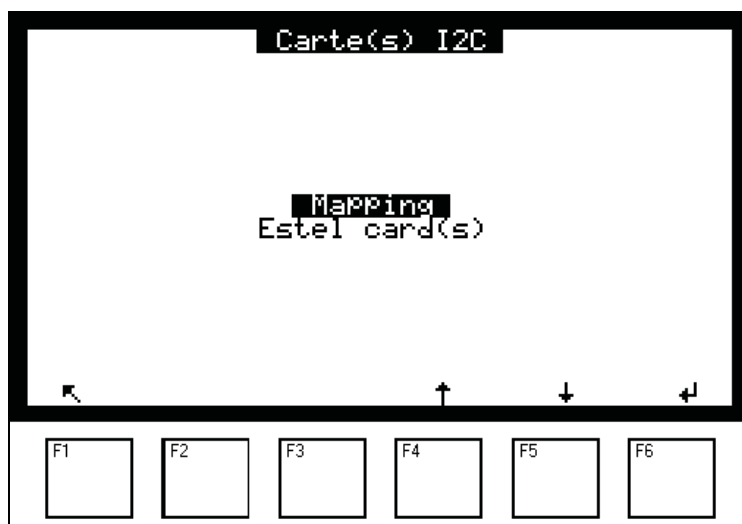
- Modification of gain resistance on ESTEL board

We carry it out if the user (customer) does not accept the other solutions.

1.4 PROGRAMMATION

The ESTEL board programming is carried out from the « ESTEL board » menu of the « Carte(s) I2C » screen.

This menu enables to visualize the effective communications of the various modules and to configure the various ESTEL boards.



The analyzer automatically detects the presence of one or several ESTEL boards and offers menus enabling the user to adjust and configure each board.

1.4.1 ESTEL CARD(S) ⇒ Analog output

To access the various screens of the ESTEL board, select the current function and choose the wanted function using the [↑], [↓] keys.

Estel card(s)			
No: 12--			
Function: Analog output			
Signal	Ran (1 à 4)	Hx + B	Test
1 NO	1 10	0	0000
PPB	100 1000	0	
2 NOX	1 10	1	0000
PPB	100 1000	0	
3 NO2	1 10	1	0000
PPB	100 1000	0	
4 CO	1 10	1	0000
PPM	100 1000	0	
			Points 4000

Navigation: ← * ↑ ↓

Function keys: F1 F2 F3 F4 F5 F6

« Analog output » Function

This screen enables to assign the parameters to the analog outputs for the ESTEL board whose n° indicated in the field « No » is highlighted. These parameters are:

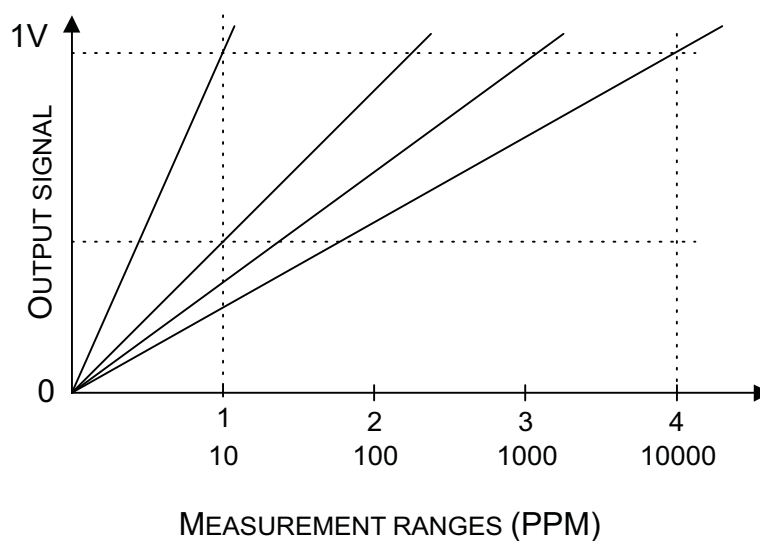
- Concentration of the analyzed gases,
- Auxiliary channels (multiplexer),
- Analog inlets.

REMINDER : On an ESTEL board, analog outputs can be configured into: 0–1 Volt, 0–10 Volt, 0–20 mA, 4–20 mA.

Four ranges are available and correspond to the full scale of the analog output, units are those of the parameters displayed in the « Signal » column.

When signal value is higher than the full scale of the current range, the analyzer switches to the next higher range. It switches again to the lower range when measurement again passes under 85%.

When the user assigns several measurement ranges to only one analog output, he can change the metrological resolution as the here-below curve shows it.



To avoid the ranges switching, the user must assign the same value to the 4 ranges of the parameter he will send to an analog output.

The $Ax+B$ calibration curve is used to adjust the mV signal of the taken-into-account analog output.

The « Test » column is used to test the 5 analog outputs and to adjust the points number.

For a range 1 :

- 0 point (lower scale of output) $\Rightarrow$ 0 volt obtained at output,
- 4000 points (higher scale of output) $\Rightarrow$ 1 volt obtained at output.

F6 key [Points 4000] enables to force the full scale on the whole analog outputs.

1.4.2 ESTEL CARD(S) ⇒ Analog input

Estel card(s)					
No: 112--					
Function Analog input					
	Name	Unit	mV	A	B
1	1-1 Ana.	mV	, 1324 x	1324 1+	0
2	1-2 Ana.	mV	, 1307 x	1307 1+	0
3	1-3 Ana.	mV	, 1542 x	1542 1+	0
4	1-4 Ana.	mV	, 1423 x	1423 1+	0

⏮ * ⏭ ⏴ ⏵

F1	F2	F3	F4	F5	F6
----	----	----	----	----	----

Each ESTEL board has 4 analog inputs : this screen is used to program characteristics of these analog inputs.

- "Name" fields are used to enter a name of 8 alphanumeric digits.
- "Unit" fields are used to select the unit by : none, ppt, ppb, ppm, $\mu\text{g}/\text{m}^3$, mg/m^3 , gr/m^3 , $\mu\text{g}/\text{Nm}^3$, mg/Nm^3 , gr/Nm^3 , $\mu\text{g}/\text{Sm}^3$, mg/Sm^3 , gr/Sm^3 , %, μgr , mgr, gr, mV, U, °C, °K, hPa, mb, b, l, NI, SI, m^3 , l/min, NI/min, SI/min, m^3/h , Nm^3/h , Sm^3/h , m/s ou km/h, in the toggle menu.
- The "Ax + B" fields enable to adjust the calibration curve of each parameter.

1.4.3 ESTEL CARD(S) ⇒ Relay

The screenshot shows a terminal window titled "Estel card(s)". It displays the configuration for card No: 12, with the function set to "Relay". Below this, there is a table with three columns: "Relays", "Type", and "Test". The table lists six relays, all currently set to "Disable", "N.O.", and "OFF". At the bottom of the screen, there are six function keys labeled F1 through F6, each with a corresponding icon.

Relays	Type	Test
1 Disable	N.O.	OFF
2 Disable	N.O.	OFF
3 Disable	N.O.	OFF
4 Disable	N.O.	OFF
5 Disable	N.O.	OFF
6 Disable	N.O.	OFF

Navigation icons: ← * ↑ ↓ ON

Function keys: F1, F2, F3, F4, F5, F6

"Relays" fields are used to control relays according to the following conditions:

Disable	⇒ Relay not assigned
General alarm	⇒ Any operating fault triggers the relay
Range over-range	⇒ Scale 2 over range triggers the relay
Flow rate	⇒ Abnormal flow rate triggers the relays
Temperature	⇒ Abnormal temperature in the analyzer triggers the relay
Pressure	⇒ Barometric pressure in chamber
Zero Air	⇒ On Zero, relay is triggered
Span	⇒ On Span, relay is triggered
Zero-Ref	⇒ On Zero-Ref, relay is triggered
Auto Span	⇒ On Auto-Span, relay is triggered
Warm-Up	⇒ On Warm-up, relay is triggered
Stop mode	⇒ In Stop mode, relay is triggered
Alarm control	⇒ Control detection during threshold over range, relay is triggered.
Alarm or Control	⇒ Relay triggered
Module alarm	⇒ Alarm detected on module, relay triggered
Measure	⇒ Relay triggered
Maintenance	⇒ In Maintenance mode, relay is triggered

- The "Type" fields are used to control (NC) or not (NO) the relays when alarms are OFF.
- "Test" fields are used to manually control these relays.

1.4.4 ESTEL CARD(S) ⇒ Remote controls

The screenshot shows a terminal window titled "Estel card(s)". Inside, it displays "No:112--" and "Function Remotes control". Below this is a table with the heading "Remote controls" and a "Test" column. The table lists four entries, all set to "Disable" with a "Test" value of "OFF". At the bottom of the terminal window are navigation icons: a cursor, an asterisk, and up/down arrows. Below the terminal window is a row of six function buttons labeled F1 through F6.

	Remote controls	Test
1	Disable	OFF
2	Disable	OFF
3	Disable	OFF
4	Disable	OFF

F1 F2 F3 F4 F5 F6

This screen displays the assignment of remote control inlets.

The available assignment choice are : « Inactive », « Stop mode », « Zero Ref. », « Zero », « Span », « Auto span ».

« Test » column is used to display the value read at remote control inlet, for the selected assignment.

1.5 INSTALLATION AND REPLACEMENT OF ESTEL BOARD

- Switch off the analyzer and unplug the mains cable before any maintenance work of the analyzer,
- Respect connection of ESTEL board / MODULE board at J20 when reassembling.

1.5.1 Switch off the analyzer

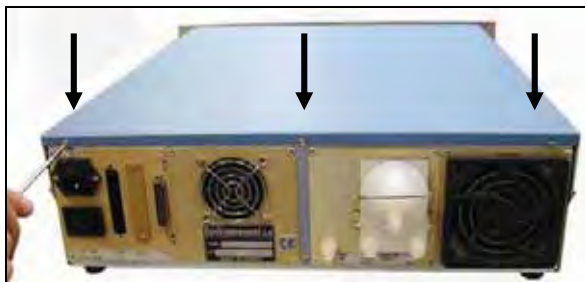


1.5.2 Unplug the mains cable

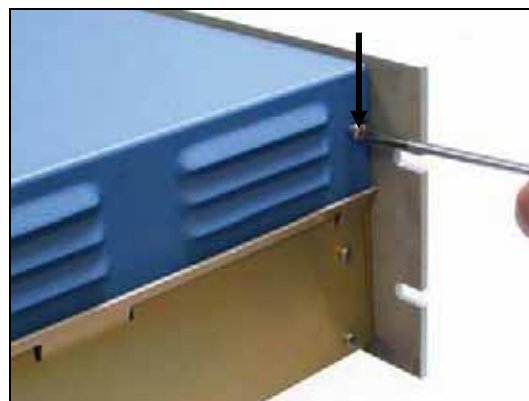


1.5.3 Put off the cover

(1) Unscrew the screws located on the rear panel of the analyzer



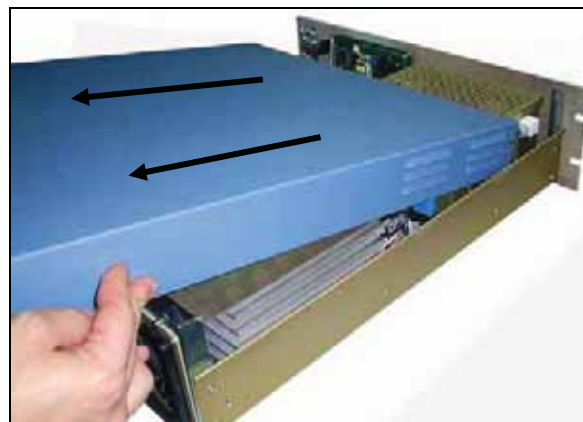
(2) Unscrew the screws located on lateral sides



(3) Lift up the cover



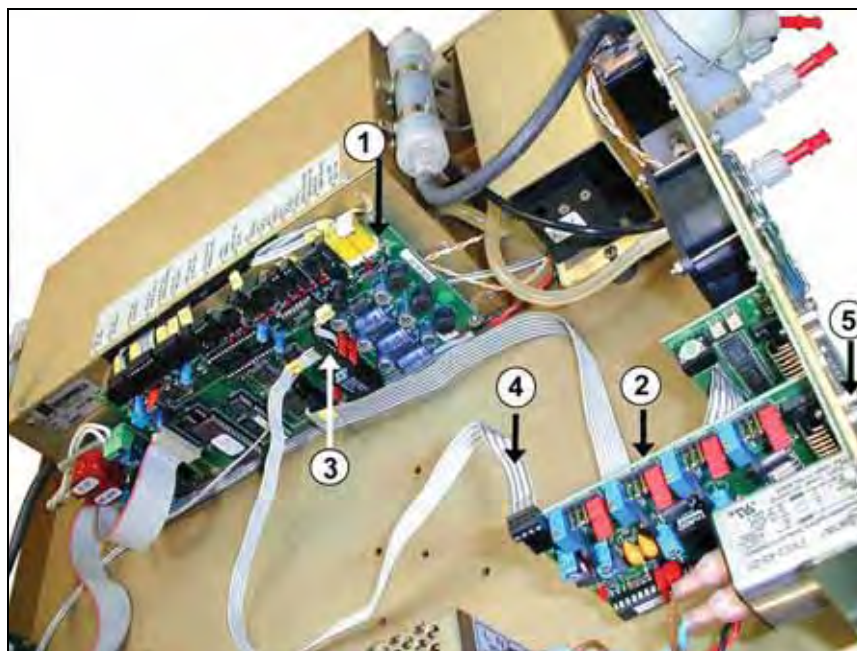
(4) Remove the cover by pulling it backward



If the analyzer is already equipped with an ESTEL board, follow step [1.5.4](#).

If the analyzer is not equipped with ESTEL board, follow step [1.5.5](#).

1.5.4 Dismount ESTEL board



- (1) Module board
- (2) ESTEL board
- (3) J20 connector on Module board
- (4) Connecting cable between Estel board / Module board
- (5) Fixing screw of Estel board on rear panel of the analyzer

Disconnect the connecting cable between ESTEL board (4) / Module board (3).

Unscrew the fixing screws (5) of ESTEL board on rear panel of the analyzer.

Remove ESTEL board.

Configure jumpers of the new board making functionality correspondences according to Table 1 or Table 2.

Re-assemble the board.

1.5.5 Unrivot the back plate (6) of rear panel of analyzer



Then, install at the same place, the new plate (7) delivered with the board

1.5.6 Installation of board inside the analyzer

**ANALYZER
SWITCHED OFF**

(1) Vertically insert the board inside its slot.



(2) Re-screw the board on the slot



(3) Fit again connector on ESTEL board



(4) Then, re-connect on Module board at J20



(5) Replace cover on the analyzer. See [1.5.3](#) .

(6) Connect mains cable and switch on the analyzer. See [1.5.2](#) and [1.5.1](#).

1.6 OPTION OF EXTERNAL CONNECTION

Five different options of ESTEL external connection are available:

DESIGNATION	REFERENCE	MARK
Option of external Estel connection	P10-1337-A	Figure 3
• Cable	• D02-INF-37-37M-M-A	(1)
• Tie-point block interface board	• C10-0012-A	(2)
• DIN track	• G13-IB-18066	(3)

DESIGNATION	REFERENCE	MARK
Option of external Estel connection + 4 insulated outputs.	P10-1338-A	Figure 4
• Cable	• D02-INF-37-37M-M-A	(1)
• Tie-point block interface board	• C10-0012-A	(2)
• Symmetrical DIN track Limit stop	• G13-IB-18066 D03-103-002-26	(3)
• 2-way galvanic insulator	• I11-Jk2000-2	(4)

DESIGNATION	REFERENCE	MARK
Option of external Estel connection + 1 insulated output	P10-1350-A	Figure 4
• Cable	• D02-INF-37-37M-M-A	(1)
• Tie-point block interface board	• C10-0012-A	(2)
• Symmetrical DIN track Limit stop	• G13-IB-18066 D03-103-002-26	(3)
• 1-way galvanic insulator	• I11-Jk2000-1	(4)

DESIGNATION	REFERENCE	MARK
Option of external Estel connection + 2 insulated outputs	P10-1351-A	Figure 4
• Cable	• D02-INF-37-37M-M-A	(1)
• Tie-point block interface board	• 10-0012-A	(2)
• Symmetrical DIN track Limit stop	• G13-IB-18066 D03-103-002-26	(3)
• 2-way galvanic insulator	• I11-Jk2000-2	(4)

DESIGNATION	REFERENCE	MARK
Option of external Estel connection + 3 insulated outputs	P10-1352-A	Figure 4
• Cable	• D02-INF-37-37M-M-A	(1)
• Tie-point block interface board	• C10-0012-A	(2)
• Symmetrical DIN track Limit stop	• G13-IB-18066 D03-103-002-26	(3)
• 2-way galvanic insulator 1-way galvanic insulator	• I11-Jk2000-2 I11-JK2000-1	(4)

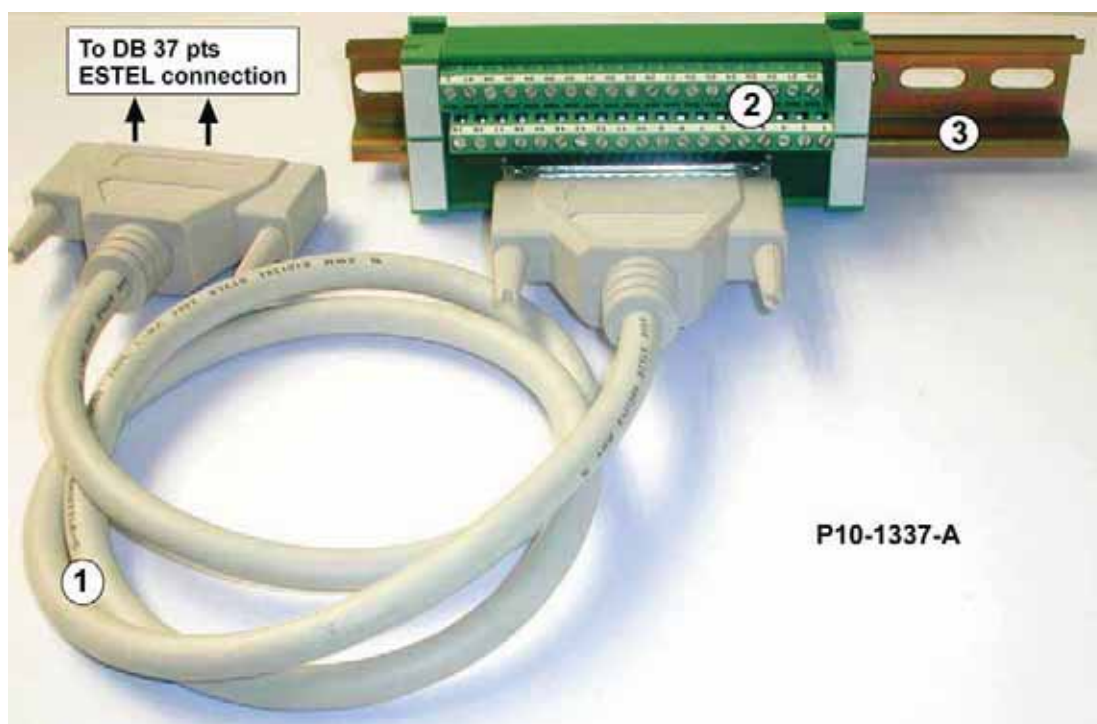


Figure 3 – Option of external connection P10-1337-A

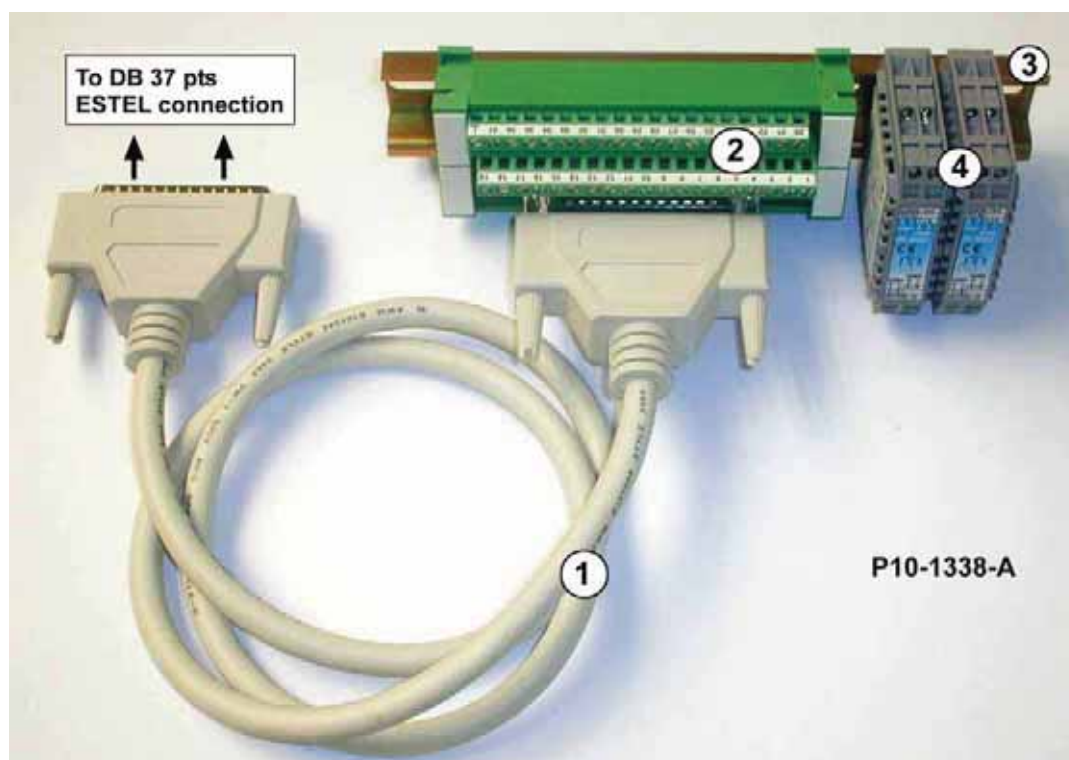


Figure 4 – Option of external connection + 4 insulated outputs P10-1338-A

SOREL Board

BOARD OF LOGIC INPUTS / OUTPUTS

OPTION OF 2M ANALYZERS

- April 2010 -

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SOREL BOARD

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1. SOREL BOARD

SOREL board is a universal board of logic inputs/outputs for the 2M analyzers. It is optional. It is possible to install up to 2 SOREL boards in an analyzer.

1.1 FUNCTION AND USE

SOREL board has 2 functions:

- Relays control (4 in all)
- Remote controls (4 inputs)

SOREL board communicates with measurement module through Bus i2C and relieves it of the Inputs/Outputs functions. It enables the remote control and/or the remote signaling of certain functions as: "measurement", "zero", "calibration" and "alarm".

1.2 TECHNICAL CHARACTERISTICS

Management by specialized micro controller:

- only one power supply of 24 volts,
- 4 logic inputs,
- 4 contacts for remote signaling, potential configurable by the user,
- Visualization of i2C communication using a LED.

Electric connection:

- 4-point connector for link with module boards of 2M series,
- Inputs / Outputs centralized on plug-in connector. This connector is screwed on the rear panel of the analyzer using a back-plate.

Voltage and current on relays:

- Maximum voltage by relay contact: 50 volts D.C.
- Maximum current by relay contact : 1 Ampere at 24 V D.C. (resistive load)

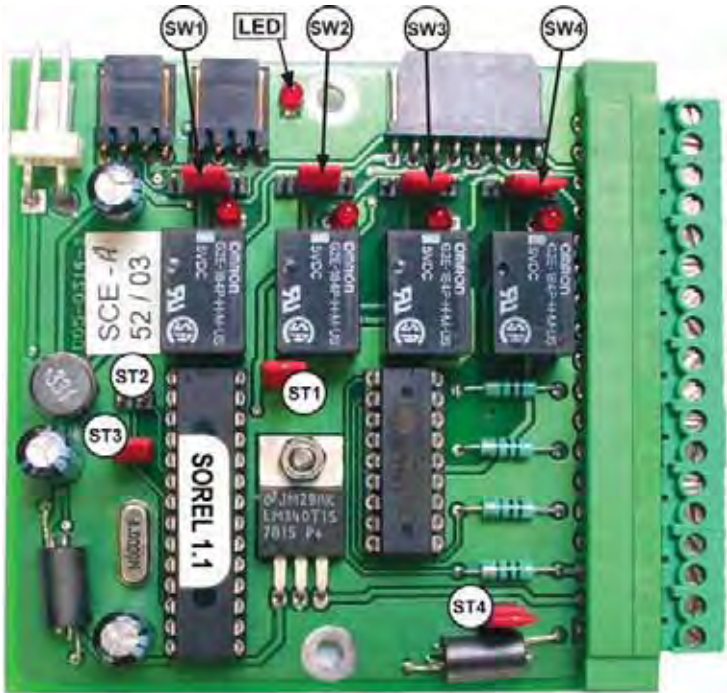
Voltage at logic inputs:

- Maximum voltage 24 V D.C.

1.3 CONFIGURATION

Table 1 - Configuration of SOREL board

JUMPERS MARK	SYMBOLS	NATURE OF OPERATIONS
ST1, ST2, ST3		SOREL selection, board N° 1
		SOREL selection, board N° 2
		SOREL selection, board N° 3
		SOREL selection, board N° 4
		SOREL selection, board N° 5
		SOREL selection, board N° 6
		SOREL selection, board N° 7
		SOREL selection, board N° 8
ST4		0 V to ground (default)
		Floating 0 V
SW1 Relay nb 1 SW2 Relay nb 2 SW3 Relay nb 3 SW4 Relay nb 4		Potential free contact
		Referenced contact to 0 V and 24V



18	Relay contact #4 (-)
17	Relay contact #4 (+)
16	Relay contact #3 (-)
15	Relay contact #3 (+)
14	Relay contact #2 (-)
13	Relay contact #2 (+)
12	Relay contact #1 (-)
11	Relay contact #1 (+)
10	GND
9	Remote control #4
8	GND
7	Remote control #3
6	GND
5	Remote control #2
4	GND
3	Remote control #1
2	+15V
1	+24V

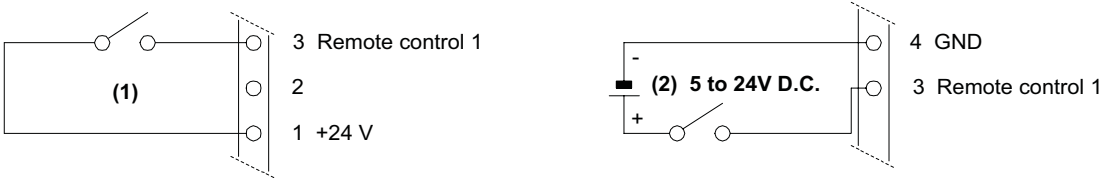


Figure 1 – SOREL board

NOTE: Output relay contacts are normally open when analyzer is switched off.

1.3.1 Programmation

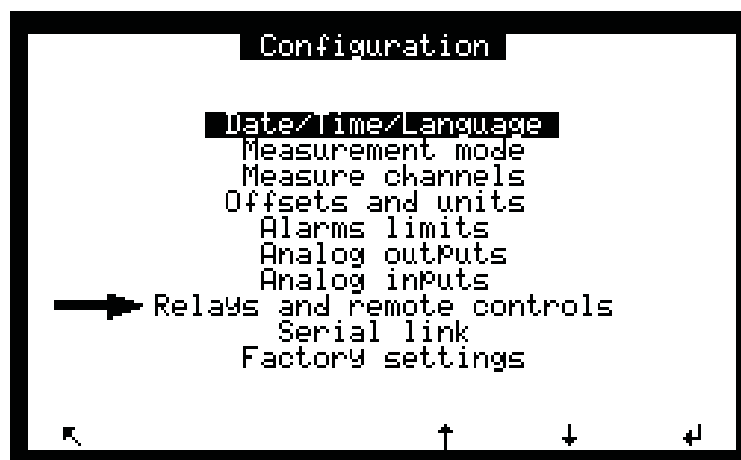


The hereafter screens (§ 1.3.1 à § 1.3.3) are given as example.

Refer to technical manual of the analyzer in which SOREL board is installed.

The analyzer automatically detects the presence of one or several SOREL and/or ESTEL boards and offers menus enabling the user to adjust and configure each board.

- In CONFIGURATION menu of main software program, the "Analog outputs", "Analog inputs", "Relays and remote controls" items are only displayed if the SOREL and/or ESTEL board option is available. **Only, the sub-menu "Relay and remote controls is necessary to program the SOREL board.**



- In the TESTS menu of the main program, the item "ESTEL card" is displayed if one SOREL board at least is detected.

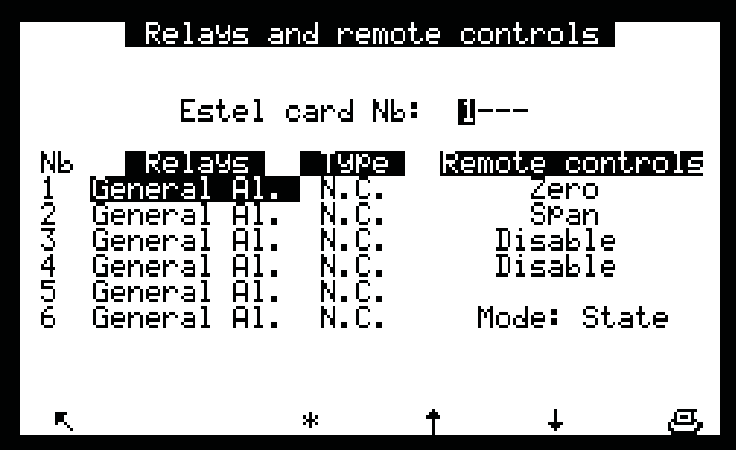
The same screen as for ESTEL board must be used, but it is necessary not to take into account of data about analog inputs and outputs.



1.3.2 CONFIGURATION ⇒ Relays and remote controls

This screen enables to configure function of each input / output of SOREL and/or ESTEL board(s).

- **SOREL board is displayed as an ESTEL board,**
- The "ESTEL card Nb: " is used to select what board to configure.
- "Relays" fields are used to control the relays according to each analyzer: refer to *CONFIGURATION ⇒ Relays and remote controls* paragraph of the technical manual of your analyzer.



Relays and remote controls			
Estel card Nb: 1---			
Nb	Relays	Type	Remote controls
1	General Al.	N.C.	Zero
2	General Al.	N.C.	Span
3	General Al.	N.C.	Disable
4	General Al.	N.C.	Disable
5	General Al.	N.C.	
6	General Al.	N.C.	Mode: State

- The "Type" fields are used to program relays into "normally closed" (NC) or "normally open" (NO) when alarms are OFF.
- The "Mode" field is used to configure the working mode of remote controls.

Two different modes are possible:

"State" mode: control is activated as long as remote control is active (closed contact).

"Rise" mode: control is activated when state modification of remote control is detected. When it is down, control remains active. A new modification of state de-activates control.

1.3.3 TESTS ⇒ ESTEL card

This screen is used to check operation of remote controls and relays.

Analog functionalities are not active for SOREL board.

The screenshot shows a terminal window titled "Estel Card(s)". It displays a table with 6 rows of data. The first row is a header with labels: "Nb", "Hx + B", "Out", "ADC", and "Rem". The subsequent rows show data for cards 1 through 6. Card 1 has "Hx + B" value 1, "Out" value 0, "ADC" value 0586, and "Rem" value OFF. Cards 2 through 6 have "Hx + B" value 1, "Out" value 0, "ADC" values 0489, 0835, 0538, and "Rem" value OFF. Below the table, there are labels "(Points)" and "(mv)". At the bottom, there are navigation keys: a left arrow, an asterisk, an up arrow, a down arrow, and "0/OFF".

Nb	Hx + B	Out	ADC	Rem
1	1	0	0586	OFF
2	1	0	0489	OFF
3	1	0	0835	OFF
4	1	0	0538	OFF
5		ON		
6		ON		

(Points) (mv)

← * ↑ ↓ 0/OFF

The "Estel card Nb:" field is used to select the board to be tested.

The "Out" fields are used to control the relays manually.

The "Rem." fields are used to know state of these logic inputs.

Definition of the specific keys of this screen:

Opens all the relay contacts.

Closes all the relay contacts.

1.4 INSTALLATION OR REPLACEMENT OF SOREL BOARD

- Switch off the analyzer and unplug the mains cable before any maintenance work in the analyzer,
- Respect connection of SOREL board / MODULE board at J20 when reassembling.

1.4.1 Switch off the analyzer



1.4.2 Unplug the mains cable



1.4.3 Put off the cover

(1) Unscrew the screws located on the rear panel of the analyzer



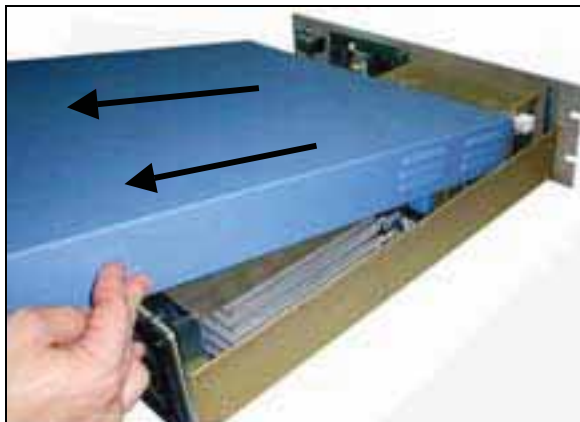
(2) Unscrew the screws located on lateral sides



(3) Lift up the cover



(4) Remove the cover by pulling it backward



If the analyzer is already equipped with a SOREL board, follow step 1.4.4

If the analyzer is not equipped with ESTEL board, follow step 1.4.5

1.4.4 Dismount SOREL board



- (1) Module board
- (2) SOREL board
- (3) Connector J20 on Module board
- (4) Connecting cable between Sorel board / Module board
- (5) Fixing screw of Sorel board on rear panel of the analyzer

Disconnect the connecting cable between SOREL board (4) / Module (3) board.

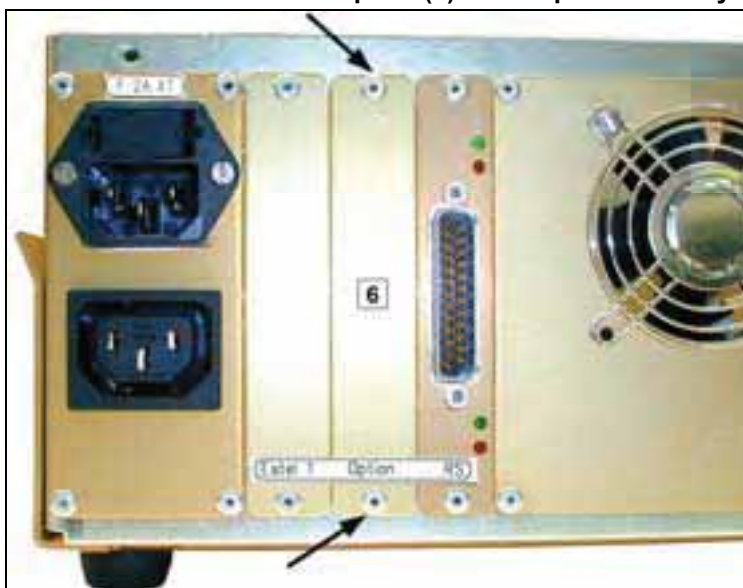
Unscrew the fixing screws (5) of SOREL board on rear panel of the analyzer.

Remove SOREL board.

Configure jumpers of the new board carrying out correspondence of functionalities according to Table 1.

Re-assemble the board.

1.4.5 Unrivet the back plate (6) of rear panel of analyzer



Then, install at the same place the new plate (7) delivered with the board

1.4.6 Installation of board inside the analyzer

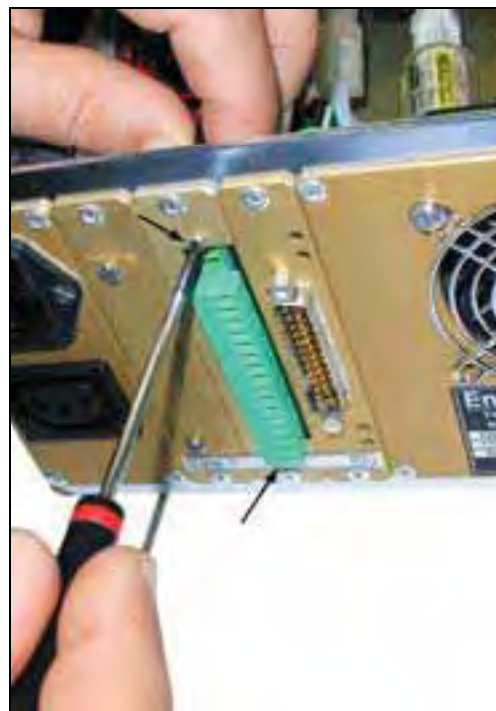


**ANALYZER
SWITCHED OFF**

(1) Vertically insert the board inside its slot.



(2) Re-screw the board on back plate



(3) Fit again connector on SOREL board



(4) Then, re-connect on Module board at J20



(5) Replace cover on the analyzer. See 1.4.3.

(6) Connect mains cable and switch on the analyzer. See 1.4.2 and 1.4.1.